2017, Fall

1. (a) Let n be a positive integer. Determine all automorphisms of $\mathbb{Z}_n$; that is, determine all isomorphisms from $\mathbb{Z}_n$ onto itself.

   (b) Let n be a positive integer and let p be a prime. Determine the number of automorphisms of $\mathbb{Z}_{p^n}$.

Proof. a). Let $\varphi: \mathbb{Z}_n \to \mathbb{Z}_n$ be the endomorphism.

Given $k \in \mathbb{Z}_n$, $\varphi(k) = \varphi(k \cdot 1) = \varphi(\underbrace{1+1+\cdots+1}_{k \text{ times}}) = k \cdot \varphi(1)$.

Therefore, the endomorphism is determined by the value of $\varphi(1)$.

And, since $\mathbb{Z}_n$ is finite, φ is an automorphism $\Leftrightarrow$ φ is surjection.

That is, $\Leftrightarrow$ $\varphi[\mathbb{Z}_n] = \{\varphi(1), \varphi(2), \cdots, \varphi(n-1)\} = \varphi(1) \cdot \mathbb{Z}_n = \mathbb{Z}_n$.

$\Leftrightarrow$ for some $x \in \mathbb{Z}$, $\varphi(1)x \equiv 1 \pmod n$ ($\Leftrightarrow 1 \in \varphi(1) \cdot \mathbb{Z}_n$)

$\Leftrightarrow \gcd(\varphi(1), x) = 1$.

Therefore, $Aut(\mathbb{Z}_n) = \{\varphi: \mathbb{Z}_n \to \mathbb{Z}_n \mid \varphi \text{ is Homomorphism s.t. } \gcd(\varphi(1), n) = 1\}$.

b). Notice that: for $1 \le a \le p^n$, $\gcd(a, p^n) \neq 1 \Leftrightarrow p \mid a$.

That is, $a \in \{k \cdot p \mid 1 \le k \le p^{n-1}\}$, thus $\varphi(p^n) = p^n - p^{n-1}$.

Therefore, $|Aut(\mathbb{Z}_{p^n})| = p^n - p^{n-1}$. ☒

2017, Fall

3. Let E/F be a Galois extension whose Galois group G is a noncyclic group of order p^2.

(a) For every intermediate field K with $F \subseteq K \subseteq E$, prove that there exists $\alpha \in E$ such that $K = F(\alpha)$.

(b) Determine the number of fields K satisfying $F \subsetneq K \subsetneq E$.

Proof. Notice that: every group of order p^2 is $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$, because: by the Lagrange's theorem, the center of G satisfies that $|Z(G)| \in \{1, p, p^2\}$. But, if $|Z(G)| = 1$, then by the class equation,

$$p^2 = |Z(G)| + \sum |G : C_G(g_i)| = 1 + pk \text{ for some } k \in \mathbb{N},$$

which is contradiction, by $p \mid p^2 - pk$ but $p \nmid 1$.
Now, $|Z(G)| \in \{p, p^2\}$, therefore $|G/Z(G)| \in \{1, p\}$, cyclic, thus G is abelian. By the fundamental theorem of finitely generated abelian group, $G \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$, but by the assumption, $G \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

a). Since E/F is finite extension, so all intermediate field K is also finite extension, and since $Gal(E/F)$ is abelian, as isomorphic to $\mathbb{Z}_p^2$, therefore the intermidate K is Galois over F, so separable.
By the primitive element theorem, finite separable extension K/F is simple extension, so proved.

b). By the fundamental theorem of Galois, there is one-to-one correspondence between

$$\{H \leq Gal(E/F)\} \longleftrightarrow \{F \leq K \leq E\}.$$

Since $\mathbb{Z}_p \times \mathbb{Z}_p$ has the subgroups

$\langle (1,0) \rangle, \langle (1,1) \rangle, \langle (1,2) \rangle, \cdots, \langle (1, p-1) \rangle, \langle (0,1) \rangle$

and $\{(0,0)\}$, $\mathbb{Z}_p \times \mathbb{Z}_p$ itself, so $p+1$ is the number we desired. □

I2017. Fall.

6. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Suppose that

$$f(0) = \frac{\pi}{2}, \qquad g(x) = \cos(f(x))$$

is monotonically decreasing. Prove that f is uniformly continuous.

Proof. Suppose that: for some $x \in \mathbb{R}$, $f(x) < 0$. If $x < 0$, then $f(x) < 0 < f(0) = \frac{\pi}{2}$ implies for some $y \in (x, 0)$, $f(y) = 0$. But, it implies $x < y < 0$ and

Note: Bounded Continuous is not necessarily uniformly continuous.
Counter example: $f(x) = \sin \frac{1}{x+1}$ on $(-1, 1)$.

[2017. Fall]

7. Let $A(t) = (a_{ij}(t))$ be an $n \times n$ matrix that is nonsingular for every $t \in \mathbb{R}$. Compute

(a) $\dfrac{d}{dt}(A(t)^{-1})$;

(b) $\dfrac{d^2}{dt^2}(A(t)^{-1})$.

Prof. Since $A(t) \cdot A(t)^{-1} = I$

2017. Fall

15. Prove or disprove each of the following statements.

(a) A product of metric spaces is metrizable.

(b) Every connected open subset of $\mathbb{R}^n$ is path connected.

Proof.

a). Let I be an uncountable index set, and define

$$X = \prod_{\alpha \in I} \mathbb{R} = \mathbb{R}^I.$$

Suppose that X is Metrizable. Then, it is first-countable, therefore for $0 = (0)_{\alpha \in I} \in X$, there exists a countable local basis

$$\mathcal{B} = \{N_k \mid k \in \mathbb{N}\}.$$

For each $k \in \mathbb{N}$, let B_k be a basis element such that $0 \in B_k \subseteq N_k$. Then, by definition of product topology,

$$\pi_\beta[B_k] = \mathbb{R} \quad \text{all } \beta \in I \text{ but finitely many indices } I_k \subset I.$$

where $\pi_\beta : X \to \mathbb{R}_\beta : (x)_{\alpha \in I} \mapsto x_\beta$, projection.

Since $\bigcup_{k \in \mathbb{N}} I_k$ is countable union of finite sets, so countable.

choose $\gamma \in I \setminus \left(\bigcup_{k \in \mathbb{N}} I_k\right)$. That is, for all $k \in \mathbb{N}$, $\pi_\gamma[B_k] = \mathbb{R}$.

Now, a set

$$V := \{(x)_\alpha \in X \mid |x_\gamma| < 1\}$$

is an open neighbor of x, but for all $k \in \mathbb{N}$, $B_k \not\subset V$ so that $N_k \not\subset V$, which is contradiction.

b). Let $C \subset \mathbb{R}^n$ be a connected open set. Let $x_0 \in C$ be fixed.
Define $A = \{x \in C \mid$ there is a path from x_0 to $x\}$.

Given $x \in A$, there is $r > 0$ such that $B_r(x) \subseteq C$ since C is open, and since the ball $B_r(x)$ is path-connected, so that $B_r(x) \subseteq A$.
Therefore, A is open.
Given $x \in C \setminus A$, there is $r > 0$ such that $B_r(x) \subseteq C$, similarly.
But, if $B_r(x) \cap A \neq \emptyset$, then there is a path from x_0 to x, so that $x \in A$, it contradicts. Thus $B_r(x) \subseteq C \setminus A$.

Now, since $A \neq \emptyset$, if $C \setminus A \neq \emptyset$, then $\{A, C \setminus A\}$ is the separation, so $C \setminus A = \emptyset$, that is $A = C$. By definition, A is path-connected, thus $C = A$ is path-connected. ∎

2017. Fall.

16. (a) Show that the one-point compactification of $\mathbb{R}^2$ is homeomorphic to S^2.

(b) Let $X = \mathbb{C}^2 \setminus \{(0,0)\}$ and define a relation $\sim$ on X by

$$(x_1, y_1) \sim (x_2, y_2) \iff x_1 = tx_2,\ y_1 = ty_2 \quad \text{for some } t \in \mathbb{C} \setminus \{0\}.$$

Show that the quotient space $X/\sim$ is homeomorphic to S^2.

2018. Fall.

1. Let $A \in M_{n\times n}(\mathbb{R})$ such that $A^2 = 0$. Prove

$$2 \cdot \operatorname{rank} A \le n.$$

Proof. Given $\alpha \in \mathbb{R}^n$, $A(A\alpha) = A^2\alpha = 0$, so $\operatorname{Im} A \subseteq \ker A$. So, $\dim \operatorname{Im} A \le \dim \ker A$. By the dimension theorem,

$$n = \text{nullity } A + \operatorname{rank} A \ge 2\operatorname{rank} A$$

□

2. Determine over $\mathbb{R}$,

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

is diagonalizable.

Proof. The characteristic polynomial of A is

$$\det \begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2\cdot(t-2) = (t-2)(t^2-3t+2) = (t-1)(t-2)(t-2).$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to characteristic polynomial, so we need to check that $(A-I)(A-2I) = 0$,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of A is $(t-1)(t-2)$, it implies A is diagonalizable, because the minimal polynomial splits over $\mathbb{R}$ and separable. Explicitly,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \begin{cases} x = -2z \\ x+y+z = 0 \end{cases} \Rightarrow \ker(A-I) = \operatorname{span}\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}\right\}$$

$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \{x + z = 0 \Rightarrow \ker(A-2I) = \operatorname{span}\left\{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right\},$$

Therefore, $A = UDU^{-1}$ where $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 2 & & \\ & 1 & \\ & & 0 \end{pmatrix}$.

[2018. Fall,

3. Let R be a commutative ring and let $x \in R$ such that $x^{2018} = 0$. Prove that $1+x$ is unit.

Proof. observe that

$$(1+x)\cdot(-x^{2017} + x^{2016} - x^{2015} + \cdots - x + 1)$$

$$= 1 - x + x^2 + \cdots + x^{2016} - x^{2017}$$
$$+ x - x^2 + \cdots - x^{2016} + x^{2017} - x^{2018}$$
$$= 1, \quad \text{since } x^{2018} = 0.$$

So, $1 - x + \cdots + x^{2016} - x^{2017}$ is inverse of $1+x$.

4. Prove that A_4 does not have a subgroup of order 6.

Proof. Since $|S_4 : A_4| = 2$, $|A_4| = 4 \cdot 3 = 12$.

Suppose that A_4 has a subgroup H of order 6.

Then, $|A_4 : H| = 12/6 = 2$, so $gH = Hg$ for any $g \in A_4$, it means $H \trianglelefteq A_4$ is normal.

Consider the afforded Homomorphism of A_4 acts A_4/H as left multiplication:

$$\varphi : A_4 \to S_{A_4/H} ; \quad g \mapsto \left(xH \mapsto gxH \right)$$

And, since every 3-cycle $\sigma \in A_4$ satisfies

$$\left(\varphi(\sigma)\right)^3 = \varphi(\sigma^3) = \varphi(id) = id$$

and $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$ so $\varphi(\sigma) = id$. It implies all 3-cycles are contained in $\ker \varphi$. Meanwhile, observe that $(1\,2\,3), (1\,2\,4) \in A_4$ satisfies

$$(1\,2\,3)^{-1}(1\,2\,4)(1\,2\,3) = (1\,3\,2)(1\,2\,4)(1\,2\,3) = (1\,4\,3)$$

$(1\,2\,3)(1\,4\,3) = (1\,4)(2\,3)$, ... in this way, A_4 is generated by all 3-cycles in A_4, so $\ker \varphi = A_4$. By the first isomorphism theorem, $A_4/\ker \varphi \cong \operatorname{Im} \varphi$ is trivial so the action is trivial.

But, the left coset action is transitive, it cannot be trivial. Contradiction. □

[Similar problems.
1. Show that A_5 has no subgroup of order 15.
Prof. $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$.
Suppose that there is a subgroup $H < A_5$ of order 15.
By the Sylow Theorem, $\#Syl_3(H) \in \{1\}$ and $\#Syl_5(H) \in \{1\}$, that is, there are order 3 subgroup $P < H$ and order 5 subgroup $Q < H$, and both are normal subgroup of H. Since $P \cong C_3$ and $Q \cong C_5$ both are cyclic, let $x \in P$ and $y \in Q$ such that $P = \langle x \rangle$ and $Q = \langle y \rangle$.
Consider the afforded Homomorphism by H acts P by conjugation

$$\varphi : H \to S_P : h \mapsto (k \mapsto hkh^{-1}).$$

Then, $\ker \varphi = \{h \in H : hkh^{-1} = k \text{ for all } k \in P\} = C_H(P)$ and $H/\ker \varphi \cong \operatorname{Im} \varphi \leq Aut(P)$. But, since $Aut(P) \cong \mathbb{Z}/2\mathbb{Z}$, and $|H|$ has no factor 2, odd number, so $|\ker \varphi| = |H|$, that is, $C_H(P) = H$. Now, x and y commute, therefore $|xy| = 15$.
But, A_5 cannot have order 15 element, because:
order of permutation in S_5 is determined by the least common multiplier of disjoint cycle decomposition, but $lcm\ 5 = 5$, $lcm(4,1) = 4$, $lcm(3,2) = 6$, $lcm(2,1,1,1) = 2$, ...
the lcm cannot be 15, so it contradicts.

[Similar problem
Let $G_1 = S_3 \times C_2$.
1. Find all normal subgroup of G_1.
2. Prove that G_1 has no normal subgroup of order 4.
Proof. Define three maps:

$$\varphi_1 : G_1 \to S_3 : (\sigma, a) \mapsto \sigma$$

$$\varphi_2 : G_1 \to C_2 : (\sigma, a) \mapsto a$$

$$\varphi_3 : G_1 \to \{\pm 1\} : (\sigma, a) \mapsto \operatorname{sgn}\sigma \cdot \pi(a) \quad \text{where } C_2 \overset{\pi}{\cong} \{\pm 1\}.$$

$$\varphi_4 : G_1 \to \{\pm 1\} \times C_2 : (\sigma, a) \mapsto (\operatorname{sgn}\sigma, a).$$

$$\varphi_5 : G_1 \to \{\pm 1\} : (\sigma, a) \mapsto \operatorname{sgn}\sigma$$

Then, φ_1 and φ_2 are Homomorphisms, being the Coordinatively projection, and φ_3 is also Homomorphism, because Sgn, π are Homomorphism and $\{\pm 1\}$ is abelian. Precisely,

$$\varphi_3((\sigma_1, a_1)\cdot(\sigma_2, a_2)) = \varphi_3(\sigma_1\sigma_2, a_1 a_2) = \operatorname{sgn}\sigma_1 \operatorname{sgn}\sigma_2 \pi(a_1)\pi(a_2)$$

$$= \operatorname{sgn}\sigma_1 \pi(a_1) \operatorname{sgn}\sigma_2 \pi(a_2) = \varphi_3(\sigma_1, a_1)\cdot\varphi_3(\sigma_2, a_2).$$

And, φ_4 is Homomorphism, being entrywise Homomorphism, and φ_5 is Homomorphism, being composition of projection and Sgn.

Now, the kernel

$\ker\varphi_1 = \{id\} \times C_2$, $\ker\varphi_2 = S_3 \times \{x^0\}$, $\ker\varphi_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$

$\ker\varphi_4 = A_3 \times \{x^0\}$, $\ker\varphi_5 = A_3 \times C_2$.
are we desired.

Are these all possibilities?
Observe that: if $N \subseteq G_1 \times G_2$ is normal, then the image of projections $\pi_i : G_1 \times G_2 \to G_i : (g_1, g_2) \mapsto g_i$, $\pi_i[N]$ is also normal of $\pi_i[G_1 \times G_2] = G_i$, because image of normal subgroup is normal in the range, and the projections are onto.
But, as we know, normal subgroup of S_3 is only $\{id\}$, A_3, S_3 and normal subgroup of $\{\pm 1\}$ is $\{1\}$ and $\{\pm 1\}$.

2018. Fall.

7. Consider the identity $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$.

a). Determine the sequence $(C_n) \subseteq \mathbb{R}$ and the range of $x \in \mathbb{R}$ for the identity hold.

b). Show that for each $n \geq 0$,

$$\sum_{k=0}^{n} \frac{C_k}{(n-k)!} = 1.$$

Proof. a),

Notice that $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$ for all $x \in \mathbb{R}$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for all } |x|<1.$$

Therefore, the coefficient C_n is determined by the Cauchy Product

$$C_n = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in $\mathbb{R} \cap (-1,1) = (-1,1)$. ∎5m.

b). Since

$$\frac{1}{1-x} = e^x \cdot \sum_{n=0}^{\infty} C_n x^n, \text{ So we can write for } x \in (-1,1)$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each $n \geq 0$, because these are infinitely differentiable, So the coefficients of Powe series are determined uniquely by the taylor Theorem. Thus,

$$1 = \sum_{k=0}^{n} \frac{C_k}{(n-k)!} \quad \text{for all } n \geq 0. \text{ ∎4m.}$$

2018. Fall.

8. Find a continuous and bounded function $f:(0,1] \to \mathbb{R}$:

a) which does not extend to a continuous function on $[0,1]$.

b). which has neither maximum nor minimum.

Proof. a). Let $f(x) = \sin\left(\frac{1}{x}\right) \quad (x \in (0,1])$.

Then, f is continuous, being composition of continuous map, and is bounded, since the sine function is bounded.

But, let $x_n = \frac{1}{2n\pi}$ and $y_n = \frac{1}{2n\pi + \frac{\pi}{2}}$, for each $n \in \mathbb{N}$.

Then, $x_n, y_n \to 0$ as $n \to \infty$ but

$f(x_n) \to 0$ and $f(y_n) \to 1$ as $n \to \infty$,

So, if there is continuous extension of f on $[0,1]$, then $f(0) = 0 = 1$, it contradiction. So, $x \mapsto \sin(1/x)$ is that we desired.

b). Let $g(x) = (1-x) \cdot \sin\left(\frac{1}{x}\right)$. Then, similarly, g is continuous and bounded, but $g[(0,1]] = (-1,1)$, so it has no maximum and minimum.

More concretely, consider the $\{y_n\}$ in a). Then,

$$g(y_n) = (1 - y_n) \to 1 \text{ as } n \to \infty,$$

So $\sup\{g(x) \mid 0 < x \leq 1\} \geq 1$. But, there is no $x \in (0,1]$ such that $g(x) = 1$, so g cannot have maximum, and similarly cannot have minimum.

□

[2018. Fall.

9. (10 pts) Suppose that a sequence of continuous real-valued functions $(f_n)_{n=1}^{\infty}$ on $[0,1]$ converges uniformly to a function f on $[0,1]$. Prove that $(f_n^2)_{n=1}^{\infty}$ converges to f^2 uniformly on $[0,1]$.

Proof. Let $\varepsilon=1$ be fixed. Then, there exists $N\in\mathbb{N}$ such that

$n\geq N \Rightarrow |f_n(x)-f(x)|<\varepsilon=1$, for any $x\in[0,1]$.

Meanwhile, since each f_n is continuous and $f_n\to f$ uniformly, f is also continuous uniformly on a compact set $[0,1]$, so $f[[0,1]]$ is also compact, thus f is bounded. Now, for $n\geq N$,

$$|f_n(x)|\leq|f(x)|+|f_n(x)-f(x)|\leq \sup_{x\in[0,1]}|f(x)|+1 \quad \text{for all } x\in[0,1],$$

therefore, $\{f_n\}$ is uniformly bounded by

$$M:=\max\left(\left\{\sup_{x\in[0,1]}|f_n(x)| \;\middle|\; n=1,2,\cdots,N\right\}\cup\left\{\sup_{x\in[0,1]}|f(x)|+1\right\}\right)$$

Now, given $\varepsilon>0$, let $N\in\mathbb{N}$ such that

$$n\geq N \Rightarrow \sup_{x\in[0,1]}|f_n(x)-f(x)|<\frac{\varepsilon}{2M}.$$ Then, for $n\geq N$,

$$|f_n(x)^2-f(x)^2|=|f_n(x)+f(x)||f_n(x)-f(x)|\leq 2M\cdot\frac{\varepsilon}{M}<\varepsilon$$

for all $x\in[0,1]$, so proved. ☺

[2019, Spring.

1. (10 pts.) Let M be an $n \times n$ matrix over a field. The (i,j)-minor of M, denoted by $\widehat{M}_{ij}$, is the determinant of the $(n-1) \times (n-1)$ submatrix of M that is given by removing the i-th row and the j-th column from M. The adjoint of M, denoted by $\mathrm{adj}(M)$, is the $n \times n$ matrix whose (i,j)-entry is given by $(-1)^{i+j}\widehat{M}_{ji}$. Prove that if A and B are invertible $n \times n$ matrices over a field, then $\mathrm{adj}(AB) = \mathrm{adj}(B)\,\mathrm{adj}(A)$. (Hint: $M\,\mathrm{adj}(M) = \det(M)\,I_n$.)

Proof. Observe that

$$\mathrm{adj}(AB) = |B^{-1}A^{-1}AB| \cdot \mathrm{adj}(AB) = |B^{-1}A^{-1}| \cdot \det(AB) \cdot I_n = |B^{-1}A^{-1}| \cdot \det A \cdot \det B$$

$$= |B^{-1}| \cdot \det B \cdot A^{-1} \cdot \det A = \mathrm{adj}(B) \cdot \mathrm{adj}(A)$$

So, proved.

2019. Spring.

2. (10 pts.) Let $V = \mathcal{P}_2(\mathbb{R}) = \{f(x) \in \mathbb{R}[x] \mid \deg f(x) \leq 2\}$ and let T be the linear operator on V defined by $T(f(x)) = f(0) + f(1)(x + x^2)$. Determine whether T is diagonalizable, and, if T is diagonalizable, find a basis $\mathfrak{B}$ for V such that $[T]_{\mathfrak{B}}$ is a diagonal matrix.

Proof. Let $A = \{1, x, x^2\}$ be the standard basis for V. Then,

$$[T]_A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \text{ so that the characterstic is } \det([T]_A + tI)$$

$$= t(t-1)(t-2)$$

Since the characteristic polynomial is seperable and splits over $\mathbb{R}$, so the minimal polynomial equals to the characteristic, and so the T is diagonalizable. Moreover, observe that

$$\ker T = \text{Span}\{x^2 - x\}$$

$$\ker(T - id) = \text{Span}\{-1 + x + x^2\}$$

$$\ker(T - 2\cdot id) = \text{Span}\{x^2 + x\}$$

Therefore, for $\mathfrak{B} = \{x^2 - x, x^2 + x - 1, x^2 + x\}$,

$$[T]_{\mathfrak{B}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

□

2019 Spring

3. Rationalize the denominator of $\dfrac{1}{3+2\cdot\sqrt[3]{2}+\sqrt[3]{4}}$

Proof Denote that $\alpha=\sqrt[3]{2}$. Then, $\alpha^3-2=0$.

Meanwhile, we wanna to find that the $\mathbb{Q}$-linear combination for

$$(3+2\cdot\sqrt[3]{2}+\sqrt[3]{4})^{-1}=(\alpha^2+2\alpha+3)^{-1}.$$

By the division algorithm,

$$\alpha^3-2=(\alpha^2+2\alpha+3)\cdot(\alpha-2)+\alpha+4$$

$$\alpha^2+2\alpha+3=(\alpha+4)(\alpha-2)+11.$$

$$\Rightarrow(\alpha^2+2\alpha+3)+(\alpha^2+2\alpha+3)(\alpha-2)^2=11 \qquad (\alpha^3-2=0)$$

$$\Rightarrow(\alpha^2+2\alpha+3)(\alpha^2-4\alpha+5)=11.$$

Therefore,

$$(\alpha^2+2\alpha+3)^{-1}=\frac{1}{11}\cdot(\alpha^2-4\alpha+5)=\frac{\sqrt[3]{4}-4\cdot\sqrt[3]{2}+5}{11}.$$

[2019. spring.
4. Let R be the Ring $\mathbb{Z}[\sqrt{-5}]$ prove or disprove that
a). $1+\sqrt{-5}$ and $1-\sqrt{-5}$ are irreducible in R.
b). $1+\sqrt{-5}$ and $1-\sqrt{-5}$ are prime in R.
Proof. Notice that

$$N: \mathbb{Z}[\sqrt{-5}] \to \mathbb{N} \cup \{0\}; a+b\sqrt{-5} \mapsto a^2+5b^2$$

is a multiplicative norm, and $N(\alpha)=1 \iff \alpha = \pm 1$.

a). Suppose that $1+\sqrt{-5}$ is reducible. Then,
$1+\sqrt{-5} = \alpha\beta$ for some non-unit $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$. So,

$N(1+\sqrt{-5}) = 6 = N(\alpha) \cdot N(\beta)$. But, since α and β are non-unit, $\{N(\alpha), N(\beta)\} = \{2, 3\}$, but $a^2+5b^2=2$ has no integer solution and $a^2+5b^2=3$ also has no integer solution, it contradicts.
Similarly, since $N(1-\sqrt{-5})=6$, so we obtain $1 \pm \sqrt{-5}$ are irreducible.

b). observe that: $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$.
Therefore, if $1+\sqrt{-5}$ is a prime number, then $1+\sqrt{5}$ divides either 2 or 3. If it divides 2, then

$$2 = \alpha \cdot (1+\sqrt{-5}) \text{ for some } \alpha \in \mathbb{Z}[\sqrt{-5}]$$

But $4 = N(2) = N(\alpha) \cdot N(1+\sqrt{-5}) = 6 \cdot N(\alpha)$

There is no such α, so contradiction. If it divides 3, similarly

$$9 = 6 \cdot N(\alpha) \text{ for some } \alpha,$$ also there is no such α.

So $1+\sqrt{-5}$ is not prime, and $N(1+\sqrt{-5}) = N(1-\sqrt{-5})$, so we can obtain the same result for $1-\sqrt{-5}$. ∎

2019. Spring.

5. Show that every non-commutative group of order 12 is solvable.

Proof. Let G be a non-commutative group of order of 12.

By the Sylow theorem, the number of Sylow 2-subgroup $n_2 \equiv 1 \bmod 2$ and $n_2 | 3$, and $n_3 \equiv 1 \bmod 3$ and $n_3 | 4$, so that

$$n_2 \in \{1, 3\} \text{ and } n_3 \in \{1, 4\},$$

If $n_3 = 1$, then there exists a normal subgroup of order 3, denote H. $|G/H| = 4$, but order of 4 subgroup must be $\mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$,
So $1 \triangleleft H \triangleleft G$ is a series such that the factors are all abelian.

Now, assume $n_3 \neq 1$. Then, $n_3 = 4$. So that G has $4 \times 2 = 8$ numbers distinct order 3 elements. If $n_2 = 3$, then there are at least $4 \times 3 - 2 \times 3 + 1 = 7$ elements, so $8 + 7 > 12$, it contradicts,
Thus $n_2 = 1$. Now, there is a normal subgroup of order 4, denote A.
$|G/A| = 3$, so cyclic, abelian. And, similarly, $A \cong \mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$,
So $1 \triangleleft A \triangleleft G$ is a series, all factors are abelian. So proved ☐

[2016, spring]

6. For each $n \in \mathbb{N}$, let

$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that $(f_n)_{n\geq 1}$ does not converges uniformly.

b) For each fixed $q > 1$, Show that (f_n) converges uniformly on $\{x \in \mathbb{R} \mid |x| \geq q\}$ and $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$

a). Given $x \in \mathbb{R}$,

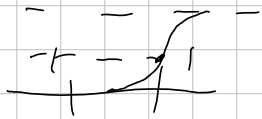
if $|x| < 1$, then $f_n(x) \to 0$ as $n \to \infty$, since $|x|^n \to 0$.

if $|x| = 1$, then $f_n(x) = \frac{1}{2}$ for all $n \in \mathbb{N}$, since $|x|^n = 1$.

if $|x| > 1$, then $f_n(x) = \frac{1}{1/|x|^n + 1} \to 1$ as $n \to \infty$, since $1/|x|^n \to 0$ as $n \to \infty$.

Therefore, $f_n \to f$ pointwise where

$$f(x) = \begin{cases} 0 & |x| < 1 \\ \frac{1}{2} & |x| = 1 \\ 1 & |x| > 1. \end{cases}$$



But, for $\varepsilon = 1/4$, and for all $n \in \mathbb{N}$,

$$\sup_{0 \leq x < 1} |f_n(x) - f(x)| = \sup_{0 \leq x < 1} |f_n(x)| = \sup_{0 \leq x < 1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$

because given $n \in \mathbb{N}$, we can take $x \in [0, 1)$ such that $\frac{x^n}{1+x^n} \geq \frac{1}{4}$

Since $\lim_{x \to 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$, so, $\sup_{x \in \mathbb{R}} |f_n - f| \geq \sup_{0 \leq x < 1} |f_n - f|$ implies $f_n \to f$ but not uniformly. //om.

b). If $|x| \geq q > 1$, then $f(x) = 1$. So, enough to show that $\sup |f_n(x) - f(x)| \to 0$.

Since $\sup_{|x| \geq q} |f_n(x) - f(x)| = \sup_{|x| \geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+|q|^n} \to 0$ as $n \to \infty$, so $f_n \to f$ uniformly on $|x| \geq q$. Similarly, for $|x| \leq \frac{1}{q}$, $f(x) = 0$ and so

$$\sup_{|x| \leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x| \leq \frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{|q|^n + 1} \to 0 \text{ as } n \to \infty,$$

So proved all cases. //om. ☺

2014, Spring

8. (10 pts.) Let $f : (0,1) \to \mathbb{R}$ be a convex function.

(a) Show that for any $0 < x < y < 1$ we have $f'_+(x) \le f'_-(y)$, where

$$f'_+(x) = \lim_{h \to 0^+} \frac{f(x+h) - f(x)}{h} \text{ and } f'_-(y) = \lim_{h \to 0^+} \frac{f(y) - f(y-h)}{h}.$$

(b) Show that the set $\{x \in (0,1) \mid f \text{ is not differentiable at } x\}$ is at most countable.

Pf. Notice that: Since f is convex, given $0<x_1<x_2<x_3<1$,

$$\frac{f(x_2)-f(x_1)}{x_2-x_1} \le \frac{f(x_3)-f(x_1)}{x_3-x_1} \le \frac{f(x_3)-f(x_2)}{x_3-x_2} \quad \cdots (*)$$

a). Let $x \in (0,1)$ be given. Then, for fixed $t \in (0,x)$,

$$\frac{f(x+h)-f(x)}{h} \ge \frac{f(x)-f(t)}{x-t} \text{ for all } 0<h<|x-t|$$

and since $\frac{f(x+h)-f(x)}{h}$ is decreasing as $h \to 0^+$, so the limit $f'_+(x)$ exists.

Similarly, $f'_-(y)$ also exists. And, for any $\delta \in (0, \frac{|y-x|}{2})$,

$$\frac{f(x+h)-f(x)}{x+h-x} \le \frac{f(y)-f(y-h)}{y-(y-h)}, \text{ so } f'_+(x) \le f'_-(y).$$

b). Let $x \in (0,1)$ be given with f is not differentiable at x.
Since $f'_+(x)$ and $f'_-(x)$ exist, by a), so that $f'_+(x) \ne f'_-(x)$.
And, since $*$, $f'_-(x) \le f'_+(x)$. So, let $r_x \in \mathbb{Q} \cap (f'_-(x), f'_+(x))$.
Then, given $y \in (x,1)$, we can take similarly r_y, but by a),

$$r_x < f'_+(x) \le f'_-(y) < r_y$$

so $r_x \ne r_y$, and similarly for $y \in (0,x)$, $r_y \ne r_x$.
Now, the map $\{x \in (0,1) \mid f \text{ is not differentiable at } x\} \to \mathbb{Q}$ is injective, so the domain set is at most countable. ∎

[2019, Spring]

9. Evaluate the integral $\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2}\,dx$.

Proof. Let $f(z)=1/(1+z^2)^2$, and a curve $C_R: t \to Re^{it}$ $(0 \le t \le \pi)$, and $C_L: t \to t$ $(-R \le t \le R)$. Then the composition $C_R * C_L$ is a closed, and $f(z)$ has only one singularity at $z=i$, as a pole.

To obtain the Residue at $z=i$, observe that

$$f(z)=\frac{1}{(1+z^2)^2}=\frac{1}{(z+i)^2\cdot(z-i)^2}$$

So, therefore, $z=i$ is a pole of order 2, so

$$\operatorname{Res}(f,i)=\lim_{z\to i}\frac{d}{dz}\left((z-i)^2\cdot f(z)\right)$$

$$=\lim_{z\to i}\frac{1}{(z+i)^2}=\lim_{z\to i}-2\cdot\frac{1}{(z+i)^3}=\frac{1}{4i}.$$

By the Residue theorem, for any $R>1$,

$$\int_{C_R * C_L} f = \operatorname{Res}(f,i)\cdot 2\pi i=\frac{\pi}{2}.$$

Meanwhile, $\int_{C_R} f(z)\,dz=\int_0^{\pi}\frac{iRe^{it}}{(1+R^2e^{2it})^2}\,dt$, so, for $R>2$,

$$\left|\int_{C_R} f\right| \le \int_0^{\pi}\frac{R}{(R^2-1)^2}\,dt \to 0 \text{ as } R\to\infty,$$ now,

$$\lim_{R\to\infty}\int_{C_R * C_L} f=\int_{-\infty}^{\infty}\frac{1}{(1+x^2)^2}\,dx=\frac{\pi}{2}. \quad \blacksquare$$

2019. Spring.

10. Let $\Omega = \{z \in \mathbb{C} \mid \operatorname{Re} z, \operatorname{Im} z > 0\}$ and $U = \{z \in \mathbb{C} \mid |z| < 1\}$. Find a one-to-one conformal mapping f from Ω onto U such that $f(e^{\frac{\pi i}{4}}) = 0$.

Proof. Since

$$g : \Omega \to \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\} : z \mapsto z^2$$

is one-to-one conformal map and

$$h : \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\} \to U : z \mapsto \frac{i - z}{i + z}$$

is one-to-one conformal map, so

$$f : \Omega \to U : z \mapsto \frac{i - z^2}{i + z^2}$$

is a one-to-one conformal map and

$f(e^{\frac{\pi}{4} i}) = 0$ since $(e^{\frac{\pi}{4} i})^2 = e^{\frac{\pi}{2} i} = i$. ∎

2019. Spring.

11. (10 pts.) Suppose that $f : (0,1) \to \mathbb{R}$ is bounded and continuous. Does there exist a sequence $(p_n)_{n\geq 1}$ of polynomials such that $(p_n)_{n\geq 1}$ converges uniformly to f on $(0,1)$? Justify your answer.

Proof. Let $f:(0,1)\to\mathbb{R} : x \mapsto \sin\frac{1}{x}$.

Then, f is bounded continuous function.
Suppose that there is such a sequence $\{p_n\}$ such that

$$p_n \to f \text{ uniformly on } (0,1).$$

Since 0 is a limit point of $(0,1)$, so that

$$\lim_{x\to 0+} f(x) = \lim_{x\to 0+} \lim_{n\to\infty} p_n(x) = \lim_{n\to\infty} \lim_{x\to 0+} p_n(x) = \lim_{n\to\infty} p_n(0)$$

exists, so that there exists a continuously extension of f to $[0,1)$.
But, the f cannot have continuous extension at $x=0$, because:
for each $n\in\mathbb{N}$, define

$$x_n = \frac{1}{2n\pi + \frac{\pi}{2}} \quad \text{and} \quad y_n = \frac{1}{2n\pi + \frac{3}{2}\pi}$$

Then, $x_n \to 0$ and $y_n \to 0$ as $n\to\infty$, but
$f(x_n)=1$ and $f(y_n)=-1$ for all $n\in\mathbb{N}$,

So, f cannot continuous at the zero. ☐

2016, Spring.

12. For $a, b \in \mathbb{Z}$, let $A_{a,b} = \{a + n \cdot b \mid n \in \mathbb{Z}\}$.

a). Prove that $\mathcal{A} = \{A_{a,b} \mid a, b \in \mathbb{Z}\}$ is a basis for a topology on $\mathbb{Z}$.

b). Is $\mathbb{Z}$ compact under the topology generated by $\mathcal{A}$?

Proof. Given $a \in \mathbb{Z}$, $a \in A_{a,0} = \{a\} \subseteq \mathbb{Z}$, so $\bigcup A_{a,b} = \mathbb{Z}$.

And, let $A_{a,b}$ and $A_{c,d}$ be given and let $k \in A_{a,b} \cap A_{c,d}$ be given.

If $A_{a,b} \cap A_{c,d} \neq \emptyset$, then for some $n_1, n_2 \in \mathbb{Z}$ such that

$k = a + n_1 b = c + n_2 d$

2020, Spring.

17. Let $S=\{(x,y,z)\in\mathbb{R}^3 \mid z=x\cdot f(\frac{y}{x}),\ x\neq 0\}$ where $f:\mathbb{R}\to\mathbb{R}$ is differentiable. Show that the tangent plane at any point of S passes through the origin.

Proof By the definition of S, given $(x_0,y_0,z_0)\in S$, a map

$X:\mathcal{O}\to S:(u,v)\mapsto(u,v,u\cdot f(\frac{v}{u}))$ where $\mathcal{O}$ is an open set containing (x_0,y_0)

is a parametrization of S at (x_0,y_0,z_0). The differential of X at (x_0,y_0) is:

$$dX=\begin{bmatrix}1 & 0\\ 0 & 1\\ f(\frac{v}{u})-\frac{v}{u}\cdot f'(\frac{v}{u}) & f'(\frac{v}{u})\end{bmatrix}\Bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of S at (x_0,y_0) is generated by

$$\left\{\begin{pmatrix}1\\0\\f(\frac{y_0}{x_0})-\frac{y_0}{x_0}\cdot f'(\frac{y_0}{x_0})\end{pmatrix},\begin{pmatrix}0\\1\\f'(\frac{y_0}{x_0})\end{pmatrix}\right\}.$$

Now, for $t_1,t_2\in\mathbb{R}$, the coordinate of a point of the tangent plane can be described by

$$(x_0,y_0,x_0\cdot f(\tfrac{y_0}{x_0}))+(t_1,0,t_1 f(\tfrac{y_0}{x_0})-t_1\cdot\tfrac{y_0}{x_0}f'(\tfrac{y_0}{x_0}))$$
$$+(0,t_2,t_2\cdot f'(\tfrac{y_0}{x_0}))$$

is zero when $t_1=-x_0$ and $t_2=-y_0$, So proved ☐

2019. Fall.

1. Classify the following groups according to the FTFGA:

$$\frac{\mathbb{Z}\times\mathbb{Z}}{\langle(n,n)\rangle}$$

Proof. Define a map for each $n\in\mathbb{N}$:

$$\varphi:\mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}_n\times\mathbb{Z}\ ;(a,b)\mapsto(a \bmod n,\ a-b).$$

Since

$$\begin{aligned}\varphi((a_1,b_1)+(a_2,b_2)) &= \varphi(a_1+a_2,\ b_1+b_2)\\ &= (a_1+a_2 \bmod n,\ a_1+a_2-b_1-b_2)\\ &= (a_1 \bmod n,\ a_1-b_1)+(a_2 \bmod n,\ a_2-b_2)\\ &= \varphi(a_1,b_1)+\varphi(a_2,b_2)\end{aligned}$$

So φ is Homomorphism, and given $(c,d)\in\mathbb{Z}_n\times\mathbb{Z}$, take $a=c$ and $b=c-d$, then

$$\varphi(a,b)=(a,a-b)=(c,d)$$

So φ is onto. And,

$(a,b)\in\ker\varphi \iff a\equiv 0 \bmod n$ and $a-b=0$

So, $\ker\varphi=\langle(n,n)\rangle$. Now, by the first 'isomorphism theorem

$$\mathbb{Z}\times\mathbb{Z}/\ker\varphi\cong\operatorname{Im}\varphi=\mathbb{Z}_n\times\mathbb{Z}.$$ ∎

2019, Fall.

2. Let p be the prime number and $\alpha \in \mathbb{F}_p$. Show that $x^p + \alpha$ is reducible over $\mathbb{F}_p$.

Proof. If $\alpha = 0$, then x^p is clearly reducible. Assume that $\alpha \neq 0$.

Since $\mathbb{F}_p^\times$ is a cyclic group of order of $p-1$, so $\alpha^{p-1} = 1$, so $\alpha^p = \alpha$.

Therefore, observe that

$$(x+\alpha)^p = \sum_{n=0}^{p} \binom{p}{n} x^{p-n} \cdot \alpha^n = x^p + \alpha^p = x^p + \alpha$$

and $x + \alpha \in \mathbb{F}_p[x]$,

So, $x^p + \alpha$ is reducible

Note: $\mathbb{F}_p^\times \simeq C_{p-1}$.

[2019. Fall.
3. Show that $\alpha = 3^{\frac{1}{3}} + 5^{\frac{1}{5}}$ is algebraic over $\mathbb{Q}$ and find $[\mathbb{Q}(\alpha):\mathbb{Q}]$.
Proof. Since $3^{\frac{1}{3}}$ and $5^{\frac{1}{5}}$ both are roots of x^3-3 and x^5-5, respectively, therefore both are algebraic over $\mathbb{Q}$. And, since sum of algebraic numbers is algebraic, so $3^{\frac{1}{3}}+5^{\frac{1}{5}}$ is algebraic.

By the eisenstein criterion, both x^3-3 and x^5-5 are irreducible over $\mathbb{Q}$, so that $[\mathbb{Q}(\sqrt[3]{3}):\mathbb{Q}]=3$ and $[\mathbb{Q}(\sqrt[5]{5}):\mathbb{Q}]=5$. Therefore, $[\mathbb{Q}(\sqrt[3]{3},\sqrt[5]{5}):\mathbb{Q}]$ is divided by 3 and 5, so $[\mathbb{Q}(\sqrt[3]{3},\sqrt[5]{5}):\mathbb{Q}]\geq 15$, and x^3-3 has a root as $\sqrt[3]{3}$ over $\mathbb{Q}(\sqrt[5]{5})$, so $[\mathbb{Q}(\sqrt[3]{3},\sqrt[5]{5}):\mathbb{Q}(\sqrt[5]{5})]\leq 3$, so that $[\mathbb{Q}(\sqrt[3]{3},\sqrt[5]{5}):\mathbb{Q}]\leq 15$, so the degree is 15.

Since $3^{\frac{1}{3}}+5^{\frac{1}{5}}\in\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}})$, $\mathbb{Q}(3^{\frac{1}{3}}+5^{\frac{1}{5}})\subseteq\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}})$. And, since $\mathbb{Q}(3^{\frac{1}{3}}+5^{\frac{1}{5}},3^{\frac{1}{3}})=\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}})$, so, since $(3^{\frac{1}{3}})^3-3=0$, $[\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}}):\mathbb{Q}(3^{\frac{1}{3}}+5^{\frac{1}{5}})]\leq 3$, so, $=1$ or 3.

Corti.

To find the degree, notice that $[\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}}):\mathbb{Q}(5^{\frac{1}{5}})]=3$, because x^3-3 has no root in $\mathbb{Q}(5^{\frac{1}{5}})$, so the tower lemma gives that $[\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}}):\mathbb{Q}]=[\mathbb{Q}(\alpha):\mathbb{Q}(5^{\frac{1}{5}})]\cdot[\mathbb{Q}(5^{\frac{1}{5}}):\mathbb{Q}]=3\cdot 5=15$.
(x^5-5 is also irreducible over $\mathbb{Q}$, by the eisenstein criterion)
Meanwhile, $\mathbb{Q}(3^{\frac{1}{3}},5^{\frac{1}{5}})\supseteq\mathbb{Q}(3^{\frac{1}{3}}+5^{\frac{1}{5}})$ is trivial

a^p-b^p

Similar problem.

4. Determine the degree over $\mathbb{Q}$ of $2+\sqrt{3}$ and of $1+\sqrt[3]{2}+\sqrt[3]{4}$.
5. Let $F=\mathbb{Q}(i)$. Prove that x^3-2 and x^3-3 are irreducible over F.
6. Prove directly from the definitions that the field $F(\alpha_1,\alpha_2,\ldots,\alpha_n)$ is the composite of the fields $F(\alpha_1), F(\alpha_2),\ldots,F(\alpha_n)$.
7. Prove that $\mathbb{Q}(\sqrt{2}+\sqrt{3})=\mathbb{Q}(\sqrt{2},\sqrt{3})$ [one inclusion is obvious, for the other consider $(\sqrt{2}+\sqrt{3})^2$, etc.]. Conclude that $[\mathbb{Q}(\sqrt{2}+\sqrt{3}):\mathbb{Q}]=4$. Find an irreducible polynomial satisfied by $\sqrt{2}+\sqrt{3}$.

4. $2+\sqrt{3}=x \Rightarrow x^2-4x+1=0$, it has no root in $\mathbb{Q}$, so that $[\mathbb{Q}(2+\sqrt{3}):\mathbb{Q}]=2$.

Since $(\sqrt[3]{2}-1)\cdot(\sqrt[3]{4}+\sqrt[3]{2}+1)=2-1=1$, so that

$$\sqrt[3]{4}+\sqrt[3]{2}+1=\frac{1}{\sqrt[3]{2}-1}$$

Therefore,

$$\mathbb{Q}(\sqrt[3]{4}+\sqrt[3]{2}+1)=\mathbb{Q}\left(\frac{1}{\sqrt[3]{2}-1}\right)=\mathbb{Q}(\sqrt[3]{2}-1)=\mathbb{Q}(\sqrt[3]{2})$$

So, the degree is 3, since x^3-2 is irreducible, by the eisenstein.

[2019. Fall.

5. Show that a symmetric $n \times n$ matrix over $\mathbb{R}$ has a real eigenvalue.

Proof. Let $f(x)$ be the characteristic polynomial of the given matrix.
By the Fundamental Theorem of Algebra, f has a root λ in $\mathbb{C}$.
Meanwhile, observe that for the given matrix A, $Ax = \lambda x$ for some $x \in \mathbb{C}^n \setminus \{0\}$.
Now, $\lambda \|x\|^2 = \langle \lambda x, x \rangle = \langle Ax, x \rangle = \langle x, A^* x \rangle = \langle x, A^T x \rangle = \langle x, Ax \rangle = \overline{\langle Ax, x \rangle} = \overline{\lambda} \|x\|^2$.
Since $\|x\| > 0$, so $\lambda = \overline{\lambda}$, it proves the λ is real root of $f(x)$, so
A has a real eigenvalue. □

Note: $f(\lambda) = 0 \iff \det(A - \lambda I) = 0$
$\Rightarrow A - \lambda I$ is non-singular, not invertible,
$\Rightarrow \ker(A - \lambda I) \neq \{0\}$.
선형대수학 수업시간에 시간나면 한 줄 정도 써줘.

2019. Fall.

6. Find the general solution

$$y' + xy = xy^2.$$

Proof If $y \equiv 0$, then it is the solution. By the uniqueness theorem, suppose $y \neq 0$ for all x. Then, the given equation can be formed:

$$\frac{y'}{y^2} + \frac{x}{y} = -\left(\frac{1}{y}\right)' + x \cdot \left(\frac{1}{y}\right) = x.$$

Multiplying $u(x) = \exp(-\frac{1}{2}x^2)$, then

$$x e^{-\frac{1}{2}x^2} = -e^{-\frac{1}{2}x^2} \cdot \left(\frac{1}{y}\right)' + x \cdot e^{-\frac{1}{2}x^2} \cdot \left(\frac{1}{y}\right)$$

$$= \left(-e^{-\frac{1}{2}x^2} \cdot \frac{1}{y}\right)', \text{ So that}$$

$$-e^{-\frac{1}{2}x^2} \cdot \frac{1}{y} = \int x e^{-\frac{1}{2}x^2} + C$$

$$= -e^{-\frac{1}{2}x^2} + C$$

Consequently, the general solution is

$$y = \frac{-e^{-\frac{1}{2}x^2}}{C - e^{-\frac{1}{2}x^2}}, \quad C \in \mathbb{R}.$$

[2016a, Fall.

8. Let $(b_n)_{n\in\mathbb{N}}$ be a decreasing sequence of positive real numbers with $b_n \to 0$. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of complex numbers such that $\left(\left|\sum_{n=1}^{N} a_n\right|\right)_{N\in\mathbb{N}}$ is bounded sequence.

a). Show that $\sum_{n=1}^{\infty} a_n b_n$ Converges.

b). Show $\sum_{n=2}^{\infty} \frac{\sin(nx)}{\log n}$ converges for all $x \in [-\pi, \pi]$.

Proof. Denote $A_n = \sum_{k=1}^{n} a_k$ and $A_0 = 0$. And let $M > 0$ be a bound for $\sum a_n$.

$$\sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N} [A_n - A_{n-1}]\cdot b_n = \sum_{n=1}^{N} A_n b_n - \sum_{n=1}^{N} A_{n-1}\cdot b_n \quad \left(\sum_{n=1}^{N} A_{n+1} b_n = \sum_{n=0}^{N-1} A_n b_{n+1}\right)$$

$$= \sum_{n=1}^{N} A_n b_n - \left(\sum_{n=1}^{N} A_n \cdot b_{n+1} + A_0 b_1 - A_N b_{N+1}\right)$$

$$= \sum_{n=1}^{N} A_n (b_n - b_{n+1}) + A_N b_{N+1}.$$

Therefore, for any $m > n \geq 1$,

$$\sum_{k=n}^{m} a_k b_k = \sum_{k=n}^{m} A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_{n-1} b_n$$

Now, using the hypothesis,

$$\left|\sum_{k=n}^{m} a_k b_k\right| \leq \sum_{k=n}^{m} |A_k|\cdot|b_k - b_{k+1}| + |A_m|\,|b_{m+1}| + |A_{n-1}|\cdot|b_n|$$

$$\leq M\cdot\left(\sum_{k=n}^{m} |b_k - b_{k+1}| + |b_{m+1}| + |b_n|\right)$$

$$= M\cdot\left(\sum_{k=n}^{m} (b_k - b_{k+1}) + b_{m+1} + b_n\right)$$

$$= M\cdot 2b_n \to 0 \text{ as } n\to\infty, \text{ so proved.}$$

b). Since $1/\log n$ is decreasing and positive real numbers, so by a), enough to show that $|\sum \sin(nx)|$ is bounded for all $x \in [-\pi, \pi]$.

observe that: $2\sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

Therefore, $2\sin x \cdot \sum_{n=1}^{N} \sin(nx) = \sum_{n=1}^{N} \cos[(n-1)x] - \cos[(n+1)x] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos((N+1)x|$

≤ 4

So for $x \in [-\pi, \pi]\setminus\{0, \pi, -\pi\}$, $\left|\sum_{n=2}^{N} \sin nx\right| \leq \frac{2}{|\sin x|}$ for any $N\in\mathbb{N}$, bounded.

for $x = 0$ or $\pm\pi$, then the sum is just zero. ☑

[2010a, Fall,

a. Let $\varphi : [0,1] \to \mathbb{R}$ be a continuous map. Evaluate

$$\lim_{\varepsilon \to 0+} \int_0^1 \left[\frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

Therefore, the given integral is

$$2i\varepsilon \int_0^1 \frac{\varphi(x)}{x^2 + \varepsilon^2} dx.$$

Let $x = \varepsilon t$. Then, $dx = \varepsilon\, dt$, so that

$$2i\varepsilon \int_0^1 \frac{\varphi(x)}{x^2 + \varepsilon^2} dx = 2i\varepsilon \cdot \int_0^{\frac{1}{\varepsilon}} \frac{\varphi(\varepsilon t)}{\varepsilon^2 t^2 + \varepsilon^2} \varepsilon \cdot dt = 2i \int_0^{\frac{1}{\varepsilon}} \frac{\varphi(\varepsilon t)}{t^2 + 1} dt$$

$$= 2i \cdot \int_0^{\infty} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{\left[0, \frac{1}{\varepsilon}\right]}(t)\, dt$$

where $\chi_{\left[0, \frac{1}{\varepsilon}\right]}$ is the characteristic function. By the dominated convergence theorem,

$$\lim_{\varepsilon \to 0+} \left(2i \cdot \int_0^{\infty} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{\left[0, \frac{1}{\varepsilon}\right]}(t)\, dt \right)$$

$$= 2i \cdot \int_0^{\infty} \lim_{\varepsilon \to 0+} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{\left[0, \frac{1}{\varepsilon}\right]}(t)\, dt$$

$$= 2i \cdot \int_0^{\infty} \frac{\varphi(0)}{t^2 + 1} dt = i\pi \cdot \varphi(0).$$

2019, Fall.

10. (10 pts.) Let $I \subseteq \mathbb{R}$ be a non-empty open interval. Let $f : I \to \mathbb{R}$ be a real analytic function, i.e. for any $x_0 \in I$ there is an open interval $J \subseteq I$ containing x_0 and $(a_n)_{n=1}^{\infty} \subseteq \mathbb{R}$ such that $f(x) = \sum_{n=1}^{\infty} a_n (x - x_0)^n, \forall x \in J$. Let $y_0 \in I$ be a point such that $f'(y_0) \neq 0$.

(a) Show that there is an open interval $J \subseteq I$ containing y_0 such that $f'(x) \neq 0, \forall x \in J$.

(b) Show that there is an open interval $J' \subseteq I$ containing y_0 and a constant $C > 0$ such that

$$|f(x) - f(y_0)|^{\frac{1}{2}} \leq C|f'(x)|, \quad \forall x \in J'.$$

Proof. a). Since every analytic function is C^∞, and the derivative at y_0 is

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-y_0)^{n-1} = a_1 + \sum_{n=2}^{\infty} n a_n (x-y_0)^{n-1}. \quad (x \in J)$$

By the assumption, $f'(y_0) = a_1 \neq 0$, and since $\sum n a_n (x-y_0)^{n-1}$ converges uniformly,

so $\lim_{x \to y_0} \sum_{n=2}^{\infty} n a_n (x-y_0)^{n-1} = \sum_{n=2}^{\infty} \lim_{x \to y_0} n a_n (x-y_0)^{n-1} = 0$

therefore for $\varepsilon = |a_1|/2$, there is $\delta > 0$ such that

$$|x - y_0| < \delta \Rightarrow |f'(x) - a_1| < \frac{|a_1|}{2}.$$

So, $J_0 = (y_0 - \varepsilon, y_0 + \varepsilon)$ is the open interval we desired.

b) Since $f(x) \to f(y_0)$ as $x \to y_0$ by the continuity and $|f'(x)| \geq |a_1|/2$ for all $x \in J_0$ by a), so we can take $\delta' > 0$ such that

$$|x - y_0| < \delta' \Rightarrow |f(x) - f(y_0)|^{\frac{1}{2}} < \frac{|a_1|}{2} \leq f'(x),$$

So proved.

Practice in the Next.

Debris.

Since f is analytic, for given $y_0 \in I$, there is an open interval J_0 containing y_0 such that $f(x) = \sum_{n=0}^{\infty} a_n (x-y_0)^n$, for some $\{a_n\}$.

Moreover, $f \in C^\infty$ in J_0, so the derivative f' is continuous on J_0.

Since the set

$$K = \{x \in J_0 \mid f'(x) = 0\} = (f')^{-1}[\{0\}] \cap J_0$$

is closed set of J_0, therefore $J_0 \setminus (f')^{-1}[\{0\}]$ is an open set containing y_0, thus for some open interval $J \subset J_0$, $f'(x) \neq 0$ for all $x \in J$.

b). By the mean-value theorem, for any $x \in I$, there is $x_c \in I$ such that

$$|f(x) - f(y_0)| = |f'(x_c)| \cdot |x - y_0|.$$

Meanwhile, the given condition equals to

$$|f(x) - f(y_0)| \leq C^2 \cdot |f'(x)|^2,$$

So combining the above condition then

$$|f'(x) - f'(y)|$$

. By the assumption, for some open interval J containing y_0, for any $x \in J_0$,

$$f(x) = \sum_{n=0}^{\infty} a_n (x-y_0)^n.$$

So, $|f(x) - f(y_0)| = |f(x) - a_0| = \left|\sum_{n=1}^{\infty} a_n (x-y_0)^n\right|$ and

$$|f'(x)| = \left|\sum_{n=1}^{\infty} n a_n (x-y_0)^{n-1}\right|.$$

Problem. Let $I \subset \mathbb{R}$ be an open interval, and let $f: I \to \mathbb{R}$ be real analytic. Suppose that for some $m \geq 2$, and $a \in I$,

$$f(a) = f'(a) = \cdots = f^{(m-1)}(a) = 0 \quad \text{and} \quad f^{(m)}(a) \neq 0.$$

1). Show that there exist an open interval $J \subset I$ containing $a \in I$. and real analytic $g: J \to \mathbb{R}$ such that

$$f(x) = (x-a)^m \cdot g(x) \quad (x \in J) \quad \text{and} \quad g(x) \neq 0 \quad (x \in J).$$

2). Show that there exist $c_1, c_2 > 0$ and open interval $J' \subset I$ containing a such that $c_1 |x-a|^m \leq |f(x)| \leq c_2 |x-a|^m \quad (x \in J')$

3). Show that there exists $J'' \subset J'$ and $C > 0$ such that

$$|f(x)|^{\frac{m-1}{m}} \leq C \cdot |f'(x)| \quad (x \in J').$$

[2019, Fall.

11. (9 pts.) Consider topological spaces X and Y, and suppose that $f : X \to Y$ is a homotopy equivalence.

(a) The mapping cylinder of f is defined by

$$M(f) = (X \times [0,1]) \coprod Y / \sim,$$

where $\sim$ is given by $(x,1) \sim f(x)$ for $(x,1) \in X \times [0,1]$. Show that $M(f)$ deformation retracts onto Y.

(b) Show that $M(f)$ is homotopy equivalent to X.

(c) Is the homotopy equivalence in (b) also a deformation retract? Prove or disprove.

Proof. (a). Let $\pi : (X \times [0,1]) \sqcup Y \to M(f)$ be the natural projection.
Define a retraction

$$r : M(f) \to Y : \begin{cases} [(x,t)] \mapsto f(x) & \text{if } [(x,t)] \in \pi[X \times [0,1]] \\ [y] \mapsto y & \text{if } [y] \in \pi[Y]. \end{cases}$$

Since $(x,1) \sim f(x)$ for all $x \in X$, so that $[(x,1)] = [f(x)]$, thus

$$r([(x,1)]) = f(x) = r([f(x)]),$$ so this r is well-defined.

Then, for the natural inclusion $i : Y \to M(f)$, $r \circ i = id_Y$ and $i \circ r \simeq id_{M(f)}$ by the deformation

$$H : M(f) \times I \to M(f) : \begin{matrix} [(x,t)] \times s \mapsto [(x, (1-s)\cdot t + s)] \\ [y] \times s \mapsto [y] \end{matrix}$$

Then, $H(\cdot, 0) \equiv id_{M(f)}(\cdot)$, and $H(\cdot, 1) = i \circ r(\cdot)$

To show $M(f)$ deformation retracts onto Y, we want to find a retraction $r : M(f) \to Y$ such that $i \circ r \simeq id_{M(f)}$ where $i : Y \to M(f)$ is the inclusion.
Define $r : M(f) \to Y : \begin{cases} f(x) & \text{if } (x,t) \in X \times I, \\ y & \text{if } y \in Y \end{cases}$

Consider the map $\tilde{f} : X \times \{1\} \to Y : (x,t) \mapsto f(x)$.

Then, we can consider the quotient map

$$\pi : \left(X \times [0,1]\right) \amalg Y \to M(f).$$

Since f is homotopy equivalence, by definition, there is a continuous map $g : Y \to X$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$.

(a).

2020 spring.

2. Let $A = \begin{pmatrix} 0 & I_n \\ 2I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{R})$

a). Determine A is diagonalizable over $\mathbb{R}$.

b). Find the characteristic and the minimal polynomial of A.

Pf. a). Observe that

$A^2 = \begin{pmatrix} 2I_n & 0 \\ 0 & 2I_n \end{pmatrix} = 2 \cdot I_{2n}$, so $A^2 - 2I = 0$, with $A \neq \pm\sqrt{2} \cdot I_{2n}$.

therefore, the minimal polynomial of A is $t^2 - 2 = (t-\sqrt{2}) \cdot (t+\sqrt{2})$,

which is separable and splits over $\mathbb{R}$, so A is diagonalizable.

b). The characteristic polynomial is:

$$\det(tI - A) = \det\begin{pmatrix} tI_n & -I_n \\ -2I_n & tI_n \end{pmatrix} = \det\left(t^2 I_n - 2I_n\right) = (t^2-2)^n$$

And the minimal is already obtained in a), so it is done. ☒

2020 spring.

3. Show that no group of order 48 is simple.

Proof. Notice that $48 = 2^4 \cdot 3$. By the Sylow theorem, for a group G of order 48, $n_2 = \#Syl_2(G) \in \{1, 3\}$ and $n_3 = \#Syl_3(G) \in \{1, 4, 16\}$.
If either $n_2 = 1$ or $n_3 = 1$, then G has a non-trivial proper normal subgroup, so it is non-simple. Now, suppose that $n_2 = 3$ and $n_3 \in \{4, 16\}$.
Consider the afforded Homomorphism

$$\varphi : G \to Sym(Syl_2(G)) : g \mapsto (P \mapsto gPg^{-1})$$

By the Sylow theorem, this action is transitive on $Syl_2(G)$, so $\ker \varphi \neq G$.
And, $Sym(Syl_2(G)) \cong S_3$, so $|G| = 48 > 6 = |S_3|$ implies that φ cannot be injective, so $\ker \varphi \neq \{e\}$. Now, $\ker \varphi \triangleleft G$ is the non-trivial normal subgroup, so G is non-simple. ☐

Note: By the Sylow theorem, all elements in $Syl_p(G)$ are conjugate.
Thus, the map $\varphi : G \to Sym(Syl_p(G)) : g \mapsto (P \mapsto gPg^{-1})$ is not trivial,
because assume $n_p > 1$, then
φ is trivial $\overset{def}{\iff} \ker \varphi = G \iff$ for all $g \in G$ and $P \in Syl_p(G)$, $gPg^{-1} = P$

And it contradicts to the Sylow Theorem. So $\ker \varphi < G$.
Meanwhile,

$$\ker \varphi = \{e\} \iff \varphi \text{ is injective} \Rightarrow G \cong \varphi[G] \leq Sym(Syl_p(G)) \cong S_{n_p}$$

Therefore, by the Contraposition,

$$|G| \nmid n_p! \Rightarrow \ker \varphi > \{e\}.$$

⌟

Trebrls

$$|P_1 \cup P_2 \cup P_3| = |P_1| + |P_2| + |P_3| - |P_1 \cap P_2| - |P_2 \cap P_3| - |P_1 \cap P_3| + |P_1 \cap P_2 \cap P_3|$$
$$= 24 + |P_1 \cap P_2 \cap P_3|$$

And, since $1 \leq |P_1 \cap P_2 \cap P_3|$, $25 \leq |P_1 \cup P_2 \cup P_3| \leq 32$

If $n_3 = 16$, then there are 16 numbers distince order 3 subgroups of G, so there are $16 \cdot 2 = 32$ distince order 3 elements, by the Lagrange theorem. Meanwhile, since $25 \leq |P_1 \cup P_2 \cup P_3|$, and $P_1 \cup P_2 \cup P_3$ does not contain the order 3 elements, so 25 + 32 elements are lay in G, it is contradiction.

Since $n_2 = 3$, there are three Sylow 2-subgroups P_1, P_2, P_3.
For $i \neq 3$,

$$|P_i P_3| = \frac{|P_i| \cdot |P_3|}{|P_i \cap P_3|} = \frac{256}{|P_i \cap P_3|} \leq 48,$$

so $4 < 256/48 \leq |P_i \cap P_3|$. Meanwhile, by the Lagrange's Theorem, $|P_2 \cap P_3|$ is a multiple of 2, so $|P_i \cap P_3| = 8$. Now, let $Q = P_1 \cap P_2$.
Since $|P_1 : Q| = 2$, by the Lagrange's theorem, so $Q \triangleleft P_1$ and $Q \triangleleft P_2$.
Therefore, $N_G(Q)$ contains P_1 and P_2, so

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 32 - 8 = 24 \text{ implies that}$$

$|N_G(Q)| \geq 24$. So, either $|N_G(Q)| = 24$ or 48.
If $N_G(Q) = 24$, then $|G : N_G(Q)| = 2$, so $N_G(Q) \triangleleft G$.
If $N_G(Q) = 48$, then $N_G(Q) = G$ and so $Q \triangleleft G$, so G is non-simple.
So proved ☐

[2020 spring.
4. Let R be a commutative Ring with 1.
Show that if $I \subset R$ is prime ideal then the radical of I

$$\sqrt{I} := \{a \in R \mid a^m \in I \text{ for some } m \in \mathbb{N}\}$$

is also a prime ideal.
Proof. Given $a, b \in \sqrt{I}$, $a^m, b^n \in I$ for some $m, n \in \mathbb{N}$. Thus,

$$(a-b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} a^k \cdot (-b)^{m+n-k} \in I,$$

since for $0 \le k \le m$, $b^{m+n-k} \in I$ and for $m+1 \le k \le m+n$, $a^k \in I$,
thus by definition of ideal, each terms are contained in I,
so $(a-b)^{m+n} \in I$. And, clearly $(ab)^{mn} \in I$, so $a-b$, $ab \in \sqrt{I}$.
It proves $\sqrt{I}$ is a subring. And, given $r \in R$, $(ra)^m = r^m \cdot a^m \in I$,
thus $ra \in \sqrt{I}$, so the radical is an ideal.
And, if $1 \in \sqrt{I}$, then $1^n = 1 \in I$ for some n, contradiction, so $\sqrt{I} \subsetneq R$, proper.
Suppose that $ab \in \sqrt{I}$, for some $a, b \in R$. Then, $(ab)^n \in I$ for some n.
Since $(ab)^n = a^n b^n$ and I is prime, either $a^n \in I$ or $b^n \in I$.
If $a^n \in I$, then, $a \in \sqrt{I}$, and if $b^n \in I$, then $b \in \sqrt{I}$.
Thus, either $a \in \sqrt{I}$ or $b \in \sqrt{I}$, so $\sqrt{I}$ is prime ideal. ☐

Note: trivially, $I \subseteq \sqrt{I}$, and given $a \in \sqrt{I}$, $a^n \in I$ for some $n \in \mathbb{N}$,
then $a \in I$ or $a^{n-1} \in I$. If $a \notin I$, then $a^{n-1} \in I$, again,
$a \in I$ or $a^{n-2} \in I$, ..., thus, $a \in I$.
Therefore, in actually, $I = \sqrt{I}$, prime.

[2020 Spring.
5. Describe the Galois group of the polynomial $x^5-3 \in \mathbb{Q}(\zeta)[x]$ over $\mathbb{Q}(\zeta)$, where ζ is a primitive 5^{th} root of unity.
Proof. Let $\alpha = \sqrt[5]{3}$. Then, $\{\alpha\zeta^k \mid 0 \le k \le 4\}$ are exactly all roots of the given polynomial, so

$$K = \mathbb{Q}(\{\alpha\zeta^k \mid 1 \le k \le 4\}) = \mathbb{Q}(\alpha, \zeta)$$ is the splitting field.

Meanwhile, $[\mathbb{Q}(\alpha):\mathbb{Q}] = 5$, since x^5-3 is irreducible over $\mathbb{Q}$, by eisenstein, $[\mathbb{Q}(\zeta):\mathbb{Q}] = 4$, since $x^4+x^3+x^2+x+1$ is irreducible over $\mathbb{Q}$, since $\frac{(x+1)^5-1}{(x+1)-1} = x^4+5x^3+10x^2+10x+5$ is irreducible, by eisenstein.

Now, $[K:\mathbb{Q}] \le [\mathbb{Q}(\alpha):\mathbb{Q}] \cdot [\mathbb{Q}(\zeta):\mathbb{Q}]$ and since $\gcd(5,4)=1$, $[K:\mathbb{Q}] \ge 20$ so that $[K:\mathbb{Q}] = 20$.

By the tower lemma,

$$[K:\mathbb{Q}(\zeta)] = 5,$$ prime, so $Gal(K/\mathbb{Q}(\zeta)) \cong C_5$.

More concretely, $K/\mathbb{Q}(\zeta)$ is Galois since K is the splitting field of separable polynomial x^5-3. ☐

Note; $[\mathbb{Q}(\alpha,\zeta):\mathbb{Q}(\zeta)] \le [\mathbb{Q}(\alpha):\mathbb{Q}]$ since $\mathbb{Q} \subseteq \mathbb{Q}(\zeta)$ so that $[K:\mathbb{Q}] \le [\mathbb{Q}(\alpha):\mathbb{Q}] \cdot [\mathbb{Q}(\zeta):\mathbb{Q}]$.

[2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = (r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(x) &= \int_{x_0}^{x} \begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}}\, dt \\ &= \int_{x_0}^{x} \frac{e^{t+2x} - e^{2t+x}}{e^{3t} + e^{2t}}\, dt \\ &= e^{2x} \int_{x_0}^{x} \frac{1}{e^{2t}+e^t}\, dt - e^x \cdot \int_{x_0}^{x} \frac{1}{e^t+1}\, dt \end{aligned}$$

Meanwhile, since $\int \frac{1}{e^t+1}\, dt = \int \frac{e^{-t}}{1+e^{-t}}\, dt = -\log(1+e^{-x})$

and $\int \frac{1}{e^{2t}+e^t}\, dt = \int \frac{e^{-t}}{e^t+1}\, dt = \int \frac{-u}{\frac{1}{u}+1} \frac{1}{u}\, du = \int \frac{-u}{1+u}\, du$

$$= \int -1 + \frac{1}{1+u}\, du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$y_p + C_2 e_1 + C_3 e_3 = \left[-e^x + e^{2x}\log(1+e^{-x}) + e^x \cdot \log(1+e^{-x}) \right]$$

$$+ C_2 \cdot e^x + C_3 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R},$$

[2020 Spring.

7. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of non-negative real numbers and consider

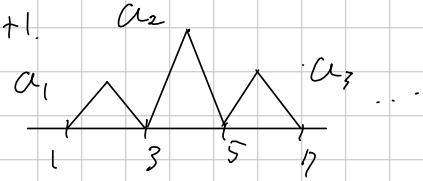
$$f(x)=\sum_{n=1}^{\infty} a_n \cdot \max\{1-|x-2n|,0\},\quad x\in[0,\infty)$$

Give a necessary and sufficient condition on (a_n) such that
a). the series to converge uniformly on $[0,\infty)$.
b). f is continuous uniformly.

Proof. Notice that: given $x\geq 0$,

$$1-|x-2n|\geq 0 \iff 2n-1\leq x\leq 2n+1.$$

Therefore, f can be described as



a). The given series converges uniformly if and only if

$$\sup_{x\geq 0}\left|f(x)-\sum_{n=1}^{N} a_n\cdot\max\{1-|x-2n|,0\}\right|\to 0 \text{ as } N\to\infty.$$

Meanwhile, the supremum term above equals to

$$\sup_{k\geq N+1} a_k$$

Therefore, $\lim_{n\to\infty}\sup a_n=0$ is the condition we desired.
(Since a_n is non-negative, $\lim_{n\to\infty}\sup a_n=\lim_{n\to\infty} a_n=0$)

b) If (a_n) is bounded by $M>0$, then, given $\varepsilon>0$, take $\delta=\frac{1}{M}$. Then

$$|x-y|<\delta=\frac{1}{M}\Rightarrow|f(x)-f(y)|\leq M\cdot|x-y|<\varepsilon,$$

so continuous uniformly.

Conversely, if (a_n) is unbounded, then

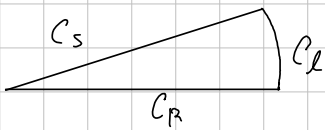
$\forall\varepsilon>0$, $\exists\delta>0$ such that $|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon$

is false, because for $\varepsilon=1$, for any $\delta>0$, there exises x_n, y_n such that $|f(x_n)-f(y_n)|=a_n\cdot|x_n-y_n|=a_n\cdot\frac{\delta}{2}>1$, we take $n\in\mathbb{N}$ such that $a_n>\frac{2}{\delta}$.
So, the condition that we desired is (a_n) is bounded. ∎
(Technically, we can take $x_n=2n$, $y_n=2n+\delta$.)

2020 Spring.

8. For each $n=2,3\cdots$ evaluate

$$\int_0^\infty \frac{1}{1+x^n}\,dx.$$



Proof. Let $f(z)=\dfrac{1}{1+z^n}$. Take a contour C as a composition of:

$C_R: t\mapsto t \quad (0\le t\le R)$, $C_\ell: t\mapsto R\cdot e^{\frac{it}{n}} \quad (0\le t\le 2\pi)$,

$C_s: t\mapsto t\cdot e^{\frac{2\pi i}{n}} \quad (0\le t\le R)$, that is, $C=C_R * C_\ell * (-C_s)$.

Inside C, $1+z^n$ has only one zero, $\exp(i\pi/n)$, so f has only one singularity inside C, the simple pole $\exp(i\pi/n)$, by the Residue theorem,

$$\int_C f = 2\pi i\cdot \mathrm{Res}(f,\exp(i\pi/n)) = \lim_{z\to\exp(i\pi/n)} (z-\exp(i\pi/n))\cdot\frac{1}{1+z^n}\cdot 2\pi i$$

$$= \frac{2\pi i}{n\cdot\exp(i\pi/n)^{n-1}} = -\frac{\exp(i\pi/n)}{n}\cdot 2\pi i$$

Meanwhile,

$$\left|\int_{C_\ell} f\right| = \int_0^{2\pi}\left|\frac{1}{1+R^n\cdot e^{it}}\frac{Ri}{n}\cdot e^{\frac{it}{n}}\right|dt$$

$$= \int_0^{2\pi}\frac{R/n}{|1+R^n\cdot e^{it}|}\,dt \le \int_0^{2\pi}\frac{R/n}{|R^n-1|}\,dt \to 0 \quad \text{as } R\to\infty.$$

and

$$\int_{C_s} f = \int_0^R \frac{e^{2\pi i/n}}{1+t^n e^{2\pi i}}\,dt = e^{\frac{2\pi i}{n}}\cdot\int_0^R\frac{1}{1+t^n}\,dt. \text{ So that}$$

$$-\frac{2\pi i}{n}\cdot\exp(i\pi/n) = \lim_{R\to\infty}\int_C f = \int_0^\infty\frac{1}{1+x^n}\,dx - \int_0^\infty\frac{1}{1+x^n}\,dx\cdot e^{\frac{2\pi i}{n}}$$

$$= \left(1-e^{\frac{2\pi i}{n}}\right)\cdot\int_0^\infty\frac{1}{1+x^n}\,dx$$

$$\Rightarrow \int_0^\infty\frac{1}{1+x^n}\,dx = -\frac{2\pi i}{n}\cdot\exp(i\pi/n)\Big/\left(1-e^{\frac{2\pi i}{n}}\right).$$

∎

[2020 spring.
9. Suppose that f is an entire and $\mathrm{Re} f(z) \neq 0$ for all $z \in \mathbb{C}$.
Show that f is constant.
Proof. Since f is entire, $z \mapsto \mathrm{Re} f(z)$ is continuous.
Therefore, $\mathrm{Re} f > 0$ or $\mathrm{Re} f < 0$, because if not, then the intermidate value property gives for some $z \in \mathbb{C}$, $\mathrm{Re} f(z) = 0$, it is contradiction.
Suppose $\mathrm{Re} f > 0$. Define

$$g(z) = e^{-f(z)}.$$

Then, $|g(z)| = e^{-\mathrm{Re} f(z)} \leq e^0 = 1$, therefore g is bounded entire, so g is constant, thus $g'(z) = -f'(z) \cdot e^{-f(z)} = 0$, $f'(z) = 0$.
Therefore, $f(z)$ is a constant.

And, if $\mathrm{Re} f < 0$, then put $g(z) = e^{f(z)}$, then we obtain same result.
□

2020 Spring

10. Let $f(x)=\sum_{n=1}^{\infty}\frac{x^{\frac{3}{2}}\cdot\cos nx}{n^{\frac{5}{2}}}$, $x\in(0,\infty)$

Prove or disprove that

a). The f is differentiable on $(0,\infty)$

b). The series converges uniformly on $(0,\infty)$.

Proof. Put $f_n(x)=\frac{x^{\frac{3}{2}}\cdot\cos nx}{n^{\frac{5}{2}}}$ And, let $x_0\in\mathbb{R}$ be given.

Let $M>x_0$ be fixed. Then, for each $N\in\mathbb{N}$,

$\sum_{n=1}^{N}|f_n(M)|\le\sum_{n=1}^{N}\frac{M^{\frac{3}{2}}}{n^{\frac{5}{2}}}$ and since $\sum_{n=1}^{\infty}\frac{1}{n^{\frac{5}{2}}}$ converges by p-test,

so $\sum_{n=1}^{N}f_n(M)$ converges absolutely and so that it converges.

Meanwhile, given $x\in[0,M]$,

$$|f_n'(x)|=\left|\frac{3}{2}\cdot\frac{x^{\frac{1}{2}}\cdot\cos nx}{n^{\frac{5}{2}}}-\frac{x^{\frac{3}{2}}\cdot\sin nx}{n^{\frac{3}{2}}}\right|$$

$$\le\frac{3}{2}\cdot\frac{M^{\frac{1}{2}}}{n^{\frac{5}{2}}}+\frac{M^{\frac{3}{2}}}{n^{\frac{3}{2}}}\quad\text{and}\quad\sum_{n=1}^{\infty}\left(\frac{3}{2}\cdot\frac{M^{\frac{1}{2}}}{n^{\frac{5}{2}}}+\frac{M^{\frac{3}{2}}}{n^{\frac{3}{2}}}\right)\text{ converges,}$$

the Weierstrass M-test gives $\sum_{n=1}^{\infty}f_n'$ converges absolutely on $(0,M]$.

Now, by the theorem for uniformly convergence and differentiation,

f is differentiable at $x_0\in(0,M]$ with

$$f'(x_0)=\lim_{N\to\infty}\sum_{n=1}^{N}f_n'(x_0).$$

Since $x_0>0$ was arbitrary, so proved.

b). Assume $\sum_{n=1}^{N}f_n(x)\to f$ uniformly. Then, equivalently,

given $\varepsilon>0$, there is $N\in\mathbb{N}$ such that $m>n\ge N$ implies

$$\sup_{x>0}\left|\sum_{k=n}^{m}f_k(x)\right|<\varepsilon,$$ so that $f_n(x)\to0$ uniformly on $(0,\infty)$.

But, given $n\in\mathbb{N}$, $\sup_{x>0}|f_n(x)|=\infty$ because; for $x_k=2\pi k/n$, $f_n(x_k)\to\infty$.

So, it contradicts to the assumption. ☒

Let $f_N(x)=\sum_{n=1}^{N}\frac{x^{\frac{3}{2}}\cdot\cos nx}{n^{\frac{5}{2}}}$

a). Let $x>0$ be given. And, $M>x$ be fixed.

Then, $\sum_{n=1}^{N}\left|\frac{M^{\frac{3}{2}}\cdot\cos nM}{n^{\frac{5}{2}}}\right|\le\sum_{n=1}^{N}\frac{M^{\frac{3}{2}}}{n^{\frac{5}{2}}}$ and it converges by p-series test,

and so that the $f_N(M)$ converges absolutely, so converges.

Meanwhile, for all $x\in[0,M]$, and each $n\in\mathbb{N}$,

$$\left|\frac{3}{2}\cdot\frac{x^{\frac{1}{2}}\cdot\cos(nx)}{n^{\frac{5}{2}}}-\frac{x^{\frac{3}{2}}\cdot\sin(nx)}{n^{\frac{3}{2}}}\right|\le\frac{3}{2}\cdot\frac{\sqrt{M}}{n^{\frac{5}{2}}}+\frac{(M)^{\frac{3}{2}}}{n^{\frac{3}{2}}}\quad\text{and since}$$

$$\sum_{n=1}^{\infty}\left(\frac{3}{2}\cdot\frac{\sqrt{M}}{n^{\frac{5}{2}}}+\frac{(M)^{\frac{3}{2}}}{n^{\frac{3}{2}}}\right)<\infty,\quad\text{so that}$$

by the Weierstrass M-test, f_N converges uniformly on $(0,M]$.

Now, by the theorem for differentiation of uniformly convergence,

f is differentiable at $x\in\mathbb{R}$ with

$$f'(x)=\lim_{N\to\infty}f_N'(x).$$

b). But, the series does not converge uniformly, because:

given $N\in\mathbb{N}$, $\sup_{x>0}|f(x)-f_N(x)|=\sup_{x>0}\left|\sum_{n=N}^{\infty}\frac{x^{\frac{3}{2}}\cdot\cos nx}{n^{\frac{5}{2}}}\right|$

[2020 Spring.

11. (10 pts.) Let $\ell^1(\mathbb{N}) = \{ (a_n)_{n\in\mathbb{N}} : a_n \in \mathbb{R}, \sum_{n\in\mathbb{N}} |a_n| < \infty \}$, and define the distance functions d, $d_* : \ell^1(\mathbb{N}) \times \ell^1(\mathbb{N}) \to [0,\infty)$ by

$$d(a,b) = \sum_{n\in\mathbb{N}} |a_n - b_n|, \quad d_*(a,b) = \sup_{n\in\mathbb{N}} |a_n - b_n|, \quad a = (a_n),\ b = (b_n) \in \ell^1(\mathbb{N}).$$

(a) Show that the metric space $(\ell^1(\mathbb{N}), d)$ is complete, i.e. every Cauchy sequence converges in $\ell^1(\mathbb{N})$.

(b) Find the completion of the metric space $(\ell^1(\mathbb{N}), d_*)$.

Prof. a),

12020. Spring

15. Let $r = r(\theta)$ be a curve in $\mathbb{R}^2$ defined in polar coordinates (r, θ) with $r > 0$, $0 \le \theta < 2\pi$.

a). Express the arc length of the curve using $r(\theta), r'(\theta)$.

b). Express the curvature at $r(\theta)$ using $r(\theta), r'(\theta), r''(\theta)$.

Pf. Denote $x = r\cdot\cos\theta$ and $y = r\cdot\sin\theta$. $\therefore \alpha(\theta) = (x, y)$.

a). $\frac{dx}{d\theta} = r'\cos\theta - r\sin\theta$, $\frac{dy}{d\theta} = r'\sin\theta + r\cos\theta$.

therefore, $\int_0^{2\pi} \left|\frac{d\alpha}{d\theta}\right| d\theta = \int_0^{2\pi} \sqrt{r'^2 + r^2}\, d\theta$.

b). Since the tangent vector with respect to the arc length s

$$t(\theta) = \frac{d\alpha}{ds} = \frac{1}{\frac{ds}{d\theta}} \cdot \frac{d\alpha}{d\theta} = \left|\frac{d\alpha}{d\theta}\right|^{-1} \cdot \frac{d\alpha}{d\theta} = \frac{1}{\sqrt{r'^2+r^2}} \cdot (r'\cos\theta - r\sin\theta,\ r'\sin\theta + r\cos\theta)$$

and since the α is plane curve, the normal vector is

$$n(\theta) = \frac{1}{\sqrt{r'^2+r^2}} \cdot (-r'\sin\theta - r\cos\theta,\ r'\cos\theta - r\cdot\sin\theta)$$

Since the curvature is

$$\begin{aligned} k(\theta) &= \left\langle \frac{dt}{ds}, n \right\rangle = \left\langle \frac{d\theta}{ds}\cdot\frac{dt}{d\theta}, n \right\rangle \\ &= \left|\frac{d\theta}{d\alpha}\right| \cdot \frac{1}{(\sqrt{r'^2+r^2})^2} \cdot \Big\langle \big(r''\cos\theta - r'\sin\theta - r'\sin\theta - r\cos\theta,\ r''\sin\theta + r'\cos\theta + r'\cos\theta \\ &\qquad - r\sin\theta\big),\ (-r'\sin\theta - r\cos\theta,\ r'\cos\theta - r\sin\theta)\Big\rangle \\ &= \frac{1}{(r'^2+r^2)^{\frac{3}{2}}} \cdot \Big\langle \big((r''-r)\cos\theta - 2r'\sin\theta,\ (r''-r)\sin\theta + 2r'\cos\theta\big), \\ &\qquad (-r'\sin\theta - r\cos\theta,\ r'\cos\theta - r\sin\theta)\Big\rangle \\ &= \frac{1}{(r'^2+r^2)^{\frac{3}{2}}} \cdot \left(r''r + r^2 + 2r'^2\right) \end{aligned}$$

Comment. 미분기하는 사라져야 한다.

Pebris

observe that

$\frac{d\alpha}{d\theta} = \left(\frac{dx}{d\theta}, \frac{dy}{d\theta}\right)$, so $t(\theta) = \frac{1}{\sqrt{r'^2+r^2}} \cdot (r'\cos\theta - r\cdot\sin\theta,\ r'\sin\theta + r\cos\theta)$

$n(\theta) = \frac{1}{\sqrt{r'^2+r^2}} \cdot (-r'\sin\theta - r\cdot\cos\theta,\ r'\cos\theta - r\cdot\sin\theta$

$$t'(\theta) = \left(-\frac{1}{2}\cdot\frac{2r'r''+2rr'}{\sqrt{r'^2+r^2}}\right)\cdot\sqrt{r'^2+r^2}\cdot t(\theta) + \frac{1}{\sqrt{r'^2+r^2}}\cdot\Big(r''\cos\theta - r'\sin\theta - r'\sin\theta - r\cos\theta,\ r''\sin\theta + r'\cos\theta + r'\cos\theta - r\sin\theta\Big)$$

Now, the Curvature

$$\begin{aligned} k(\theta) = \langle t', n\rangle &= \frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\Big\langle \big((r''-r)\cos\theta - 2r'\sin\theta,\ (r''-r)\sin\theta + 2r'\cos\theta\big), \\ &\qquad (-r'\sin\theta - r\cos\theta,\ r'\cos\theta - r\sin\theta)\Big\rangle \\ &= \frac{1}{(r'^2+r^2)^{\frac{3}{2}}}\cdot\left(r^2 - rr'' + 2r'^2\right) \end{aligned}$$

[2020, Fall.

1. Let p be a prime. Find the number of subgroups H of $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$, $|H| = p^2$

Proof. [By the fundamental theorem of finitely generated abelian group, either $H \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.]

If $H \cong \mathbb{Z}_p \times \mathbb{Z}_p$, then for all $x \in H$, $p \cdot x = 0$. Therefore,
$H \subseteq \{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\}$.
Meanwhile, $pa = 0$ and $pb = 0$ if and only if $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$ and $b \in \langle p^2 \rangle \subseteq \mathbb{Z}_{p^3}$,
so $\{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$.
Since $\langle p \rangle \times \langle p^2 \rangle$ has already order p^2, so $H = \langle p \rangle \times \langle p^2 \rangle$.]

If $H \cong \mathbb{Z}_{p^2}$, that is, H is generated by order p^2 element,
Since $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ has order $p^2 \iff p^2 \cdot x = 0$ and $px \neq 0$.
Every element $a \in \mathbb{Z}_{p^2}$ satisfies $p^2 a = 0$, and $b \in \mathbb{Z}_{p^3}$ satisfies $p^2 \cdot b = 0$
if and only if $b \in p\mathbb{Z}_{p^3}$ and $\mathbb{Z}_{p^3}$ has number of p^2 of p-multiples,
so there are $p^2 \cdot p^2 = p^4$ elements satisfying $p^2 \cdot x = 0$.
And, $px = 0$ if and only if $a \in p\mathbb{Z}_{p^2}$ and $b \in p^2\mathbb{Z}_{p^3}$, where $x = (a,b)$.
So, there are p^2 elements satisfying $px = 0$.
Consequently, there are $x^4 - x^2$ elements x such that $H = \langle x \rangle$.
Moreover, the cyclic group of order p^2 has $\varphi(p^2) = p^2 - p$ generators,
So the possible $H \cong \mathbb{Z}_{p^2}$ are

$$(p^4 - p^2)/(p^2 - p)$$

numbers.

Finally, the number of H is $1 + (p^4 - p^2)/(p^2 - p)$ □

Note: for a prime p, $\varphi(p^n) = p^n - p^{n-1}$.

[2020 Fall.

2. For each $n \in \mathbb{N}$, let $a_n = \cos(2\pi/n)$.

a). Prove that a_n is algebraic over $\mathbb{Q}$

b). Calculate $\deg_{\mathbb{Q}}(a_5)$.

Proof. a). Let $\xi = \exp(2\pi i/n)$. Then, $\xi^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$ gives that $a_n = \cos(2\pi/n) = (\xi + \xi^{-1})/2$.

Meanwhile, since $\xi^n = \exp 2\pi i = 1$, so $\xi^n - 1$ gives ξ is algebraic over $\mathbb{Q}$.

Therefore, ξ^{-1} is also algebraic, thus $a_n = \dfrac{\xi + \xi^{-1}}{2}$ is algebraic over $\mathbb{Q}$.

b). Let $w = \exp(2\pi i/5)$. Since w is the primitive 5th root of unity, $w^4 + w^3 + w^2 + w + 1 = 0$. And, dividing by w^2 then $2a_5 = w + w^{-1}$ implies

$$\begin{aligned} w^4 + w^3 + w^2 + w + 1 = 0 &\Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0 \\ &\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0 \\ &\Rightarrow 4a_5^2 + 2a_5 - 1 = 0. \end{aligned}$$

Since $4x^2 + 2x - 1$ is irreducible over $\mathbb{Q}$ by the discriminant $2^2 + 16 = 20$ but $\sqrt{20} \notin \mathbb{Q}$, so $\deg_{\mathbb{Q}}(a_5) = 2$. ☑

[Fail.

a) Notice that De Moivre's formular gives

$$e^{i \cdot nz} = (\cos nz + i \cdot \sin nz) = (\cos z + i \cdot \sin z)^n$$

Therefore

$$\begin{aligned} \cos n \cdot \frac{2\pi}{n} = 1 &= \mathrm{Re}\left(\cos\frac{2\pi}{n} + i \cdot \sin\frac{2\pi}{n}\right)^n \\ &= \mathrm{Re}\left(\sum_{k=0}^{n} \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k} \cdot \binom{n}{k}\right) \end{aligned}$$

Meanwhile, $\sin\left(\frac{2\pi}{n}\right)^2 = 1 - \cos\left(\frac{2\pi}{n}\right)^2$ and the real part of $\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k}$ yield only $n-k$ is even, since i^{n-k} is real $\Leftrightarrow n-k$, so $\mathrm{Re}\left(\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos\frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin\frac{2\pi}{n}\right)^{n-k}\right) - 1 = 0$ is evaluated $\cos\left(\frac{2\pi}{n}\right)$ for some polynomial over $\mathbb{Q}$, so a_n is algebraic over $\mathbb{Q}$.

b). From above argument,

$$\begin{aligned} 1 &= 1 \cdot a_5^0 \cdot 0 + 5 \cdot a_5^1 \cdot (1 - a_5^2) - 10 \cdot a_5^3 \cdot (1 - a_5^2)^2 + a_5^5 \\ &\qquad k=0 \qquad\qquad k=1 \qquad\qquad k=3 \qquad\quad k=5 \\ &= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5 \\ &= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7. \end{aligned}$$

So, $10a_5^7 - 11a_5^5 + 15a_5^3 - 5a_5 + 1 = 0$.

[2020, Fall]

3. Let $\mathbb{R}$ be a field. Show that $(x^3-y^5) \subseteq \mathbb{R}[x,y]$ is prime ideal.

Proof. Since $\mathbb{R}[x,y]$ is U.F.D., every irreducible element is prime.
Thus, enough to show that x^3-y^5 is irreducible.
Notice that: $\mathbb{R}[x,y] = \mathbb{R}[y][x]$ by definition, so $x^3-y^5 \in \mathbb{R}[y][x]$.
Since $\gcd(1, y^5) = 1$, x^3-y^5 is primitive, by the Gauss Lemma,
x^3-y^5 is irreducible over $\mathbb{R}[y] \Leftrightarrow x^3-y^5$ is irreducible over $\mathbb{R}(y)$.
But, since x^3-y^5 has degree 3, x^3-y^5 is reducible only if it has a linear factor, in other say, x^3-y^5 has a root in $\mathbb{R}(y)$.
Suppose that for some $f(y)/g(y) \in \mathbb{R}(y)$,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then, $f(y)^3 = y^5 \cdot g(y)^3$, But $\deg f(y)^3 \equiv 0 \mod 3$, and
$\deg y^5 \cdot g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \mod 3$.
Therefore, there is no such a root, so x^3-y^5 is irreducible over $\mathbb{R}[y]$,
So it is irreducible over $\mathbb{R}[x,y]$.

Note:

$$K[x,y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test}}.$$

[Modified

Prove that $\gcd(m,n)=1$ implies $x^m-y^n \in F[x,y]$ is irreducible.

[2020. Fall.

4. Let $W \subseteq V = \mathbb{R}^n$ be a subspace. Show that there exists a homogeneous system of linear equation whose solution space is W.

Proof. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_k\}$ be a basis for W.

Then, we can extend the basis $\mathcal{B}$ to $\overline{\mathcal{B}} = \mathcal{B} \cup \{\alpha_{k+1}, \cdots, \alpha_n\}$ for V.

Now, consider the map:

$$f_j : V \to \mathbb{R} ; \sum c_i \alpha_i \mapsto c_j$$

Then, $W = \ker f_{k+1} \cap \cdots \cap \ker f_n$, so for the standard basis S for V,

$A = \begin{pmatrix} [f_{k+1}]_S \\ \vdots \\ [f_n]_S \end{pmatrix}$ satisfies $\ker A = W$, so

($[f]_S$ is row vector representation).

$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is the system we desired.

[2020 Fall.
5. Let A be an invertible $n \times n$ matrix and let $B = adj(A)$.
Show that the characteristic polynomials $\phi_A(t)$ and $\phi_B(t)$ of A and B, respectively, satisfy the formal identity

$$(-t)^n \cdot \phi_A\left(\frac{\det(A)}{t}\right) = \det(A) \cdot \phi_B(t)$$

Proof. Since $adj(A) \cdot A = BA = \det A \cdot I_n$,

and by the characteristic polynomial

$$(-t)^n \phi_A\left(\frac{\det(A)}{t}\right) = (-t)^n \det\left(\frac{\det A}{t} \cdot I_n - A\right) = \det(tA - \det A \cdot I_n) \text{ and}$$

$$\det(A)\phi_B(t) = \phi_B(t) \cdot \det(A) = \det(tA - BA) = \det(tA - \det A \cdot I_n),$$

so proved. ∎

2020 Fall.

6. Let f be a C^2-function defined on $(0,\infty)$ such that

$$xf''(x)+f'(x)=0,\ \forall x>0,\quad f(1)=0,\ f'(1)=\frac{1}{2\pi}.$$

Find f as explicitly.

Proof. Since $xf''(x)+f'(x)=(xf'(x))'$, therefore for some C,

$$xf'(x)=C \Rightarrow f(x)=C\cdot\log x+C_0.$$

Now, using boundary conditions: $f(1)=C_0=0$, $f'(1)=C=\frac{1}{2\pi}$, therefore

$$f(x)=\frac{1}{2\pi}\cdot\log x \qquad \square$$

[2020 Fall,
7. For $n=1,2,3,\cdots$, let $P_n(z)=\sum_{k=0}^{n}\frac{z^k}{k!}$ where $z\in\mathbb{C}$.

Show that: for every $R>0$, there is a positive integer $N\in\mathbb{N}$ such that
$\forall n\geq N$ and $|z|<R$, $P_n(z)\neq 0$.

Proof. Let $R>0$ be fixed. Then, for $\sup=\sup_{|z|<R}$,
$\sup|P_n(z)-e^z|=\sup\left|\sum_{k=n+1}^{\infty}\frac{z^k}{k!}\right|\leq\sum_{k=n+1}^{\infty}\frac{|z|^k}{k!}\leq\sum_{k=n+1}^{\infty}\frac{R^k}{k!}\to 0$ as $n\to\infty$,
So $P_n(z)\to e^z$ uniformly on $|z|<R$.
Therefore, given $\varepsilon=e^{-R}/2$, take $N\in\mathbb{N}$ such that
$n\geq N\Rightarrow|P_n(z)-e^z|<\varepsilon$ for all $|z|<R$, so that
$|P_n(z)|\geq\frac{e^{-R}}{2}>0$, So proved. ☒

[Debris
Let $R>0$ be fixed. Then, for each $n\in\mathbb{N}$,
$P_n(z)=z^n\cdot\left(\frac{1}{n!}+\frac{1}{(n-1)!}\cdot\frac{1}{z}+\cdots+\frac{1}{z^{n-1}}+\frac{1}{z^n}\right)\ (0<|z|<R)$
By the triangle inequality, for $0<|z|<R$,
$|P_n(z)|\geq|z|^n\cdot\left|\frac{1}{n!}-\left|\frac{1}{(n-1)!\cdot z}+\cdots+\frac{1}{1!\cdot z^{n-1}}+\frac{1}{z^n}\right|\right.$
Now, for $|z|>1$,
$\left|\frac{1}{(n-1)!z}+\cdots+\frac{1}{z^n}\right|\leq\left|\frac{1}{(n-1)!z}\right|+\cdots+\left|\frac{1}{z^n}\right|\leq\frac{1}{|z|^n}\cdot\left(\frac{1}{(n-1)!}+\cdots+\frac{1}{0!}\right)$
$\to 0$ as $n\to\infty$

[2020 Fall.

8. Define a sequence of functions $(g_n)_{n=1}^{\infty}$ by

$$g_n(x) = \frac{x^{n+1} + 2\cdot\cos x}{(n^2+1)\cdot(x^n+1)}$$

Prove that $\sum_{n=1}^{\infty} g_n$ converges to a continuous function $[0,\infty)$.

Proof. Let $f_n(x) = \frac{x^{n+1}}{(n^2+1)\cdot(x^n+1)}$ and $h_n(x) = \frac{2\cdot\cos x}{(n^2+1)\cdot(x^n+1)}$ for each n.

First, since $|h_n(x)| \le \frac{2}{n^2}$ for all $x\in[0,\infty)$ and $\sum_{n=1}^{\infty}\frac{2}{n^2}$ converges, by the Weierstrass M-test, $\sum_{n=1}^{N} h_n$ converges uniformly and so that $\sum_{n=1}^{\infty} h_n(x)$ is continuous, being limit of continuous functions.

Meanwhile, let $x_0\in[0,\infty)$ be given, and $M > x_0$ be fixed.

Then, since $\frac{x^{n+1}}{x^n+1} = x\cdot\frac{x^n}{x^n+1} < x$, for each n and x, so that

$|f_n(x)| \le \frac{x}{n^2+1} \le \frac{M}{n^2+1}$ for all $x\in[0,M]$, and $\sum_{n=1}^{\infty}\frac{M}{n^2+1}$ converges, so by the Weierstrass M-test again, similarly,

$\sum_{n=1}^{\infty} f_n(x)$ is continuous on $[0,M]$, so at x_0.

Since $x_0\in[0,\infty)$ was arbitrary, given $x\ge 0$,

$$\sum_{n=1}^{\infty} g_n(x) = \sum_{n=1}^{\infty} f_n(x) + h_n(x) = \sum_{n=1}^{\infty} f_n(x) + \sum_{n=1}^{\infty} h_n(x)$$

is continuous at x. So proved. □

Note: In actually, for any $M>0$,

$$|g_n| \le \frac{M+2}{n^2+1}.$$

[2020 Fall.
14, Show that a compact subset S of a Hausdorff space X is closed.
And, determine the statement is true without the Hausdorff condition.

[2021. Spring.

1. Show that no group of 2020 is simple.

Proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p-subgroups.
By the Sylow theorem, $n_{101} \equiv 1 \bmod 101$ and $n_{101} \mid 20$, so n_{101} must be 1,
Therefore, the Sylow 101-subgroup is normal so proved.

[Without using Sylow theorem.
By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.
Denote $A = \{gHg^{-1} \mid g \in G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \to \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\#A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \to A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H-orbit of this action is $Orb_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\#Orb_H(K) = |H : Stab_H(K)|, \ Stab_H(K) \leq H \Rightarrow |Stab_H(K)| \in \{1, 101\}.$$

So, $\#Orb_H(K) \in \{1, 101\}$ and $Orb_H(K) \leq A$, so $\#Orb_H(K) = 1$.
therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.
Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{H\}$. That is, $H \trianglelefteq G$. ∎

2021 spring

2. (10 pts.) Let A and B be non empty subsets of a ring R. Denote by $A+B$ the set $\{a+b \mid a \in A, b \in B\}$ and by AB the set of all finite sums $\{a_1b_1 + \ldots + a_nb_n \mid n \in \mathbb{Z}^+; a_i \in A; b_i \in B\}$. Show that if A and B are ideals of R, then $A+B$ and AB are ideals of R.

Proof. Let $a_1, a_2 \in A$ and $b_1, b_2 \in B$ be given. Then,

$$(a_1+b_1)-(a_2+b_2)=(a_1-a_2)+(b_1-b_2) \in A+B$$

and $0 = 0+0 \in A+B$, so $A+B$ is a subring. Moreover, given $r \in R$ and $a \in A$, $b \in B$,

$$r\cdot(a+b)=ra+rb \in A+B \quad \text{and} \quad (a+b)\cdot r = ar+br \in A+B$$

since both A and B are ideals, so $A+B$ is ideal.

Let $\sum_{k=1}^{n} a_k b_k \in AB$ and $\sum_{k=1}^{m} a_k' b_k' \in AB$ be given, where $a_i, a_i' \in A$ and $b_i, b_i' \in B$. Then,

$$\sum_{k=1}^{n} a_k b_k - \sum_{k=1}^{m} a_k' b_k' = \sum_{k=1}^{n} a_k b_k + \sum_{k=1}^{m} (-a_k') b_k' \in AB, \text{ and } 0\cdot 0 \in AB,$$

so AB subring.

And given $r \in R$,

$$r\cdot\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n} (ra_k) b_k \in AB \quad \text{and} \quad \left(\sum_{k=1}^{n} a_k b_k\right)\cdot r = \sum_{k=1}^{n} a_k (b_k \cdot r) \in AB,$$

since A and B are ideals, so AB is ideal. ∎

[2021. spring.
3. Let p be a prime number, and L be the Galois extension $\mathbb{Q}(\zeta)$ where ζ is a primitive p^2th root of unity. 55.
a). Show that all $\mathbb{Q}$-automorphisms of L are of the form

$$\zeta \mapsto \zeta^k, \quad p \nmid k, \quad k \in \mathbb{Z}.$$

b). Show that L contains a subfield whose Galois group is $\mathbb{Z}/p\mathbb{Z}$.
Proof.
a). By definition, $\{\zeta^0, \zeta^1, \cdots, \zeta^{p^2-1}\}$ is a set of roots of the polynomial

$$x^{p^2}-1 = (x-1)(x^{p^2-1}+x^{p^2-2}+\cdots+x+1)$$

Meanwhile, $x^{p^2-1}+x^{p^2-2}+\cdots+x+1$ has a irreducible factor $\prod_{\substack{1\le a\le p^2 \\ \gcd(a,p^2)=1}} (x-\zeta^a) = P(x)$

So, since the automorphisms maps a root to the conjugate, so every automorphism of $L/\mathbb{Q}$ maps $\zeta \mapsto \zeta^k$, $\gcd(k, p^2)=1$, so, $p \nmid k$.
Moreover, since $\mathbb{Q}(\zeta)$ is a finite dimensional vector space generated by ζ^k $(k=0,1,\cdots)$, so if $\zeta \mapsto \zeta^k$ chosen, then the automorphism is determined. it proves a),

b). Sine $\varphi(p^2) = p^2-p$, where φ is the Euler-Phi function, the Galois extension of $P(x)$ over $\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta)$ and it has the degree of p^2-p, so, this extension K satisfies $|Gal(K/\mathbb{Q})| = p^2-p$. And, by the Cauchy's theorem, it has a subgroup of order p, $G \le Gal(K/\mathbb{Q})$. Now, the fixed field $\mathbb{Q}(\zeta)^G$ is we desired.

Revised in the Next

[2021 Spring.
4. Let V be a finite dimensional vector space.
Show that every basis for V has the same number of elements.
Proof. Let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ and $\mathcal{B}' = \{\beta_1, \cdots, \beta_m\}$ be basis for V.
Assume $n > m$. Since $\mathcal{B}'$ span V, so for each $1 \le i \le n$,

$$\alpha_i = c_{i1}\beta_1 + \cdots + c_{im}\beta_m \text{ for some } c_{ij}.$$

In particular,

$$\alpha_{m+1} = c_{m+1,1}\beta_1 + \cdots + c_{m+1,m}\beta_m$$

Now, let $a_1, \cdots, a_n$ be scalars such that

$$\begin{pmatrix} c_{11} & \cdots & c_{1m} \\ c_{21} & \ddots & \vdots \\ c_{m1} & \cdots & c_{mm} \end{pmatrix}^T \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} c_{m+1,1} \\ \vdots \\ c_{m+1,m} \end{pmatrix}.$$

Since the matrix in the left term is invertible, because $\alpha_1, \cdots, \alpha_n$ are linearly independent, so such $a_1, \cdots, a_n$ exist as not all zero.
And, now,

$$\alpha_{m+1} = a_1\alpha_1 + \cdots + a_n\alpha_n \text{ is non-trivial representation,}$$

it contradicts to $\mathcal{B}$ is basis.
Similarly, $n < m$ cannot be happen, so $n = m$. ∎

[2021. spring]

7. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \cdots + a_d z^d \quad \text{over } \mathbb{C},\ a_d \neq 0.$$

8m.

Evaluate the value $\lim_{R\to\infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz$

Proof. By the fundamental theorem of Algebra, f has d-roots $z_1, \cdots, z_d \in \mathbb{C}$,

That is, $f(z) = a_d (z - z_1) \cdots (z - z_d)$

Moreover, $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \cdots + \frac{1}{z - z_d}$, since calculation.

Therefore, for $R > \max\{|z_1|, \cdots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz = \sum_{n=1}^{d} \operatorname{Res}\left(\frac{z}{z - z_n}, z_n\right) = \sum_{n=1}^{d} \lim_{z \to z_n} z = z_1 + \cdots + z_n$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \cdots + z_n) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d}\right),$$

□

[2021. Spring.]

8. Let $\{v_k\}, \{s_k\}, \{u_k\}$ be non-negative real sequences satisfying

$$v_{k+1} \le v_k - s_k + u_k \text{ for all } k \in \mathbb{N} \text{ and } \sum_{k=0}^{\infty} u_k < \infty$$

Show $s_k \to 0$ and $\{v_k\}$ has a limit as $k \to \infty$.

Proof. observe that

$$v_{k+1} - v_k + s_k \le u_k \text{ for all } k \in \mathbb{N}.$$

Therefore, given $N > 1$,

$$\sum_{k=1}^{N} (v_{k+1} - v_k + s_k) = v_{N+1} - v_1 + \sum_{k=1}^{N} s_k \le \sum_{k=1}^{N} u_k \le \sum_{k=1}^{\infty} u_k$$

thus $$\sum_{k=1}^{N} s_k \le \sum_{k=1}^{\infty} u_k + v_1 - v_{N+1} \le \sum_{k=1}^{\infty} u_k + v_1 < \infty,$$

So $\sum_{k=1}^{\infty} s_k$ converges, $s_k \to 0$ as $k \to \infty$. ∎

Meanwhile, define

$$w_k = v_k - \sum_{i=1}^{k-1} u_i \quad (k \ge 2).$$

Then, for any $k \ge 2$,

$$w_{k+1} - w_k = v_{k+1} - v_k - u_k \le -s_k \le 0, \text{ so } \{w_k\} \text{ is monotone decreasing.}$$

Moreover, given $k \ge 2$,

$$w_k = v_k - \sum_{i=1}^{k-1} u_i \ge -\sum_{i=1}^{k-1} u_i \ge -\sum_{i=1}^{\infty} u_i$$

thus, w_k is bounded below, so converges. Now,

$$v_k = \sum_{i=1}^{k-1} u_i + w_k$$

converges, being sum of convergent sequences. ∎

[2021 Spring.
9. Show that every positive harmonic function in $\mathbb{C}$ is constant.
Proof. Let $u(x,y)$ be the positive harmonic function. Then, there is an entire $f(x,y)$ such that $\operatorname{Re} f = u$. Now, let

$$g(z) = e^{-f(z)}.$$

Then, since $|g(z)| = e^{-\operatorname{Re} f}$ and since $\operatorname{Re} f > 0$,

$$\leq e^{0} = 1.$$

Now, g is entire bounded function, so by the Liouville's theorem, g is constant, so that $g' = -f' \cdot e^{-f} = 0$, thus $f' \equiv 0$.
Now, $f = C$ for some $C \in \mathbb{C}$, so $u = \operatorname{Re} f$ is also constant. □

2021 Fall.

1. Let a_i, n_i, $i=1,2,\cdots,\ell$ be Positive integers. Let G be a group defined by $G=\mathbb{Z}^{\sum_{i=1}^{\ell} a_i}/H$, where

$$H=\langle (\underbrace{k_1 n_1,\cdots,k_1 n_1}_{a_1},\underbrace{k_2 n_2,\cdots,k_2 n_2}_{a_2},\cdots,\underbrace{k_\ell n_\ell,\cdots,k_\ell n_\ell}_{a_\ell}) \mid k_i\in\mathbb{Z},\forall i\rangle$$

Classify the group G according to the FTFGA.

Proof. Define a map

$$\varphi: \mathbb{Z}^{\sum_{i=1}^{\ell} a_i} \longrightarrow n_1\mathbb{Z}\times\mathbb{Z}^{a_1-1}\times n_2\mathbb{Z}\times\mathbb{Z}^{a_2-1}\times\cdots\times n_\ell\mathbb{Z}\times\mathbb{Z}^{a_\ell-1}$$

$$:(x_1,\cdots,x_{\sum a_i})\mapsto(x_1 \bmod n_1, x_1-x_2, x_1-x_3,\cdots,x_1-x_{a_1-1},\cdots)$$

Then, it is onto Homomorphism, with $\ker\varphi=H$, therefore

$$G=\mathbb{Z}^{\sum a_i}/H=\mathbb{Z}^{\sum a_i}/\ker\varphi\cong\mathbb{Z}^{(\sum a_i)-\ell}\times n_1\mathbb{Z}\times\cdots\times n_\ell\mathbb{Z},$$

by the first isomorphism theorem and FTFGA. ∎

$\mathbb{Z}\times\mathbb{Z}/\langle(a,a)\mid a\in n\mathbb{Z}\rangle$

$\mathbb{Z}\times\mathbb{Z}\to n\mathbb{Z}\times\mathbb{Z}$

$(a,b)\mapsto(a \bmod n, a-b)$

2021 Fall.

2. (10 pts) Let R be a commutative ring with unity such that for all $r \in R$, there exists some $n > 1$ such that $r^n = r$. Show that any prime ideal of R is maximal.

Proof. There are two ways. Let $P \subset R$ be a prime ideal.

1). Let $M \subsetneq R$ be a proper ideal such that $P \subseteq M \subsetneq R$.

Assume $M \setminus P \neq \emptyset$. Take $a \in M \setminus P$. Then, by the hypothesis,

$$a^n = a \quad \text{for some } n > 1,$$ so

$a \cdot (a^{n-1} - 1) = 0 \in P$. Since P is prime ideal, either $a \in P$ or $a^{n-1} - 1 \in P$.

But, $a \in P$ is contradiction to $a \in M \setminus P$. If $a^{n-1} - 1 \in P$, then

$a^{n-1}, a \in M \Rightarrow a^n - (a^{n-1}) = 1 \in M$. So $M = R$, but it contradicts.

Thus, P is a Maximal ideal. □

2021 Fall.

3. Find the order of the splitting field of the polynomial x^5+8 over $\mathbb{Q}$.

Proof. Notice that:

$x^5+8=0$ if and only if $x=\sqrt[5]{8}\cdot\exp\left(\frac{1}{5}\pi i+\frac{k}{5}\cdot 2\pi i\right)$, $k\in\{0,1,2,3,4\}$.

Denote that $\alpha=\exp\left(\frac{1}{5}\pi i\right)$, $\beta=\left(\frac{1}{5}\cdot 2\pi i\right)$. Then, the splitting field is

$$K=\mathbb{Q}\left(\sqrt[5]{8}\cdot\alpha\cdot\beta^k \mid k=0,1,2,3,4\right)$$

$$=\mathbb{Q}(\sqrt[5]{8}\alpha,\beta)$$

observe that: $[\mathbb{Q}(\sqrt[5]{8}\alpha):\mathbb{Q}]=5$, because: $(\sqrt[5]{8}\alpha)^5+8=-8+8=0$, thus x^5+8 has $\sqrt[5]{8}\alpha$ as a root, and

$$(x+2)^5+8=\sum_{k=0}^{5}\binom{5}{k}x^{5-k}\cdot 2^k+8$$

$$=x^5+10x^4+40x^3+80x^2+80x+32+8$$

$$=x^5+10x^4+40x^3+80x^2+80x+40 \text{ is irreducible,}$$

by the eisenstein criterion for 5, so x^5+8 is irreducible.

And $[\mathbb{Q}(\beta):\mathbb{Q}]=4$, because: $\beta^5-1=0$, so

$$x^5-1=(x-1)(x^4+x^3+x^2+x+1)$$

implies the cyclotomic polynomial $x^4+x^3+x^2+x+1$ has β as a root, and it is irreducible, since $\frac{(x+1)^5-1}{x}=\frac{1}{x}\cdot(x^5+5x^4+10x^3+10x^2+5x+(1-1))$, irreducible.

Now, by the tower Lemma, $[K:\mathbb{Q}]$ is divided by 4 and 5, so $\gcd(4,5)=1$ implies $[K:\mathbb{Q}]\geq 20$, and since

$$[K:\mathbb{Q}]=[K:\mathbb{Q}(\beta)][\mathbb{Q}(\beta):\mathbb{Q}]\leq[\mathbb{Q}(\sqrt[5]{8}\cdot\alpha):\mathbb{Q}]\cdot[\mathbb{Q}(\beta):\mathbb{Q}]\leq 20$$

therefore $[K:\mathbb{Q}]=20$ ∎

Note: $[\mathbb{Q}(\alpha,\beta):\mathbb{Q}]\leq[\mathbb{Q}(\beta)(\alpha):\mathbb{Q}(\beta)]\cdot[\mathbb{Q}(\beta):\mathbb{Q}]\leq[\mathbb{Q}(\alpha):\mathbb{Q}]\cdot[\mathbb{Q}(\beta):\mathbb{Q}]$

because $m_{\alpha,\mathbb{Q}(\beta)}(x)\mid m_{\alpha,\mathbb{Q}}(x)$.

[2021 Fall.
4. Let $A \in M_{m\times m}$, $B \in M_{n\times m}$ be matrices over a field F.
Show that: If $AB = I_m$, then rank A = rank B,
Proof. Let $L_A : F^n \to F^m : X \mapsto AX$ and $L_B : F^m \to F^n : Y \mapsto BY$.
Since $\operatorname{Im} L_A \supseteq \operatorname{Im} L_A \circ L_B = \operatorname{Im} I_{F^m}$, rank $A \geq m$, so that rank $A = m$.
And, $\ker L_B \subseteq \ker L_A \circ L_B = \ker I_{F^m} = \{0\}$, so $\ker L_B = \{0\}$, thus nullity $L_B = 0$.
Now, by the dimension theorem,

$$\text{rank } B = m - \text{nullity } L_B = m,$$

So proved □

[2021 Fall.

5. Let V be a vector space over a field F, and let $\{v_1, \cdots, v_n\}$ be a basis for V.

a). Describe the dual basis $\{v_1^*, \cdots, v_n^*\}$ for V^*.
Prove that it is indeed a basis.

b). Let W be a subspace of V and let $W^\perp := \{f \in V^* \mid f[W] = 0\}$.
Prove that $\dim W^\perp = \dim V - \dim W$.

Proof. a). v_i^* is a linear functional on V, defined as

$$v_i^* : V \to F : \alpha \mapsto ([\alpha]_{\{v_1, \cdots, v_n\}})_i$$

That is, given $\alpha \in V$, $\alpha = \sum_{j=1}^{n} a_j v_j$ for some unique $a_j \in F$, and $v_i^*(\alpha) = a_i$. Since $\dim V^* = \dim V \cdot \dim F = n$, to show $\{v_1^*, \cdots, v_n^*\}$ is a basis, enough to show that the set is linearly independent.
Assume $\sum_{i=1}^{n} c_i v_i^* = 0$. Then, for each v_j,

$$0 = \left(\sum_{i=1}^{n} c_i v_i^*\right)(v_j) = c_j, \text{ so } c_1 = c_2 = \cdots = c_n = 0. \text{ So proved.}$$

b). Let $\{\alpha_1, \cdots, \alpha_k\}$ be a basis for W, and
$\{\alpha_1, \cdots, \alpha_k, \cdots, \alpha_n\}$ be an extended basis for V.
Then, $\{\alpha_{k+1}^*, \cdots, \alpha_n^*\}$ is linearly independent by a), so enough to show
$W^\perp = \text{span}\{\alpha_{k+1}^*, \cdots, \alpha_n^*\}$.

Given $f \in \text{span}\{\alpha_{k+1}^*, \cdots, \alpha_n^*\}$, $f = \sum_{i=k+1}^{n} c_i \alpha_i^*$, and so for any $1 \le j \le k$,
$f(\alpha_j) = \sum_{i=k+1}^{n} c_i \alpha_i^*(\alpha_j) = \sum_{i=k+1}^{n} c_i \delta_{ij} = 0$, so $f \in W^\perp$.
Given $f \in W^\perp$, $f = \sum_{i=1}^{n} c_i \alpha_i^*$ for some $c_i \in F$.
If $c_1, \cdots, c_k$ are not all zero, then for some $1 \le j \le k$,
$f(\alpha_j) = c_j \neq 0$, it is contradiction, by definition of $W^\perp$, so
$f \in \text{span}\{\alpha_{k+1}^*, \cdots, \alpha_n^*\}$. Thus, proved. ∎

[2021 Fall.
6. Suppose that $(x_n)_{n=1}^{\infty}$ is a Cauchy sequence in a Metric space X. Show that: if there exists a subsequence of $(x_n)_{n=1}^{\infty}$ converging to $x \in X$, then $(x_n)_{n=1}^{\infty}$ converges to $x \in X$.

Proof. Let $\varepsilon > 0$ be given. Since (x_n) is Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon/2$$

And, by the hypothesis, there is a subsequence $(x_{n_k})_{k=1}^{\infty}$ and for some $M \in \mathbb{N}$, $k \geq M \Rightarrow d(x_{n_k}, x) < \varepsilon/2$

Now, $r \geq \max\{N, M\}$,

$$d(x_r, x) \leq d(x_r, x_{n_r}) + d(x_{n_r}, x) < \varepsilon$$

(By definition of subsequence, $n_r \geq r$.) ∎

[2021 Fall.
7. Solve the following initial value problem for y

$$\begin{cases} y' = (1+y^2)\cdot x \\ y(0) = 0 \end{cases}$$

and determine the interval of definition of the solution.

Proof. Since $\frac{1}{1+y^2}\,dy = x\,dx$, we obtain

$$\frac{1}{2}x^2 + C = \tan^{-1} y, \quad \text{so} \quad y = \tan\left(\frac{1}{2}x^2 + C\right).$$

And, $y(0) = \tan C = 0$, so $C = n\pi$ $(n \in \mathbb{Z})$.

Moreover, the solution is defined when

$$-\frac{\pi}{2} + n\pi \le \frac{1}{2}x^2 + n\pi \le \frac{\pi}{2} + n\pi$$

So that $x^2 < \pi$, $(-\sqrt{\pi}, \sqrt{\pi})$ is the interval we desired. □

2021. Fall.

8. Let $w \in \mathbb{R}$. Prove that

$$\int_{-\infty}^{\infty} \cos(wt) \cdot e^{-t^2} dt = \sqrt{\pi} \cdot e^{-\frac{w^2}{4}}.$$

Proof. Since $\cos(\cdot)$ and e^{-t^2} are even functions, so

$$\int_{-\infty}^{\infty} \cos(wt) \cdot e^{-t^2} dt = 2 \cdot \int_{0}^{\infty} \cos(wt) \cdot e^{-t^2} dt$$

Meanwhile, for $R > 0$,

$$\int_0^R \cos(wt) e^{-t^2} dt = \frac{1}{w} \cdot \sin(wt) e^{-t^2} \Big|_0^R - \int_0^R \frac{1}{w} \cdot$$

If $w = 0$, then the integral $\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$, so we are done. Since $\cos(wt) = \cos(-wt)$, assume that $w > 0$. Let $f(z) = \cos(wz) \cdot e^{-z^2}$.

Consider the half-circle curve, composition of

$C_R : t \mapsto t \quad (-R \le t \le R)$ and $C_R : t \mapsto Re^{it} \quad (0 \le t \le \pi)$.

By the Cauchy's theorem, the entire f satisfies

$\int_{C_R * C_R} f = 0$, meanwhile,

[2021. Fall.
9. Suppose that f is entire and $\operatorname{Re} f(x+iy) \geq x$ for all $x, y \in \mathbb{R}$.
Show that f is a polynomial of degree one.
Proof. Notice that the given condition can be rewritten as

for all $z \in \mathbb{C}$, $\operatorname{Re} f(z) \geq \operatorname{Re} z$.

Define $g(z) = \exp(z - f(z))$. Then,

$$|g(z)| = |\exp(-\operatorname{Re} f(z) - i \operatorname{Im} f(z) + (\operatorname{Re} z + i \cdot \operatorname{Im} z))|$$
$$= \exp(-\operatorname{Re} f(z) + \operatorname{Re} z) \leq \exp(0) = 1$$

Since g is entire, being composite of entire functions,
so, the Lioville's theorem gives that g is a constant.
Therefore, $g(z) = \exp(z - f(z)) = C$ for some $C \in \mathbb{C} \setminus \{0\}$, so that

$$0 = g'(z) = (1 - f'(z)) \cdot \exp(z - f(z)) \Rightarrow f'(z) = 1$$

Consequently, $f(z) = z + a$ for some $a \in \mathbb{C}$, so proved. ∎

[2021. Fall.

10. (10 pts) We define the **Cantor set** $\mathcal{C}$ as follows:

$$C_0 := [0,1], \qquad C_n := \frac{C_{n-1}}{3} \bigcup \left(\frac{2}{3} + \frac{C_{n-1}}{3} \right), \quad n \geq 1, \qquad \mathcal{C} := \bigcap_{n=1}^{\infty} C_n.$$

Here, for a set $A \subset [0,1]$, $\frac{A}{3} := \{a/3 : a \in A\}$ and $b + A := \{b + a : a \in A\}$ for $b \in \mathbb{R}$.

Show that any $x \in [0,1]$ can be written as the difference of two elements from $\mathcal{C}$. **Hint.** First, you may show that the statement is true, when $\mathcal{C}$ is replaced with C_n with any $n \geq 1$.

Proof.

2021 Fall.

12. (10 pts) Define an equivalence relation $\sim$ on $X := \mathbb{C}^2 \setminus \{(0,0)\}$ by

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \exists t \in \mathbb{C} \setminus \{0\} \text{ such that } (x_1, y_1) = (tx_2, ty_2).$$

(a) Show that $X/\sim$ is homeomorphic to the 2-sphere S^2.

(b) Show that the 2-sphere S^2 cannot be a union of countably many circles embedded in S^2.

Proof. a). Define a map

$$f: X \to \mathbb{C} \cup \{\infty\} : (x,y) \mapsto \begin{cases} \infty & \text{if } y = 0 \\ \frac{x}{y} & \text{if } y \neq 0, \end{cases}$$

where $\mathbb{C} \cup \{\infty\}$ is the one-point compactification of $\mathbb{C}$.
Then, f is continuous, because: given an open set $U \subseteq \mathbb{C} \cup \{\infty\}$,
if $\infty \notin U$, $f^{-1}[U] = f|_{X \setminus (\mathbb{C} \times \{0\})}^{-1}[U] \subseteq \mathbb{C}$ is open since $(x,y) \mapsto x/y$ conti.,
if $\infty \in U$, $f^{-1}[\mathbb{C} \cup \{\infty\} \setminus U] = f^{-1}[\mathbb{C} \setminus U]$ is closed set,
so f is continuous. And, fix $y \in \mathbb{C} \setminus \{0\}$ and let $x = t \cdot y$ where $t \in \mathbb{C}$,
we obtain f is onto. Moreover, notice that

$$f(x_1, y_1) = f(x_2, y_2) \Leftrightarrow \frac{x_1}{y_1} = \frac{x_2}{y_2} \text{ or } y_1 = y_2 = 0 \Leftrightarrow (x_1, y_1) \sim (x_2, y_2).$$

So f induces a bijective continuous map

$$\tilde{f}: X/\sim \to \mathbb{C} \cup \{\infty\} : [(x,y)] \mapsto f(x,y).$$

satisfying $f = \tilde{f} \circ \pi$ where $\pi: X \to X/\sim : z \mapsto [z]_\sim$.
Meanwhile, $i: \mathbb{C} \cup \{\infty\} \to X : z \mapsto \begin{cases} (1,0) & \text{if } z = \infty \\ (z,1) & \text{if } z \neq \infty \end{cases}$
then $\pi \circ i$ is continuous onto map from compact set onto $X/\sim$, so
$X/\sim$ is compact. Finally, $\tilde{f}$ is closed map since $X/\sim$ compact, $\mathbb{C} \cup \{\infty\}$ T_2,
Consequently $\tilde{f}$ is homeomorphism, so that $X/\sim \cong \mathbb{C} \cup \{\infty\} \cong \mathbb{R}^2 \cup \{\infty\} \cong S^2$ ☑

Note:

$$\begin{array}{ccc} f: X & \longrightarrow & S^2 \\ \downarrow \circlearrowleft & & \uparrow \\ \mathbb{C} \cup \{\infty\} & \xrightarrow[\cong]{} & \mathbb{R}^2 \cup \{\infty\} \end{array} \quad : (x,y) \mapsto \left(\frac{2x'}{1+x'^2+y'^2}, \frac{2y'}{1+x'^2+y'^2}, \frac{x'^2+y'^2-1}{1+x'^2+y'^2} \right)$$

$$\downarrow \quad \circlearrowleft \quad \uparrow$$

$$\frac{x}{y} \mapsto \left(\mathrm{Re}\left(\frac{x}{y}\right), \mathrm{Im}\left(\frac{x}{y}\right) \right)$$

$$\| \qquad \|$$

$$x' \qquad y'$$

2021. Fall.

15. (10 pts) For a given smooth surface S, let us denote the Gaussian curvature at p by $K(p)$ and the mean curvature at p by $H(p)$.

(a) Compute the Gaussian curvature $K(p)$ of the surface $z = y^2 - x^2$ at $p = (0,0,0)$.

(b) Describe the surface S for which $K(p) = H(p)^2$ for all $p \in S$. (Justify your answer.)

Proof. Let $X: B_1(0) \to \mathbb{R}^3 : (u,v) \mapsto (u, v, v^2-u^2)$, the Monge patch.

observe $X_u = (1, 0, -2u)$, $X_v = (0, 1, 2v)$

$X_{uu} = (0,0,-2)$, $X_{uv} = (0,0,0)$, $X_{vv} = (0,0,2)$

and the normal $n = (2u, -2v, 1)/\sqrt{1+4u^2+4v^2}$

the coefficients of the first-fundamental form is

$E = 1+4u^2$, $F = -4uv$, $G = 1+4v^2$

and

$e = -2/\sqrt{1+4u^2+4v^2}$, $f = 0$, $g = 2/\sqrt{1+4u^2+4v^2}$

Now, the gaussian curvature is

$$K(p) = \left(-4/(1+4u^2+4v^2)\right) \Big/ (1+4u^2+4v^2)$$

$$= \frac{-4}{(1+4u^2+4v^2)^2},$$ so that at $p=(0,0,0)$, $K(p) = -4$.

b).

2022. Spring

1. Let G be a group of order 2021.

a). Prove that G is not simple.

b) Prove that G is cyclic

Proof. Notice that $|2021| = 43 \cdot 47$.

a). By the Sylow theorem, the number n_{43} of Sylow 43-subgroup satisfies $n_{43} \equiv 1 \mod 43$ and $n_{43} | 47$, so $n_{43} = 1$.

Therefore, $Syl_{43}(G) = \{P\}$, and $gPg^{-1} \in Syl_{43}(P)$ for all $g \in G$ implies $gPg^{-1} = P$, so P is normal. Thus G is not simple.

b). Similarly, $n_{47} = 1$, so there is a unique Sylow 47-subgroup Q.

By the Lagrange's theorem, all non-zero elements in P and Q have order of 43 and 47, respectively, so $P \cap Q = \{e\}$.

Since both 43 and 47 are prime, P and Q are cyclic, denote $P = \langle x \rangle$ and $Q = \langle y \rangle$ since $[x,y] = x^{-1}y^{-1}xy \in P \cap Q$, because P and Q both are normal, so $x^{-1}y^{-1}x \in Q$ and $y^{-1}xy \in P$, so $[x,y] = e$. Thus x and y commute. Now, xy is order of $43 \cdot 47$, so $G = \langle xy \rangle$ is cyclic.

2022 spring

2. Prove or disprove that each of following is PID:

a) $\mathbb{R}[x]$

b) $\mathbb{R}[x,y]$

c) $\mathbb{Z}[\sqrt{-1}]$

d) $\mathbb{Z}[x]$

Proof.

a) $\mathbb{R}[x]$ is PID, because since $\mathbb{R}$ is a field so $\mathbb{R}[x]$ is Euclidean domain, so for any ideal $I \subseteq \mathbb{R}[x]$, we can find a generator of I

b) $\mathbb{R}[x,y]$ is not P.I.D, because: let $(x,y) \subset \mathbb{R}[x,y]$.
If $(x,y) = (P(x,y))$ for some $P \in \mathbb{R}[x,y]$,
then $P(0,0) = 0$, that is, P has zero constant term.
But, $x, y \in (x,y) = (P(x,y))$ implies $\deg_x P, \deg_y P \leq 1$,

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3. Find the Galois group of the polynomial $x^5+3x^3-2x^2-6$ over $\mathbb{Q}$.

Proof. Notice that

$$x^5+3x^3-2x^2-6 = x^2\cdot(x^3-2)+3(x^3-2) = (x^3-2)\cdot(x^2+3).$$

Since the irreducible factor x^3-2 has three roots $\sqrt[3]{2}$, $\sqrt[3]{2}\cdot e^{\frac{2\pi i}{3}}$, $\sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}}$ and the irreducible factor x^2+3 has two roots $\pm\sqrt{-3}$,

The splitting field is

$$K = \mathbb{Q}\left(\sqrt[3]{2}, \sqrt[3]{2}\cdot e^{\frac{2\pi i}{3}}, \sqrt[3]{2}\cdot e^{\frac{4\pi i}{3}}, \sqrt{-3}, -\sqrt{-3}\right) = \mathbb{Q}\left(\sqrt[3]{2}, e^{\frac{2\pi i}{3}}, \sqrt{-3}\right)$$

Moreover, since $\mathbb{Q}$ is perfect, so $K/\mathbb{Q}$ is Galois.

Meanwhile, notice that

$$e^{\frac{2\pi i}{3}} = -\frac{1}{2}+\frac{\sqrt{-3}}{2}$$

Therefore, $K = \mathbb{Q}(\sqrt[3]{2}, e^{\frac{2\pi i}{3}})$. Finally,

$$\left[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}\right] = 3$$

and since $w = e^{\frac{2\pi i}{3}}$ satisfies w^2+w+1, so

$$\left[\mathbb{Q}(\sqrt[3]{2}, w):\mathbb{Q}(\sqrt[3]{2})\right] \le 2$$

But since $w \in \mathbb{C}\setminus\mathbb{R}$ and $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, so the degree is actually 2.

Since $Gal(K/\mathbb{Q}) \subseteq Sym(\{\sqrt[3]{2}, \sqrt[3]{2}w, \sqrt[3]{2}w^2\})$, the tower lemma gives $\left[\mathbb{Q}(\sqrt[3]{2}, w):\mathbb{Q}\right] = 6$, so $|Gal(K/\mathbb{Q})| = 6$, that is,

$$Gal(K/\mathbb{Q}) \cong S_3.$$

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6. Let $A, B \subseteq \mathbb{R}^k$. Define $A+B=\{a+b \mid a\in A,\ b\in B\}$. Show that:

a). If A is compact and B is closed, then $A+B$ is closed.

b). Even if A and B are closed, $A+B$ need not be closed.

Proof. a). Let $\{z_n\}$ be a sequence in $A+B$. By definition of $A+B$, for each $n\in\mathbb{N}$, $z_n = a_n + b_n$ for some $a_n\in A$ and $b_n \in B$.

Suppose that $\{z_n\}$ is convergent sequence which converges to z.

Since A is compact, $\{a_n\}\subseteq A$ has a convergent subsequence, denote $\{a_{n_k}\}$. Since convergent sequence is bounded,

$$|b_{n_k}| \le |a_{n_k}| + |a_{n_k}+b_{n_k}| = |a_{n_k}| + |z_{n_k}|$$

Thus $\{b_{n_k}\}$ is bounded, so it has a convergent subsequence, denote $\{b_{n_{k_s}}\}$. Since A is compact and B is closed, the limit of $\{a_{n_{k_s}}\}$ and $\{b_{n_{k_s}}\}$ are contained in A and B, respectively, and the limit of $\{a_{n_{k_s}}+b_{n_{k_s}}\}$ is same to the origin $\{z_n\}$, thus $\lim_{n\to\infty} z_n \in A+B$

b). In $\mathbb{R}$, let $A=\mathbb{N}$ and $B=\{-n+\frac{1}{n} \mid n\in\mathbb{N}\}$.

Then, both A and B are closed, because have no limit point,

Since $\{\frac{1}{n} \mid n\in\mathbb{N}\}\subseteq A+B$, but the limit point of $\{\frac{1}{n} \mid n\in\mathbb{N}\}$, $0\notin A+B$, thus $A+B$ is not closed. ☐

Note: Every sequence in Compact Metric has Convergent subsequence, in other saying, in the Metric space,

Compact $\iff$ Sequentially Compact

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7. Consider $y'' - 3y' - 4y = e^{-t}$, $t \in \mathbb{R}$.

a) Find a homogeneous solution y_h.

b) Find a particular solution and the general solution.

Proof. a). Since the characteristic is

$$r^2 - 3r - 4 = (r-4)(r+1),$$

So $y_h = e^{4t}$ and $y_h = e^{-t}$ are independent solutions.

b). Calculate the Wronskian:

$$W(e^{4t}, e^{-t}) = \det \begin{pmatrix} e^{4t} & e^{-t} \\ 4e^{4t} & -e^{-t} \end{pmatrix} = -e^{3t} - 4e^{3t} = -5e^{3t}.$$

Now, the particular solution is

$$y_p = \int_0^x \begin{vmatrix} e^{4t} & e^{4x} \\ e^{-t} & e^{-x} \end{vmatrix} \cdot b(t) \cdot /(-5e^{3t})\, dt$$

$$= \int_0^x -\frac{1}{5} \cdot e^{-4t} \cdot \left(e^{4t-x} - e^{4x-t}\right) dt$$

$$= -\frac{1}{5} \int_0^x e^{-x} - e^{4x-5t}\, dt$$

$$= -\frac{1}{5} \cdot \left(xe^{-x} - e^{4x} \cdot \left[\left(-\frac{1}{5} \cdot e^{-5x}\right) + \frac{1}{5}\right]\right)$$

$$= -\frac{1}{5} xe^{-x} - \frac{1}{25}\left(e^{-x} - e^{4x}\right)$$

Finally,

$$y = C_1 e^{4t} + C_2 e^{-t} + \left(-\frac{1}{5} te^{-t}\right) \quad (C_1, C_2 \in \mathbb{R})$$

is the general solution, we desired. □

Note: In this case, let $y_p = Ate^{-t}$.

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8. Define $f:\mathbb{R}\to\mathbb{R}; x\mapsto \sum_{n=1}^{\infty}\frac{1}{x^2+n^2}$

Then, f is differentiable on $\mathbb{R}$.

Proof. For each $m\in\mathbb{N}$, define

$$f_m(x)=\sum_{n=1}^{m}\frac{1}{x^2+n^2}.$$

At first, $f_m(0)\to f(0)$ as $m\to\infty$, because that

$$\sum_{n=2}^{m}\frac{1}{n^2}<\sum_{n=2}^{m}\frac{1}{n^2-n}=\sum_{n=2}^{m}\frac{1}{n-1}-\frac{1}{n}=1-\frac{1}{m}\to 1 \text{ as } m\to\infty,$$

so the increasing bounded sequence $f_m(0)$ converges. Now,

$$f_m'(x)=\sum_{n=1}^{m}-\frac{2x}{(x^2+n^2)^2}$$

converges uniformly, because

if $|x|\le 1$, then

$$\left|\frac{-2x}{(x^2+n^2)^2}\right|\le\frac{2}{n^4} \quad\text{and}\quad \sum\frac{1}{n^4}<\infty,$$

if $|x|>1$, then

$$\left|\frac{-2x}{(x^2+n^2)^2}\right|\le 2\cdot\left|\frac{x}{x^2+n^2}\cdot\frac{1}{x^2+n^2}\right|\le 2\cdot\frac{1}{n^2} \quad\text{and}\quad \sum\frac{1}{n^2}<\infty$$

therefore, the Weierstrass M-test gives f_m' converges uniformly.

Now, f_m converges at some point and f_m' converges uniformly on $\mathbb{R}$, the limit function f is differentiable. $\square$

~~First, for any $x\in\mathbb{R}$, $|x^2+n^2|\ge n^2$, so~~

~~$\sum_{n=1}^{\infty}\left|\frac{1}{x^2+n^2}\right|\le\sum_{n=1}^{\infty}\frac{1}{n^2}<\infty$, because $\sum_{n=2}^{m}\frac{1}{n^2}<\sum_{n=2}^{m}\frac{1}{n^2-n}=\sum_{n=2}^{m}\frac{1}{n-1}-\frac{1}{n}=1-\frac{1}{m}\to 1$,~~

~~as $m\to\infty$. Therefore, given $\varepsilon>0$, for some $N\in\mathbb{N}$, $m>n\ge N$ implies~~

~~$\sup_{x\in\mathbb{R}}\left|\sum_{k=n}^{m}\frac{1}{x^2+k^2}\right|\le\sup_{x\in\mathbb{R}}\sum_{k=n}^{m}\left|\frac{1}{x^2+k^2}\right|<\sum_{k=n}^{m}\frac{1}{k^2}<\varepsilon$.~~

~~Now, define for each $m\in\mathbb{N}$,~~

~~$$f_m(x)=\sum_{n=1}^{m}\frac{1}{x^2+n^2}.$$~~

~~We already shown $f_m\to f$ uniformly. And,~~

~~$$f_m'(x)=\sum_{n=1}^{m}-\frac{2x}{(x^2+n^2)^2}$$~~

~~Converges uniformly, because~~

2022. spring.

11. Prove or disprove that:

a). A compact subset of any topological space is closed.

b). A connected metric space having more then one point is uncountable.

Proof.

a). Let $\mathbb{Z}_3 = \{0,1,2\}$ with indiscrete topology $\mathcal{T} = \{\mathbb{Z}_3, \emptyset\}$.
Then, a singletone $\{1\}$ is not closed, but it is compact set, because every open cover of $\mathbb{Z}_3$ is finite. So, false.

b) Let $a, b \in X$ be distinct elements of a connected metric space X
Given $\lambda \in (0,1)$, if there is no point $x \in X$ such that $d(a,x) = \lambda \cdot d(a,b)$,
then $U = \{x \in X \mid d(a,x) < \lambda \cdot d(a,b)\}$ and $V = \{x \in X \mid d(a,x) > \lambda \cdot d(a,b)\}$
forms a separation of X, so it contradicts to the connectedness.
($U \cap V = \emptyset$ and U, V are open is clear, and
$U \cup V = X$ is obtained by the assumption).
So, given $\lambda \in (0,1)$, there is a point $x_\lambda \in X$ such that $d(a,x_\lambda) = \lambda \cdot d(a,b)$.
Thus, if $\lambda, \lambda' \in (0,1)$ are different, then

$$d(a,x_\lambda) \neq d(a,x_{\lambda'})$$

So the map $(0,1) \to X$; $\lambda \mapsto x_\lambda$ is well-defined injective map,
thus X is uncountable. ☐

[2022. Fall.

1. Let G be a group and H and K are normal subgroup of G.
If $H \cap K = \{e\}$, then $HK \cong H \times K$.
Proof. Define a map

$$\varphi : H \times K \to HK : (h, k) \mapsto hk.$$

It is well-defined by the definition of HK, and it is homomorphism;
given $h_1, h_2 \in H$ and $k_1, k_2 \in K$,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 k_1 k_2$$

meanwhile, given $h \in H$ and $k \in K$, $hkh^{-1}k^{-1} \in K$ and $\in H$, because
$hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$, by the normality. But, $H \cap K = \{e\}$ implies
$hkh^{-1}k^{-1} = e$, so $hk = kh$, these commutee so,

$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1))\,\varphi((h_2, k_2))$, so φ preserves the operation.
And, $\varphi[H \times K] = HK$ by just definition, φ is onto. Now, the first
isomorphism theorem gives that $H \times K / \ker \varphi \cong HK$.
And, $\ker \varphi = \{(h, k) \in H \times K \mid hk = e\} = \{e\}$,
so $H \times K \cong H \times K / \{e\}$, it proves the assertion. ☒

2022. Fall.

2. (10 pts) Let R be a commutative ring with unity, and let I be an ideal of R. Let us consider the ideal $\langle I \rangle$ of $R[x]$ generated by I. Show that if I is a prime ideal of R, then $\langle I \rangle$ is a prime ideal of $R[x]$. Does the maximality of I imply the maximality of $\langle I \rangle$? (Hint: consider a canonical map $R[x] \to (R/I)[x]$)

Proof. Notice that:

$$\pi : R[x] \to R[x]/\langle I \rangle : f(x) \mapsto f(x) + \langle I \rangle$$

is a natural projection, which is Homomorphism with $\ker \pi = \langle I \rangle$.

And $$\varphi : R[x]/\langle I \rangle \to (R/I)[x] : \sum a_n x^n + \langle I \rangle \mapsto \sum (a_n + (I)) \cdot x^n$$

is an isomorphism.

Since I is prime, so R/I is an integral domain.
The polynomial ring over an integral domain is also integral domain,
so that $(R/I)[x] \cong R[x]/\langle I \rangle$ is an integral domain, thus $\langle I \rangle$ is prime.

But, the maximality is not hereditary: let $3\mathbb{Z} \subset \mathbb{Z}$ be an ideal.
Since $\mathbb{Z}/3\mathbb{Z} \cong \mathbb{F}_3$ is a finite field of order 3, $3\mathbb{Z}$ is maximal
but since $\mathbb{F}_3[x]$ is <u>not</u> a field, so

$$\mathbb{F}_3[x] \cong (\mathbb{Z}/3\mathbb{Z})[x] \cong \mathbb{Z}[x]/\langle 3\mathbb{Z} \rangle$$

implies that $\langle 3\mathbb{Z} \rangle$ is not a maximal ideal. ▱

2022. Fall.

3. Let $\alpha=\sqrt{2+\sqrt{2}}$. Show that $\mathbb{Q}(\alpha)/\mathbb{Q}$ is Galois and describe the Galois group $Gal(\mathbb{Q}(\alpha)/\mathbb{Q})$.

Proof. Since $\alpha^4-4\alpha^2+4=2$, so x^4-4x^2+2 has α as a root, and it is irreducible, by the eisenstein criterion. Meanwhile,

$$x^4-4x^2+2=(x^2-(2+\sqrt{2}))\cdot(x^2-(2-\sqrt{2}))$$
$$=(x-\sqrt{2+\sqrt{2}})(x+\sqrt{2+\sqrt{2}})\cdot(x-\sqrt{2-\sqrt{2}})(x+\sqrt{2-\sqrt{2}})$$

and $\sqrt{2}=\alpha^2-2\in\mathbb{Q}(\alpha)$, $\sqrt{2-\sqrt{2}}=\sqrt{2}/\sqrt{2+\sqrt{2}}\in\mathbb{Q}(\alpha)$

So that $\mathbb{Q}(\alpha)$ is the splitting field of separable polynomial, thus $\mathbb{Q}(\alpha)$ is Galois over $\mathbb{Q}$.

Since $[\mathbb{Q}(\alpha):\mathbb{Q}]=\deg m_\alpha(x)=4$, so $|Gal(\mathbb{Q}(\alpha)/\mathbb{Q})|=4$.

Put $\beta=\sqrt{2-\sqrt{2}}$. Consider an automorphism $\sigma\in Gal(\mathbb{Q}(\alpha)/\mathbb{Q})$ as

$$\sigma:\alpha\mapsto\beta.$$

Then, $\sigma^2(\alpha)=\sigma(\beta)$ and since

$$\alpha\beta=\sqrt{2}=\alpha^2-2 \Rightarrow \beta=\alpha-\frac{2}{\alpha}=\frac{\alpha^2-2}{\alpha}$$

thus $\sigma(\beta)=\sigma(\alpha)-2\cdot\frac{1}{\sigma(\alpha)}=\beta-\frac{2}{\beta}=-\alpha$

So σ has order of 4, thus

$$Gal(\mathbb{Q}(\alpha)/\mathbb{Q})=\langle\sigma\rangle\cong C_4$$

$\square$

2022 Fall.

5. (10 pts) Let $f \in \mathbb{C}[a,b,c,d]$ be a polynomial satisfying $f(a,b,c,d) = f(a',b',c',d')$ whenever the two matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ are similar. Prove that there is a polynomial $g \in \mathbb{C}[x,y]$ such that

$$f(a,b,c,d) = g(a+d, ad-bc) \quad \text{whenever} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ is diagonalizable.}$$

Proof. Since the given matrix is diagonalizable,

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \sim \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$ where $\alpha+\beta = a+d = \operatorname{tr} A$, $\alpha\beta = ad-bc = \det A$.

So, $f(a,b,c,d) = f(\alpha,0,0,\beta)$. Meanwhile,

$\alpha =$

2022 Fall.

6. (10 pts) Let $C = \{z \in \mathbb{C} : |z - i| = 2\}$ denote a positively oriented (counter-clockwise) circle in $\mathbb{C}$. Evaluate the following integral and prove your answer:

$$\int_C \frac{1}{z^4+4} dz.$$

Proof. Notice that:

$$z^4+4 = (z^2+2i)(z^2-2i) = (z-(1-i))\cdot(z+(1-i))(z-(1+i))(z+(1+i))$$

And $z = 1+i$ and $z = -1+i$ are pole inside C. The Residue are:

$$\text{Res}\left(\frac{1}{z^4+4}, 1+i\right) = \left(\frac{d}{dz}(z^4+4)\Big|_{z=1+i}\right)^{-1} = \frac{1}{4(1+i)^3} = \frac{1}{8i(1+i)}$$

$$\text{Res}\left(\frac{1}{z^4+4}, -1+i\right) = \frac{1}{4(-1+i)^3} = \frac{1}{-8i(-1+i)} = \frac{1}{8}\cdot\frac{1}{(1+i)}$$

Now, by the Residue theorem,

$$\int_C \frac{1}{z^4+4} dz = 2\pi i\cdot\left(\frac{1}{1+i}\right)\cdot\left(\frac{1}{8}\left(\frac{1}{i}+1\right)\right)$$

$$= 2\pi i\cdot\left(\frac{1-i}{2}\right)\cdot\left(\frac{1}{8}\cdot(1-i)\right)$$

$$= \frac{\pi}{8} i\cdot(-2i) = \frac{\pi}{4}$$

$\square$

[2022 Fall.

7. (10 pts) Let X be a metric space and let $x \in X$. Suppose that (x_n) is a sequence in X such that every subsequence of (x_n) has a further subsequence converging to x. Prove that (x_n) converges to x.

Proof. Suppose (x_n) does not converges to x.
Then, for some $\varepsilon > 0$, for any $N \in \mathbb{N}$, there is $k \geq N$ such that

$$d(x_k, x) \geq \varepsilon.$$

For $N=1$, take x_{n_0} such that $n_0 > N=1$, and $d(x_{n_0}, x) \geq \varepsilon$.
And, for $N=n_0$, take x_{n_1} such that $n_1 > n_0$ and $d(x_{n_1}, x) \geq \varepsilon$.
Inductively, we can construct a subsequence (x_{n_k}) such that for any $k \in \mathbb{N}$, $d(x_{n_k}, x) \geq \varepsilon$.

By the construction, (x_{n_k}) has no further subsequence converging to x, it is contradiction to the hypothesis. ∎

[2022. Fall.

8. For continuous functions $g:[0,1]\to\mathbb{R}$ and $h:[-1,1]\to\mathbb{R}$, define

$$f:[0,1]\to\mathbb{R};\ x\mapsto \int_0^1 h(x-y)g(y)\,dy$$

For each $n\in\mathbb{N}$, define

$$f_n:[0,1]\to\mathbb{R};\ x\mapsto \frac{1}{n}\cdot\sum_{i=0}^{n-1} h\left(x-\frac{i}{n}\right)\cdot g\left(\frac{i}{n}\right)$$

Prove that $f_n\to f$ uniformly on $[0,1]$

Proof. Define $H:[0,1]\times[0,1]:(x,y)\mapsto h(x-y)\cdot g(y)$.

Since h and g both are continuous, therefore H is continuous and Since $[0,1]^2$ is compact set, so H is continuous uniformly.

Given $\varepsilon>0$, there exists $\delta>0$ such that for any $(x_1,y_1),(x_2,y_2)\in[0,1]^2$,

$$\|(x_1,y_1)-(x_2,y_2)\|<\delta \Rightarrow \|H(x_1,y_1)-H(x_2,y_2)\|<\varepsilon.$$

Now, given $x\in[0,1]$, and $n>\frac{1}{\delta}$, ($n>\frac{1}{\delta}$ gives $\frac{1}{n}<\delta$)

$$\begin{aligned}
|f(x)-f_n(x)| &= \left|\int_0^1 H(x,y)\,dy - f_n(x)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \frac{1}{n}\cdot\sum_{i=0}^{n-1} H\left(x,\frac{i}{n}\right)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H\left(x,\frac{i}{n}\right)dy\right| \\
&\le \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}\left|H(x,y)-H\left(x,\frac{i}{n}\right)\right|dy \\
&< \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}}\varepsilon\,dy = \varepsilon.
\end{aligned}$$

Therefore, $n>\frac{1}{\delta}\Rightarrow \sup_{x\in[0,1]}|f_n(x)-f(x)|<\varepsilon$, So, proved. □

2022 Fall.

9. (10 pts) Find all entire functions $f : \mathbb{C} \to \mathbb{C}$ satisfying

$$\lim_{R\to\infty} \frac{1}{R^3} \iint_{D_R} |f(x+iy)|\,dxdy = 0,$$

where $D_R = \{z \in \mathbb{C} : |z| < R\}$ denotes the open disk of radius $R > 0$.

Proof. Observe that:

$$\iint_{D_R} |f(x+iy)|\,dxdy = \int_0^{2\pi}\int_0^R |f(re^{i\theta})|\cdot r\,dr\,d\theta.$$

Since f is entire,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \quad \text{where} \quad a_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{2\pi i}\cdot\int_{|z|=r} \frac{f(z)}{z^{n+1}}\,dz \quad (0<r<R)$$

$$= \frac{1}{2\pi r^n}\cdot\int_0^{2\pi} f(re^{i\theta})\cdot e^{-in\theta}\,d\theta$$

Therefore, $2\pi r^n\cdot|a_n| \le \int_0^{2\pi} |f(re^{i\theta})|\,d\theta$. So

$$\int_0^R 2\pi r^{n+1}\cdot|a_n|\,dr \le \int_0^R\int_0^{2\pi} r\cdot|f(re^{i\theta})|\,d\theta\,dr, \text{ dividing } R^3,$$

$$\frac{2\pi|a_n|}{R^3}\cdot\frac{1}{n+2}\cdot R^{n+2} \le \frac{1}{R^3}\cdot\int_0^R\int_0^{2\pi} r\cdot|f(re^{i\theta})|\,d\theta\,dr \to 0 \text{ as } R\to\infty$$

So that $a_n = 0$ for all $n \ge 1$, and, every constant $f = C \in \mathbb{C}$ satisfies

$$\frac{1}{R^3}\cdot\iint_{D_R} C\,dx\,dy = \frac{1}{R^3}\cdot C\cdot\pi R^2 \to 0 \text{ as } R\to\infty,$$

Consequently, $f \equiv C$ constant map. ∎

2022 Fall.

11. (10 pts) Consider the set $\mathbb{R}$ of real numbers with the co-countable topology $\mathcal{T}$ given by

$$\mathcal{T} = \{\emptyset\} \cup \{S \subset \mathbb{R} : \mathbb{R} \setminus S \text{ is countable}\}.$$

(a) Show that $\mathcal{T}$ defines a topology.

(b) Determine if the interval $[0, 1]$ is compact with the co-countable topology. (Explain your answer.)

Proof. a).

Since $\mathbb{R}\setminus\mathbb{R} = \emptyset$ is countable, $\emptyset, \mathbb{R} \in \mathcal{T}$.

Let U_α $(\alpha \in I)$ be sets in $\mathcal{T}$. Then, (if one of $U_\alpha = \emptyset$, the trivial, let $U_\alpha \neq \emptyset$)

$$\mathbb{R}\setminus\left(\bigcup_{\alpha\in I} U_\alpha\right) = \mathbb{R} \cap \left(\bigcap_{\alpha\in I} U_\alpha^c\right) = \bigcap_{\alpha\in I} \mathbb{R}\setminus U_\alpha$$

is intersection of countable sets, so it is also countable. And,

for any finite sets $U_1, \cdots, U_n$ in $\mathcal{T}$,

$$\mathbb{R}\setminus\left(\bigcap_{k=1}^{n} U_k\right) = \mathbb{R} \cap \left(\bigcup_{k=1}^{n} U_k^c\right) = \bigcup_{k=1}^{n} \mathbb{R}\setminus U_k.$$

is finite union of countable sets, so it is countable.

Now, the set $\mathcal{T}$ satisfies the Axiom of Topology, it is a Topology.

b). For each $n\in\mathbb{N}$, define $U_n = ([0,1]\setminus K)\cup\{\frac{1}{n}\}$ where $K = \{\frac{1}{n} \mid n\in\mathbb{N}\}$.

Then, $[0,1]\setminus U_n = ([0,1]\cap K)\setminus\{\frac{1}{n}\}$, countable, so each U_n is open in $[0,1]$.

Moreover, $\bigcup_{n=1}^{\infty} U_n = [0,1]$, but $\{U_n \mid n\in\mathbb{N}\}$ has no finite subcover,

because if there is a finite subcover

$$\{U_{n_1}, \cdots, U_{n_k}\} \text{ for } [0,1],$$

then for $m = \max\{n_1, \cdots, n_k\}$,

$$U_{n_1} \cup \cdots \cup U_{n_k} \subseteq \bigcup_{n=1}^{m} U_n \subsetneq [0,1],$$

so $[0,1]$ is <u>not</u> compact. ∎

2022. Fall.

12. (10 pts)

(a) Let X be a topological space and let $Z \subset X$ be a subspace with the induced topology. Suppose that Z is connected. Determine if its closure $\overline{Z}$ is connected or not. (Explain your answer.)

(b) Show that, for $n \geq 1$, the standard sphere $S^n \subset \mathbb{R}^{n+1}$ is connected.

Proof.

a). Suppose that the closure $\overline{Z}$ is disconnected, with separation $\{U, V\}$, such that $U, V \subset X$ are open sets such that $U \cap \overline{Z} \neq \emptyset$, $V \cap \overline{Z} \neq \emptyset$, $U \cap V \cap \overline{Z} = \emptyset$, $\overline{Z} \subseteq U \cup V$.

Want to show that: $\{U, V\}$ is also separation for Z

First, if $U \cap Z = \emptyset$, then $Z \in U^c$. But, since U^c is closed and $\overline{Z}$ is the smallest closed set containing Z, so $\overline{Z} \subseteq U^c$. But it equals to $\overline{Z} \cap U = \emptyset$, it contradicts to the assumption so $U \cap Z \neq \emptyset$, and $V \cap Z \neq \emptyset$,

Similarly, Now, since $Z \subseteq \overline{Z}$,

$$U \cap V \cap Z \subseteq U \cap V \cap \overline{Z} = \emptyset \Rightarrow U \cap V \cap Z = \emptyset$$

and $Z \subseteq \overline{Z} \subseteq U \cup V$, as a result, $\{U, V\}$ is the separation for Z, it is contradiction to the hypothesis. So $\overline{Z}$ is connected.

b). Since S^n is one-point compactification of $\mathbb{R}^n$, so, $S^n \setminus \{*\} \cong \mathbb{R}^n$.

Since $\mathbb{R}^n$ is connected, so $S^n = \overline{S^n \setminus \{*\}}$ is also connected by a).

12022. Fall.

13. Let S be the subset of $\mathbb{R}^3$ given by $\{(x,y,z) \mid z^2 = x^2+y^2,\ z \geq 0\}$.
Show that S is not regular surface.

proof. Suppose that S is a regular surface. Then, for $(0,0,0) \in S$, there exists a neighbor V containing $(0,0,0)$ such that $V \cap S$ is a graph of

$$z = f(x,y) \quad \text{or} \quad y = g(x,z) \quad \text{or} \quad z = h(y,z)$$

where f, g, h are differentiable. Since projection of S into xz, yz-plain are not 1-1, g, h cannot be such a map.
Meanwhile, $f(x,y) = \sqrt{x^2+y^2}$ at locally $(0,0,0)$, but f is not differentiable at $(x,y)=(0,0)$, so it cannot be possible. Therefore, S is not regular.

(More specifically, $(0, \pm\varepsilon, \sqrt{1-\varepsilon^2}) \in S$ for any $\varepsilon \neq 0$, so there is no 1-1 yz-projection. similarly xz.)

[2023 spring.

1). Let $A=\begin{bmatrix} 0 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{bmatrix}$

a). Find the minimal polynomial of A.
b). " Jordan form "
c). " rational form "

proof.

a). First, the characteristic polynomial $\chi(t)$

$\star$: tr = sum eigen.

$$\det\begin{bmatrix} t & 3 & -7 \\ 1 & t+1 & -5 \\ 1 & 2 & t-6 \end{bmatrix} = t\cdot[(t+1)(t-6)+10] - 3\cdot[t-6+5] - 7\cdot[2-1-t]$$
$$= t\cdot[t^2-5t+4] - 3\cdot[t-1] + 7[t-1]$$
$$= t\cdot[t-4]\cdot[t-1] + 4[t-1]$$
$$= (t-1)(t-2)^2.$$

Since the minimal polynomial have same roots to χ, so we need to check that $(A-I)(A-2I)=0$. (If not, then Cayley-Hamilton theorem guarantees that χ is the minimal polynomial.).

$$(A-I)\cdot(A-2I) = \begin{bmatrix} -1 & -3 & 7 \\ -1 & -2 & 5 \\ -1 & -2 & 5 \end{bmatrix}\cdot\begin{bmatrix} -2 & -3 & 7 \\ -1 & -3 & 5 \\ -1 & -2 & 4 \end{bmatrix} \neq 0$$

So, $\chi=(t-1)(t-2)^2$.

b). $\dim\ker(A-2)=1$, since $0<\dim\ker(A-2)\leq\deg(t-1)=1$, and $\dim\ker(A-2I)\leq 2=\deg(t-2)^2$. Meanwhile, $\begin{bmatrix} 2 & -3 & 7 \\ -1 & -3 & 5 \\ -1 & -2 & 4 \end{bmatrix}$ has more than one linearly independent row vectors, so the nulity is 1. So, the Jordan form is

$$J_A=\begin{bmatrix} 1 & & \\ & 2 & 1 \\ & & 2 \end{bmatrix}.$$

c). The minimal polynomial is equal to the characteristic, so A is cyclic, thus the Rational form is

$$R_A=\begin{bmatrix} 0 & 0 & 4 \\ 1 & 0 & -8 \\ 0 & 1 & 5 \end{bmatrix}.$$

2023. spring.

2. (10 pts) Let A be an $m \times n$ real matrix of rank $r \geq 1$. Prove the following.

(a) (10 pts) There are an $m \times r$ matrix B and an $r \times n$ matrix C such that $A = BC$.

(b) If $A = BC = B'C'$ for $m \times r$ matrices B, B' and $r \times n$ matrices C, C', then there is an invertible matrix Q such that $B' = BQ^{-1}$ and $C' = QC$.

Proof. a). Since rank $A = r \geq 1$, let $\{\alpha_1, \cdots, \alpha_r\}$ be a basis for $\mathrm{Im}A$. So, for each $1 \leq i \leq n$, there exist unique scalar C such that

$$Ae_i = C_{1i}\cdot\alpha_1 + \cdots + C_{ri}\cdot\alpha_r.$$

Now, let $B = (\alpha_1 \cdots \alpha_r) \in M_{m\times r}$ and $C = (C_{ij}) \in M_{r\times n}$, then

$$A = (Ae_1 \cdots Ae_n) = (\alpha_1 \cdots \alpha_r)\cdot\begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & \ddots & & \vdots \\ C_{r1} & \cdots & & C_{rn} \end{pmatrix} = BC$$

b). Since rank $A = r$, so rank BC = rank $B'C' = r$, so necessarily rank $B \geq r$. But, by the dimension theorem, r = rank B + nulity B, so rank $B = r$. That is, the column vectors of B are linearly independent. Now, for the given B and B',

$\beta = \{Be_1, \cdots, Be_r\}$ and $\beta' = \{B'e_1, \cdots, B'e_r\}$ are basis for $\mathrm{Im}A$,

So for $Q = [I_{\mathrm{Im}A}]^{\beta}_{\beta}$, $Q^{-1}e_i = [Be_i]_{\beta'}$

$$B'\cdot e_i = BQ^{-1}e_i$$

So that $B' = BQ^{-1}$. And, $BC = B'C' = BQ^{-1}C'$

$$\Rightarrow B\cdot(C - Q^{-1}\cdot C') = 0$$

and since nulity $B = 0 = r -$ rank r, so $C - Q^{-1}C' = 0$. So proved ∎

$$B' = BQ^{-1}$$

[2023. Spring
4. Let $V=\{id, (12)(34), (13)(24), (14)(23)\} \leq S_4$, 5점,
a). Show that V is normal subgroup.
b). $S_4/V \cong S_3$,

a). Since conjugate cannot change the order of elements,
and cannot change the parity; since the sign function is Homomorphism,

$sgn(\sigma\tau\sigma^{-1}) = sgn(\sigma)\cdot sgn(\sigma^{-1})\cdot sgn(\tau) = sgn(\tau)$, for any $\sigma, \tau \in S_n$.

Now, the fact that V contains all order two even permutations in S_4,
so every conjugate of V is contained in V, so normal.] 5점

b) Consider S_3 is a subgroup of S_4, the number 4 fixed. Then,

$$\varphi : S_3 \to S_4/V : \sigma \mapsto \sigma V$$

is well-defined, and given $\sigma_1, \sigma_2 \in S_3 \leq S_4$,

$$\varphi(\sigma_1\sigma_2) = \sigma_1\sigma_2 V = \sigma_1 V \sigma_2 V = \varphi(\sigma_1)\varphi(\sigma_2)$$

So it Homomorphism. And, since all elements in S_3 fixes 4, $S_3 \cap V = \{id\}$.
So $|\ker\varphi| = \{id\}$. Now, since $|S_3| = 6 = |\ker\varphi| \cdot |\mathrm{Im}\,\varphi|$, $|\mathrm{Im}\,\varphi| = 6$,
So $\mathrm{Im}\,\varphi = S_4/V$, since $|S_4/V| = 6$, so φ is onto.
Now, the first isomorphism theorem gives $S_3 \cong S_4/V$. ☒

2023 Spring

5. Let $I=(2, x^2+x+1) \subseteq \mathbb{Z}[x]$ be an ideal

Show that I is maximal ideal and give a multiplication table for $\mathbb{Z}[x]/I$.

Pf. Define a map

$$\pi : \mathbb{Z}[x] \to \mathbb{Z}/2\mathbb{Z}[x] ; \sum_{n=0}^{n} a_n x^n \mapsto \sum_{n=0}^{n} (a_n \bmod 2)\cdot x^n$$

Then, π is an onto Homomorphism, ($\pi(0)=0$, $\pi(1)=1$) with $\ker \pi = 2\mathbb{Z}[x] = (2)$, so $\mathbb{Z}[x]/\ker \pi = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$.

Meanwhile, x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, because it is of degree 2 and $0^2+0+1=1\neq 0$, $1^2+1+1=1\neq 0$ in $\mathbb{Z}/2\mathbb{Z}$.

Now, Consider the natural Homomorphism

$$\varphi : \mathbb{Z}/2\mathbb{Z}[x] \to \left(\mathbb{Z}/2\mathbb{Z}[x]\right)/(x^2+x+1) ; p(x) \mapsto p(x)+(x^2+x+1)$$

It is trivially onto, so $\varphi\circ\pi$ is onto Homomorphism.

Moreover,

$$\ker \varphi\circ\pi = (\varphi\circ\pi)^{-1}[\{0\}] = \pi^{-1}[\varphi^{-1}[\{0\}]] = \pi^{-1}[\ker \varphi] = \pi^{-1}[(x^2+x+1)]$$

Now,

$$f \in \pi^{-1}[(x^2+x+1)] \iff \pi(f) \in (x^2+x+1)$$
$$\iff \pi(f) = (x^2+x+1)\cdot\pi(g) \text{ for some } g(x) \in \mathbb{Z}[x]$$
$$\iff \pi(f-(x^2+x+1)g) = 0$$
$$\iff f-(x^2+x+1)g \in (2)$$
$$\iff f \in (2)+(x^2+x+1) = (2, x^2+x+1).$$

So, $\mathbb{Z}[x]/\ker(\varphi\circ\pi) = \mathbb{Z}[x]/(2, x^2+x+1) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$.

And, Since $\mathbb{Z}/2\mathbb{Z}$ is field and x^2+x+1 is irreducible over $\mathbb{Z}/2\mathbb{Z}$, so $(\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$ is a field, so, the quotient is field if and only if (x^2+x+1) is maximal, So proved.

Since $\mathbb{Z}[x]/I \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2+x+1)$, Let $\alpha = x+(x^2+x+1)$.

Then $\alpha^2 = x^2+(x^2+x+1) = x+1+(x^2+x+1) = \alpha+1$, So

	1	α
1	1	α
α	α	$\alpha+1$

is the table.

Generalized in the next page.

Let $I=(a,f)\subseteq R[x]$ be an ideal generated by $a, f\in R[x]$.

Consider the diagram:

$$R[x] \xrightarrow{\varphi} (R/aR)[x] \xrightarrow{\pi_{\overline{f}}} (R/aR)[x]/(\overline{f}).$$

$$R[x] \xrightarrow{\pi} R[x]/aR[x], \quad (R/aR)[x] \xrightarrow{\simeq} R[x]/aR[x]$$

$R[x]\to R[x]/aR[x]$: $P(x)\mapsto P(x)+aR[x]$ Onto Homomorphism.

$R[x]/aR[x]\to(R/aR)[x]$: $P(x)+aR[x]\mapsto \overline{P(x)}$, where $\overline{\sum a_n\cdot x^n}=\sum(a_n+aR)\cdot x^n$. Isomorphism.

$\Rightarrow \varphi: R[x]\to(R/aR)[x]$ is Onto Homomorphism, So is $\pi_{\overline{f}}\circ\varphi$ onto Homomorphism.

Moreover,

$$\ker(\pi_{\overline{f}}\circ\varphi)=(\pi_{\overline{f}}\circ\varphi)^{-1}[\{0\}]=\varphi^{-1}[\ker\pi_{\overline{f}}]$$

$$=\{P(x)\in R[x] \mid \overline{P(x)}\in\ker\pi_{\overline{f}}\}=\{P(x)\in R[x] \mid \overline{P(x)}\in(\overline{f(x)})\}$$

$$\overset{(*)}{=}\{P(x)\in aR[x]+f(x)R[x]\}=(a)+(f)=(a,f)=I.$$

Assume $a\in R$ and $(a)\subseteq R$ is maximal, and $\overline{f}\in(R/aR)[x]$ is irreducible.

Then, R/aR is a field, and therefore $(R/aR)[x]$ is P.I.D.

So, $(\overline{f})$ is maximal, it implies that $(R/aR)[x]/(\overline{f})$ is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x]+fR[x])\cong(R/aR)[x]/\overline{f}\cdot(R/aR)[x]$$

Note: $(*)$ is obtained by: given $P(x)\in R[x]$,

$$\overline{P}(x)\in(\overline{f(x)})\iff\overline{P(x)}=\overline{g(x)}\cdot\overline{f(x)} \text{ for some } g(x)\in R[x]$$

$$\iff P(x)-g(x)f(x)\in aR[x] \text{ for some } g(x)\in R[x]$$

Another Solution.

Observe that $\mathbb{Z}[x]/(2, x^2+x+1)$.

Given $f(x) \in \mathbb{Z}[x]$, since the leading coefficient of x^2+x+1 is a unity, so we can apply the division Algorithm.

$$f(x) = g(x)\cdot(x^2+x+1) + r(x) \text{ for some } g, r \in \mathbb{Z}[x] \text{ with } \deg r \le 1.$$

Therefore, $\mathbb{Z}[x]/(2, x^2+x+1) = \{ax+b + (2, x^2+x+1) \mid a, b \in \mathbb{Z}\}$

And, for any $a, b \in \mathbb{Z}$, $ax+b = (a \bmod 2)\cdot x + (b \bmod 2) + 2nx + 2m$

for some $n, m \in \mathbb{Z}$, so $= \{ax+b + (2, x^2+x+1) \mid a, b \in \{0, 1\}\}$.

And, $1 \notin (2, x^2+x+1)$ and $x \notin (2, x^2+x+1)$ and $x+1 \notin (2, x^2+x+1)$ implies these are distince with zero, and also these are mutually distince, just simple calculation. Let $\alpha = x+1$.

Now, $\mathbb{Z}[x]/I$ has four elements, $\{0, 1, \alpha, \alpha+1\}$ where $\alpha^2+\alpha+1 = 0$,

so the table

	0	1	α	$\alpha+1$
0	0	0	0	0
1	0	1	α	$\alpha+1$
α	0	α	$\alpha+1$	1
$\alpha+1$	0	$\alpha+1$	1	α

is obtained since every non-zero element has an inverse, it is a field.

Comment. Multiplication table 요구한 부분에서 이게 더 좋게 보이게 가까운듯?

[2023. Spring.
6. Suppose that $f(x)$ is a continuous function on $[0,1]$ satisfying

$$\int_0^1 f(x)\cdot\varphi(x)\,dx = 2-\int_0^1 \varphi(x)\,dx$$

for any continuous function $\varphi:[0,1]\to\mathbb{R}$. Find all possible f.

Proof. Let φ be any continuous map on $[0,1]$.

Then, $\int_0^1 \varphi(x)(f(x)-2)\,dx = 0$.

Since f is continuous, $f(x)-2$ is also continuous. Take $\varphi = f-2$.

Then, $\int_0^1 (f(x)-2)^2\,dx = 0$. Since $(f(x)-2)^2 \geq 0$, so f must be $f\equiv 2$. ∎

2023, Spring.

7. Let $f: \mathbb{C} \to \mathbb{C}$ be an entire satisfying 10m

$$u^2(z) - 2v^2(z) = 1 \quad \text{for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. Observe that: $f(z) = 0 \iff u(z) = v(z) = 0$

But, $u^2 - 2v^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f(z)} = \frac{1}{u^2(z) + v^2(z)} = \frac{1}{1 + 3v^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, So f is also constant function. ☑

Or, $|f^2| = u^2 + v^2 = 1 + 3v^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3v^2}} \leq 1$. ☑

observe that: ~~differentiating gives~~ that

$2uu_x - 2vv_x = 0 \Rightarrow$ ~~$uu_x = vv_x = -v \cdot u_y$~~

2023 spring.

8. (10 pts) Let C be a closed subset of $\mathbb{R}$ and let $F : C \to C$. Suppose that there exists $\lambda \in (0,1)$ such that

$$|F(x) - F(y)| \le \lambda |x - y| \text{ for all } x, y \in C. \qquad 2^{\text{a}}.$$

We say that $x_0 \in C$ is a fixed point of F if it satisfies $F(x_0) = x_0$.

(a) Let $(x_n)_{n=1}^{\infty}$ be a sequence in C such that $x_{n+1} = F(x_n)$ for all $n \in \mathbb{N}$. Prove that the sequence $(x_n)_{n=1}^{\infty}$ is convergent and the limit is a fixed point of F.

(b) Prove that F has a unique fixed point.

Proof. a). Given $n \in \mathbb{N}$,

$$|x_{n+1} - x_n| = |F(x_n) - F(x_{n-1})| \le \lambda \cdot |x_n - x_{n-1}| = \lambda \cdot |F(x_{n-1}) - F(x_{n-2})| \le \cdots$$
$$\le \lambda^n \cdot |x_1 - x_0|$$

Now, for any fixed $N \in \mathbb{N}$, $m > n \ge N$ implies that

$$|x_m - x_n| \le |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \cdots + |x_{n+1} - x_n|$$
$$\le (\lambda^{m-1} + \lambda^{m-2} + \cdots + \lambda^n) \cdot |x_1 - x_0|$$
$$\le \sum_{n=N}^{\infty} \lambda^n \cdot |x_1 - x_0| \to 0 \text{ as } N \to \infty$$

Therefore, $\{x_n\}$ is a Cauchy sequence in $\mathbb{R}$, so x_n converges, and since C is closed, $x_n \to x$ for some $x \in C$. Moreover, given $\varepsilon > 0$,

$|x - y| < \frac{\varepsilon}{\lambda} \Rightarrow |F(x) - F(y)| < \frac{\varepsilon}{\lambda} \cdot \varepsilon = \varepsilon$, so F is continuous, thus

$$F(x) = F(\lim_{n\to\infty} x_n) = \lim_{n\to\infty} F(x_n) = \lim_{n\to\infty} x_{n+1} = x$$

So, the limit x is the fixed point we desired. ∎ 9m.

b). Let $x, y \in C$ be fixed point of F. Then

$$|F(x) - F(y)| = |x - y| \le \lambda |x - y|$$

Since $\lambda < 1$, so $x = y$. Thus the fixed point is unique. ∎ 3m

[2023. Spring.]

9. (10 pts) Determine the solution of the initial value problem

$$y' = -4xy + e^{-2x^2}\cos x, \quad y(0) = 1.$$

For which values of x is the solution defined? Justify your answer.

Proof. Multiplying the integration factor

$$M(x) = \exp\int 4x\,dx = e^{2x^2}$$

Then

$$e^{2x^2}y' + 4x\cdot e^{2x^2}y = (e^{2x^2}\cdot y)' = \cos x$$

So that $y = e^{-2x^2}\cdot(\sin x + C)$

and using the initial condition,

$$y(0) = 1\cdot(0+C) = 1, \quad \text{so } C = 1.$$

Thus, the solution we desired is

$$y(x) = e^{-2x^2}(\sin x + 1)$$

and the solution is defined in $x \in \mathbb{R}$. ☑

2023 Spring

10. Let $f:[0,\infty)\to[0,\infty)$ be a continuous function satisfying

$$\int_0^\infty f(x)\,dx<\infty.$$

a1. Prove or disprove that $\lim_{x\to\infty} f(x)=0$.

b1. If f is continuous uniformly, then $\lim_{x\to\infty} f(x)=0$?, Prove or disprove.

Proof.

a). Define a map on $[0,\infty)$

$$f_n(x)=\begin{cases}0 & \text{otherwise}\\ n^2(x-n) & \text{if } x\in[n,n+\frac{1}{n^2}]\\ n^2(n+\frac{2}{n^2}-x) & \text{if } x\in[n+\frac{1}{n^2},n+\frac{2}{n^2}]\end{cases}$$

Then $\int_0^\infty f_n=\frac{1}{n}$ and f_n is continuous, for each $n\in\mathbb{N}$.

Let $f=f_1+f_{1+}+\cdots+f_{n^2+}+\cdots$. Then, f is continuous and

$$\int_0^\infty f=\sum_{n=1}^{\infty}\frac{1}{n^2}<\infty.$$

(The continuousity is obtained since investigate each $[n,n+\frac{2}{n^2}]$)

But, for $\epsilon=\frac{1}{2}$, for any $\delta>0$, there is $x>\delta$ such that $|f(x)-0|=1\geq\frac{1}{2}$.

So $\lim f(x)\neq 0$.

b). Suppose $\lim_{x\to\infty} f(x)=0$ is false but f is continuous uniformly.

Let $\epsilon=2$. Then, for any $n\in\mathbb{N}$, there is $x_n>n$ such that $f(x_n)\geq 2$.

Since f is uniformly continuous, take $1>\delta>0$ such that $|x-y|<\delta$ implies $|f(x)-f(y)|<1$. Then, for each $n\in\mathbb{N}$,

$$\int_{x_n}^{x_n+\delta} f(x)\,dx\geq\delta\cdot 1$$

So, $\int_0^\infty f\geq\sum_{n=1}^{N}\int_{x_n}^{x_n+\delta} f\geq\delta\cdot N$ diverges, So contradicts.

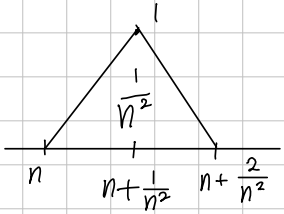
Thus $\lim f=0$. ∎

Revised in the next.

Let $f:[0,\infty)\to\mathbb{R}$ be non-negative valued continuous function,
does not converges to zero,
but the improper integral converges.

Find and convince concretely, and change the condition as f continuous uniformly.

Define $f_n(x)=\begin{cases} n^2(x-n) & \text{if } x\in[n,\, n+\frac{1}{n^2}] \\ n^2\left(n+\frac{2}{n^2}-x\right) & \text{if } x\in[n+\frac{1}{n^2},\, n+\frac{2}{n^2}] \\ 0 & \text{otherwise.}\end{cases}$



Then, f_n is continuous, with $\int_0^\infty f_n = \frac{1}{n^2}$.
Now, let $f=\sum_{n=2}^{\infty} f_n$. Since the interval $[n, n+\frac{2}{n^2}]$ are disjoint mutually, $n\geq 2$,
So the f is well-defined, but it does not uniformly converges, because

$$\sup_{x\geq 0}\left|f(x)-\sum_{n=2}^{N} f_n(x)\right| = 1 \quad \text{for any } N\in\mathbb{N}.$$

Nevertheless, $\int_0^\infty f = \int_0^\infty \sum_{n=2}^{\infty} f_n = \sum_{n=2}^{\infty}\int_0^\infty f_n = \sum_{n=2}^{\infty}\frac{1}{n^2}$, the integral converges.
But, for each $n\in\mathbb{N}$, $f(n+\frac{1}{n^2})=1$, So $f(x)$ does not tends to zero as $x\to\infty$.

Assume that $f(x)$ is continuous uniformly
If f does not converges to zero, then there is $\varepsilon>0$ such that
for each $R>0$, there is $x\geq R$ such that $f(x)\geq\varepsilon$.
Now, let $\delta>0$ such that $|x-y|<\delta \Rightarrow |f(x)-f(y)|<\frac{\varepsilon}{2}$.
And, let $\{x_n\}$ be a positive sequence such that for all $n\in\mathbb{N}$,

$$x_{n+1} > x_n + 2\delta \quad \text{and} \quad f(x_n)\geq\varepsilon.$$

Then, $\int_0^{x_n+\delta} f \geq \sum_{k=2}^{n}\frac{\varepsilon}{2}\cdot 2\delta = n\varepsilon\delta \to\infty$ as $n\to\infty$,
So it contradicts to $\int_0^\infty f<\infty$. ∎

2023, Spring

11. (10 pts) Prove or disprove the following statements.

(a) For a metric space X, a subspace $A \subset X$ and a point $x \in X$, there is a sequence of points of A converging to x if and only if x is in the closure $\overline{A}$ of A.

(b) Any two metrics on $\mathbb{R}^n$ induce the same metric topology.

Proof.

a). It is true, because that:

Suppose that there is a sequence $\{a_n\}$ in A converges to $x \in X$.

Then, for any $\varepsilon > 0$, $B_\varepsilon(x) \cap A \neq \emptyset$, because since $a_n \to x$, so for some $N \in \mathbb{N}$, $n \geq N \Rightarrow a_n \in B_\varepsilon(x)$. Thus $x \in \overline{A}$. Conversely, let $x \in \overline{A}$.

Then, for each $n \in \mathbb{N}$, $B_{\frac{1}{n}}(x) \cap A = \emptyset$, so we can construct a sequence $\{a_n\}$ such that $a_n \in B_{\frac{1}{n}}(x) \cap A$ for each $n \in \mathbb{N}$. Now, the $\{a_n\}$ is a sequence in A and it converges to x.

b). Define $d: \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R} : (x, y) \mapsto \begin{cases} 1 & x \neq y \\ 0 & x = y. \end{cases}$

Then, $d(x,x) = 0 \Leftrightarrow x = y$ and $d(x,y) = d(y,x)$ for all $x, y \in \mathbb{R}^n$,

Moreover given $x, y, z \in \mathbb{R}^n$,

$d(x,z) \leq d(x,y) + d(y,z)$, by just caseworks

(If $d(x,z) = 0$, then we are done, and if $d(x,z) = 1$, then either $d(x,y) = 1$ or $d(y,z) = 1$, so we are done)

This d generates the discrete topology, but the euclidean metric is not discrete, so it is false. ☒

2023, Spring

12. (15 pts) Let X_n be the real projective space $\mathbb{R}P^n$, i.e. the quotient space of $\mathbb{R}^{n+1} \setminus \{(0, \cdots, 0)\}$ where the equivalence relation is defined by

$$(x_1, \cdots, x_{n+1}) \sim (y_1, \cdots, y_{n+1}) \Leftrightarrow \exists t \in \mathbb{R} \setminus \{0\} \text{ such that } (x_1, \cdots, x_{n+1}) = t(y_1, \cdots, y_{n+1}).$$

(a) Show that X_1 is homeomorphic to the circle S^1.

(b) For a topological space X and $A \subset X$, a retraction of X to A is a continuous map $r : X \to A$ such that the restriction of r to A is the identity. Remove a point from X_2 and consider its retraction to a circle in X_2. (You may assume that such a retraction exists.) Using the retraction, show that X_2 is not homeomorphic to the 2-sphere S^2.

Proof. a). In X_1, that is, we are in the $\mathbb{R}^2$-plane. In $\mathbb{R}^2\setminus\{(0,0)\}$, we can correspond $(x_1,x_2) = r\cdot e^{i\theta}$, so that for $(x_1,x_2) = r_1 e^{i\theta_1}$, $(y_1,y_2) = r_2 e^{i\theta_2}$,

$(x_1,x_2)\sim(y_1,y_2) \Leftrightarrow e^{i(\theta_1-\theta_2)} = \pm 1 \Leftrightarrow \theta_1 - \theta_2 = k\pi$ for some $k\in\mathbb{Z}$.

Define a map

$$f : \mathbb{R}^2\setminus\{(0,0)\} \to S^1 : re^{i\theta} \mapsto e^{2i\theta}$$

Then, f is continuous onto map, being composition of continuous maps and for $r=1$, $0\le\theta\le\pi$ fills the codomain. Moreover,

$\theta_1 = \theta_2 + k\pi$ for some $k\in\mathbb{Z} \Leftrightarrow e^{2i\theta_1} = e^{2i\theta_2 + 2k\pi}$ for some $k\in\mathbb{Z}$

$\Leftrightarrow f(r_1e^{i\theta_1}) = f(r_2e^{i\theta_2})$

therefore, f induces a Homeomorphism,

$$\tilde{f} : \mathbb{R}P^1 \to S^1 : [(x_1,x_2)] \mapsto f(x_1,x_2).$$

which is well-defined and injective by (*), onto since f is onto, and $(\tilde{f})^{-1}\circ f = \pi : \mathbb{R}^2\setminus\{(0,0)\} \to X_1$ which is a continuous map, so $\tilde{f}^{-1}$ is conti. Consequently $\mathbb{R}P^1 \cong S^1$.

b). Let $r : X_2\setminus\{p\} \to A (\subset X_2\setminus\{p\})$ be the restriction.

[2023 Spring.
13. Let α be a geodesic on the unit sphere $S^2=\{(x,y,z)\mid x^2+y^2+z^2=1\}$. Prove that α is contained in a plane.
Pf.

2023 Spring.

14. Let $\alpha : (0,1) \to \mathbb{R}^3$ be a unit-speed curve with curvature $k > 0$.
Prove that: if $\alpha(t) \in \mathbb{R}^2 \times \{0\}$ for all $t \in (0,1)$,
then α is of a part of a circle of radius $1/k$

[2023. spring.
15. Compute the Gaussian curvature K and the mean curvature H of $S=\{(x,y,z)\in\mathbb{R}^3 \mid 2z=x^2+y^2\}$.

Proof. By definition of S, consider the monge-patch

$$X:\mathbb{R}^2\to S;(u,v)\mapsto(u,v,\frac{1}{2}(u^2+v^2))$$

and $X_u=(1,0,u)$, $X_v=(0,1,v)$,

$X_{uu}=(0,0,1)$, $X_{uv}=(0,0,0)$, $X_{vv}=(0,0,1)$

$$\begin{matrix}1&0&u\\0&1&v\end{matrix}$$

So the normal vector is

$$n(u,v)=(-u,-v,1)\cdot\frac{1}{\sqrt{1+u^2+v^2}}$$

Now, the associated matrix of 1st, 2nd fundamental form are

$$I=\begin{pmatrix}1+u^2&uv\\uv&1+v^2\end{pmatrix},\quad II=\begin{pmatrix}1&0\\0&1\end{pmatrix}\cdot\frac{1}{\sqrt{1+u^2+v^2}}$$

The Gaussian curvature is

$$K=1/(1+u^2+v^2)^2$$

and the mean curvature is

$$H=\frac{1}{(1+u^2+v^2)^{\frac{3}{2}}}\cdot(2+u^2+v^2).$$

□

[2023. Fall.

1. Suppose that $\{a_n\}$ and $\{b_n\}$ are bounded sequences of real numbers.

a). Prove that

$$\limsup_{n\to\infty}(a_n+b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

b). Give an example for the equality does not hold,

Pf

a). Since for each $n\in\mathbb{N}$,

$$\sup_{k\ge n}(a_n+b_n) \le \sup_{k\ge n} a_n + \sup_{k\ge n} b_n,$$

taking limit to the both side we obtain the result.

b). Let $a_n=(-1)^n$ and $b_n=(-1)^{n+1}$. Then,

$\limsup(a_n+b_n)=0$

but

$\limsup a_n = \limsup b_n = 1$.

2023 Fall.

2. Find all solutions to the ODE

$$xy' + y = x^2 \cdot y^2 \quad \text{and} \quad y(1) = 1.$$

Proof. Since $y(1) = 1 \neq 0$, $y > 0$ for some neighbor of $x = 1$. In this neibor,

$$xy' + y = x^2y^2 \Rightarrow \frac{1}{x} \cdot \frac{1}{y^2} \cdot y' + \frac{1}{x^2} \cdot \frac{1}{y} = 1$$

$$\Rightarrow \left(\frac{1}{x} \cdot \frac{1}{y}\right)' = -1$$

$$\Rightarrow \frac{1}{xy} = -x + c$$

$$\Rightarrow \frac{-1}{x(x-c)} = y$$

Since $y(1) = 1$, $\frac{1}{c-1} = 1 \Rightarrow c = 2$.

Finally, $y(x) = \frac{1}{x(2-x)}$ for some neibor of $x = 1$.

is the solution of the initial value problem, we desired.

Moreover, by the uniqueness theorem, it is the unique solution through $(1, 1) \in \mathbb{R}^2$.

2023 Fall.

3. Suppose that $F(x,y)$ is a continuous function defined on $R=[0,1]^2$ such that the partial derivative $\partial_x F$ exists and is continuous on $(0,1)\times[0,1]$. Let us define $f:[0,1]\to\mathbb{R}; x\mapsto \int_0^1 F(x,y)\,dy$

Prove that f is differentiable on $(0,1)$ and satisfies

$$f'(x)=\int_0^1 \partial_x F(x,y)\,dy \quad \text{for all } x\in(0,1).$$

Proof. Let $x\in(0,1)$ be given. We want to show that:

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\int_0^1 \partial_x F(x,y)\,dy.$$

Meanwhile, observe that: for any $|h|>0$,

$$\frac{1}{h}\cdot\left(f(x+h)-f(x)\right)=\frac{1}{h}\cdot\left(\int_0^1 F(x+h,y)-F(x,y)\,dy\right)$$

and since $\partial_x F$ is continuous, by the fundamental theorem of calculus,

$$F(x+h,y)-F(x,y)=\int_x^{x+h}\partial_x F(t,y)\,dt.$$

Let $\epsilon>0$ be given. Take $r>0$ such that $x\in[x-r,x+r]\subset(0,1)$.

Then, $\partial_x F$ is uniformly on the compact set $[x-r,x+r]\times[0,1]$, so it is also continuous uniformly, and so that there is $\delta>0$ such that

$$\forall t,x\in[x-r,x+r] \text{ and } y\in[0,1],\ |t-x|<\delta \Rightarrow |\partial_x(t,y)-\partial_x(x,y)|<\frac{\epsilon}{1-0}=\epsilon.$$

Now, for $0<|h|<\delta$, $\left|\frac{1}{h}\cdot\left(f(x+h)-f(x)\right)-\int_0^1\partial_x F(x,y)\,dy\right|$

$$=\left|\frac{1}{h}\cdot\int_0^1\left(\int_x^{x+h}\partial_x F(t,y)\,dt\right)dy-\int_0^1\partial_x F(x,y)\,dy\right|$$

$$=\left|\frac{1}{h}\cdot\int_0^1\left(\int_x^{x+h}\partial_x F(t,y)\,dt-h\cdot\partial_x F(x,y)\right)dy\right.$$

$$=\left|\frac{1}{h}\cdot\int_0^1\left(\int_x^{x+h}\partial_x F(t,y)\,dt-\int_x^{x+h}\partial_x F(x,y)\,dt\right)dy\right.$$

$$=\left|\frac{1}{h}\cdot\int_0^1\left(\int_x^{x+h}\partial_x F(t,y)-\partial_x F(x,y)\,dt\right)dy\right|$$

$$\le\frac{1}{h}\cdot\int_0^1\int_x^{x+h}\left|\partial_x F(t,y)-\partial_x F(x,y)\right|dt\,dy\le\frac{1}{h}\cdot h\cdot\epsilon=\epsilon.$$

So proved. ∎

2023. Fall.

4. Find all $\alpha > 0$ such that

$$\int_0^\infty \cos(x^\alpha)\,dx$$

exists as a real number.

Proof. Since $\cos(x^\alpha)$ is bounded, the given improper integral converges if and only if $\int_1^\infty \cos(x^\alpha)\,dx$ converges.

observe that: let $x^\alpha = u$, then $\alpha \cdot x^{\alpha-1}\,dx = du$ and so that

$$\int_1^t \cos(x^\alpha)\,dx = \int_1^{t^\alpha} \frac{1}{\alpha} \cdot u^{\frac{1}{\alpha}-1} \cdot \cos(u)\,du$$

If $\alpha > 1$, then $\frac{1}{\alpha} - 1 < 0$, and

$$\int_1^{t^\alpha} u^{\frac{1}{\alpha}-1} \cdot \cos(u)\,du = u^{\frac{1}{\alpha}-1}\sin(u)\Big|_1^{t^\alpha} - \int_1^{t^\alpha} \left(\frac{1}{\alpha}-1\right) \cdot u^{\frac{1}{\alpha}-2} \cdot \sin(u)\,du$$

And, $u^{\frac{1}{\alpha}-1} \cdot \sin(u)\Big|_1^{t^\alpha} = t^{(1-\alpha)} \cdot \sin(t^\alpha) - \sin(1)$ converges as $t \to \infty$,

and since $\left|u^{\frac{1}{\alpha}-2} \cdot \sin u\right| \le u^{\frac{1}{\alpha}-2}$ and $\int_1^\infty u^{\frac{1}{\alpha}-2}\,du$ exists, $\left(\frac{1}{\alpha}-2 < -1\right)$

therefore $\int_1^\infty \left(\frac{1}{\alpha}-1\right) \cdot u^{\frac{1}{\alpha}-2} \cdot \sin(u)\,du$ exists and so $\int_1^\infty \cos(x^\alpha)\,dx$ converges.

If $\alpha \le 1$, then $\frac{1}{\alpha} - 1 \ge 0$, and for large enough $n \in \mathbb{N}$,

$$\frac{1}{\alpha} \cdot \int_{2n\pi - \frac{1}{2}\pi}^{2n\pi + \frac{1}{2}\pi} u^{\frac{1}{\alpha}-1} \cos(u)\,du \ge \frac{1}{\alpha} \cdot \int_{2n\pi - \frac{1}{2}\pi}^{2n\pi + \frac{1}{2}\pi} \cos(u)\,du = \frac{2}{\alpha},$$

So by the Cauchy test for integral, it does not converges. ∎

Q. If $|f| \le g$ and $\int_1^\infty g < \infty \Rightarrow \int f$ converges?

2023 Fall,

5. (10 pts.) Let $\Omega \subset \mathbb{C}$ be a non-empty open set and let $f_n, f : \Omega \to \mathbb{C}$, $n \geq 1$, be holomorphic functions on Ω such that the sequence $(f_n)_{n=1}^{\infty}$ converges uniformly to f on every compact subset of Ω. Prove that the sequence $(f_n')_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. Given $z_0 \in G$, take $\delta > 0$ such that $\overline{B(z_0, r)} \subset G$. By the hypothesis, $f_n \to f$ uniformly on $\overline{B_\delta(z_0)}$, and so f is continuous on $B_\delta(z_0)$.
Let T be a triangular path inside $B_\delta(z_0)$. Since $f_n \to f$ uniformly,
$\int_T f_n(z)\,dz \to \int_T f(z)\,dz$. But, by the Cauchy-Goursat's theorem, each $n \in \mathbb{N}$, $\int_T f_n(z)\,dz = 0$, so $\int_T f(z)\,dz = 0$.
Now, by the Morera's theorem, f is Holomorphic on $B_\delta(z_0)$.
Since z_0 was arbitrary, so f is Holomorphic on G. To show the second assertion, let $K \subset G$ be a compact subset. Since $\mathbb{C}$ is T_4 space and G is open, there is an open U such that $K \subseteq U \subseteq \overline{U} \subseteq G$.
Now, for each $\delta > 0$, define

$$U_\delta := \{z \in U \mid \overline{D_\delta}(z) \subset U\} \text{ where } D_\delta = \{z' \in \mathbb{C} \mid |z' - z| < \delta\}.$$

Given $z \in U_\delta$, by the Cauchy's integral formula,

$$f'(z) - f_n'(z) = \frac{1}{2\pi i} \cdot \int_{C_\delta} \frac{f(w) - f_n(w)}{(w - z)^2}\,dw$$

where C_δ is a curve centered at z with radius $\delta > 0$. Therefore,

$$|f'(z) - f_n'(z)| \leq \frac{1}{2\pi} \cdot \int_{C_\delta} \frac{|f(w) - f_n(w)|}{|w - z|^2}\,|dw| \leq \frac{1}{2\pi} \cdot \frac{2\pi\delta}{\delta^2} \cdot \sup_{w \in C_\delta} |f(w) - f_n(w)|$$

$$\leq \frac{1}{\delta} \cdot \sup_{w \in \overline{U}} |f(w) - f_n(w)| \to 0 \text{ as } n \to \infty.$$

Since the bound that tends to zero is independent of choice of z, Therefore, we can choose $\delta > 0$ small enough to $K \subset U_\delta$, so $f_n' \to f'$ uniformly. ☐

[2023 Fall.
6. Let $A \in M_{n\times n}(\mathbb{R})$ be given. Put $m_A(t) \in \mathbb{R}[t]$ be the minimal polynomial and $\chi_A(t) \in \mathbb{R}[t]$ be the characteristic polynomial.
a). Does there exists a 3 by 3 matrix A such that

$$m_A(t) = t^2 - t \quad \text{and} \quad \chi_A(t) = t^3 - t^2 \ ?$$

b). If $C \in M_{5\times 5}(\mathbb{R})$ such that $C^5 = I_5$, is it necessarily true that

$$m_C(t) = t^5 - 1 \ ?$$

Proof.
a). Let $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then, $\chi_A(t) = t^2(t-1) = t^3 - t^2$.
And, $A \cdot (A - I) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$,
so $m_A(t) = t(t-1)$.
Therefore, exists.

b). if $C = I_5$, then $C^5 = I_5$ but the minimal polynomial is $t-1$.
So, not necessary. □

2023 Fall.

7. Let $K = \mathbb{F}_{p^2}$ be a finite field of order p^2.

a). Find the number of all 3×3 invertible matrices over K.

b). Let V be a 5-dimensional vector space over K and $W \subset V$ be a 2-dimensional sub space. Find the number of all K-linear maps

$$T: V \to V \quad \text{such that } T[W] \subseteq W.$$

Proof a). Since $A \in M_{3\times 3}(K)$ is invertible if and only if the rows of A are linearly independent, so the possible cases are

$$\underbrace{[(p^2)^3 - 1]}_{\text{Choose first row except zero}} \cdot \underbrace{[(p^2)^3 - p^2]}_{\text{Choose second except multiple of the first}} \cdot \underbrace{[(p^2)^3 - p^2 \cdot p^2]}_{\text{similar.}}$$

So, $|GL_3(K)| = (p^6-1)(p^6-p^2)(p^6-p^4)$.

b). Let $\{\alpha_1, \alpha_2\}$ be the basis for W.

since $T[W] \subseteq W \iff T(\alpha_1), T(\alpha_2) \in W$, so $T[W] \subseteq W$!!!

$T(\alpha_1) = a\cdot\alpha_1 + b\cdot\alpha_2$ and $T(\alpha_2) = c\cdot\alpha_1 + d\cdot\alpha_2$ for some $a, b, c, d \in K$

And, let $\{\alpha_1, \alpha_2, \cdots, \alpha_5\}$ be the extended basis for V.

$T(\alpha_i) = \sum_{j=1}^{5} a_{ij}\alpha_j$ for $i \geq 3$, so the number of such T is

$$p^2 \cdot p^2 \cdot p^2 \cdot p^2 \cdot \left((p^2)^5\right)^3.$$

[2023 Fall,

9. (10 pts.) Let F be a field of order 3^2.

(a) Classify all groups of order $|F^\times|$, where $F^\times$ is the set of units in F.

(b) Show that $F^\times$ is a group and find the group among the groups in your answer of (a) which is isomorphic to $F^\times$.

Proof. a). Notice that : $|F^\times| = 8$. If G is an abelian group, then

$$G \cong \mathbb{Z}_8 \text{ or } \mathbb{Z}_4 \times \mathbb{Z}_2 \text{ or } \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$

by the fundamental theorem of finitely generated abelian group.
And, if G is not abelian, then

$$G \cong \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle \text{ or } \langle i, j, k \mid i^2 = j^2 = k^2 = -1,\ (-1)^2 = 1 \rangle$$

(이거밖에 없다는건 어케 논증하지(..)

b). By definition, all elements of $F^\times$ are units, so these have inverse and the multiplication of units is unit, so closed under the operation. Clearly, $1 \in F^\times$, so $F^\times$ is a group and abelian, by definition of field.
So, $F^\times \cong \mathbb{Z}_8$ or $\mathbb{Z}_4 \times \mathbb{Z}_2$ or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.

2023 Fall.

10. Find the Galois group of the splitting field of the following over $\mathbb{Q}$:

a) x^p-1 for a prime p.

b) x^3-2.

Proof. a). Let $\zeta=\exp(2\pi i/p)$. Then, $\zeta^0, \zeta^1, \cdots, \zeta^{p-1}$ are distince roots of x^p-1, so that $K=\mathbb{Q}(\zeta)$ is the splitting field of x^p-1, and since all p-numbers are distinct, x^p-1 is separable, so that $K/\mathbb{Q}$ is Galois. Meanwhile, since

$$x^p-1=(x-1)(x^{p-1}+\cdots+x+1) \quad \text{and since}$$

$$(x+1)^{p-1}+\cdots+(x+1)+1=\frac{(x+1)^p-1}{(x+1)-1}=x^{p-1}+\binom{p}{1}x^{p-2}+\cdots+\binom{p}{p-1}$$

is irreducible by eisenstein, so $x^{p-1}+\cdots+x+1$ is irreducible over $\mathbb{Q}$ and has ζ as a root, so that $[K:\mathbb{Q}]=p-1$. Moreover, every automorphism of $\mathrm{Gal}(K/\mathbb{Q})$ is determined by $\zeta\mapsto\zeta^a$ $(1\le a\le p-1)$, so

$$\mathrm{Gal}(K/\mathbb{Q})=\{\zeta\mapsto\zeta^a \mid a\in\mathbb{Z}_p^{\times}\}\cong\mathbb{Z}_p^{\times}\cong\mathbb{Z}_{p-1},$$

b). Let $w=\exp(2\pi i/3)$. Then, $\sqrt[3]{2}\cdot w^k$ $(k=0,1,2)$ are roots of x^3-2. By the eisenstein, x^3-2 is irreducible over $\mathbb{Q}$ and has $\sqrt[3]{2}$ as a root, and x^2+x+1 is irreducible over $\mathbb{Q}$ and has w as a root, by a). Now, the splitting field of the separable polynomial x^3-2 is

$$K=\mathbb{Q}(\sqrt[3]{2}\cdot w^k \mid k=0,1,2)=\mathbb{Q}(\sqrt[3]{2},w)$$

satisfies $[K:\mathbb{Q}]=[K:\mathbb{Q}(\sqrt[3]{2})]\cdot[\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}]=2\cdot3=6$, the fact $[K:\mathbb{Q}(\sqrt[3]{2})]=2$ is given by $\mathbb{Q}(\sqrt[3]{2})\subset\mathbb{R}$ but $w\notin\mathbb{R}$. Thus $\mathrm{Gal}(K/\mathbb{Q})$ has order of 6, and $\sigma\in\mathrm{Gal}(K/\mathbb{Q})$ is determined by

$$\sigma:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\cdot w^a & (a=0,1,2),\\ w\mapsto w^b & (b=1,2)\end{cases}$$ So let

$$r:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\\ w\mapsto w^2\end{cases},\quad s:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\,w\\ w\mapsto w\end{cases}.$$ Then, $r^2=\mathrm{id}$, $s^3=\mathrm{id}$, and $(sr)^2=\mathrm{id}$, so $\left(sr:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\,w\\ w\mapsto w^2\end{cases}\right)$

$$\mathrm{Gal}(K/\mathbb{Q})=\langle r,s \mid r^2=s^3=(sr)^2=\mathrm{id}\rangle\cong S_3.$$ ∎

2023 Fall.

12. Consider the Parametrized surface

$$X(u,v)=\left(u-\frac{u^3}{3}+uv^2,\ v-\frac{v^3}{3}+vu^2,\ u^2-v^2\right) \quad (u,v)\in\mathbb{R}^2.$$

Compute its Gaussian Curvature.

proof Calculate the derivatives and normal:

$$X_u=(1-u^2+v^2,\ 2uv,\ 2u),\ X_v=(2uv,\ 1-v^2+u^2,\ -2v)$$

So the normal vector is

$$n(u,v)=$$

$$\left(-4uv^2\right.$$

2023, Fall.

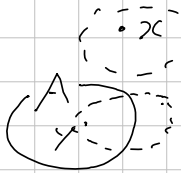
13. (10 pts.) A subspace A of a topological space X is a *retract* if there is a continuous function $f : X \to A$ such that $f(a) = a$ for all $a \in A$.

(a) Show that if A is a retract of a Hausdorff space X, then A is closed in X.

(b) Is the converse of this statement true? Justify your answer.

Proof. a). Let $f: X \to A$ be a continuous map such that $f|_A = id_A$.
Given $x \in X \setminus A$, $f(x) \in A$ and so that $x \neq f(x)$.
Since X is Hausdorff, there exist open sets U, V such that

$$x \in U, \quad f(x) \in V, \quad U \cap V = \emptyset.$$



Then, by definition of subspace, $A \cap V$ is open of A and so that $f^{-1}[A \cap V]$ is open of X, with

$$f^{-1}[A \cap V] = X \cap f^{-1}[V] = f^{-1}[V].$$

Since $f(x) \in V$, $x \in f^{-1}[V]$ and so that

$f^{-1}[V] \cap U$ is a open set containing x, and disjoint from A:

if $y \in A$ and $y \in f^{-1}[V] \cap U$, then

$f(y) = y$ and $f(y) \in V$ and $y \in U$, so $y \in U \cap V$, it contradiction. ⊠

b). Let $X = \mathbb{R}$ equipped usual topology, and $A = \{0, 1\} \subset \mathbb{R}$.
Then, $\mathbb{R}$ is Hausdorff and A is discrete. Suppose A is retract by a continuous map $f: X \to A$ such that $f(0) = 0$, $f(1) = 1$. Then,

$0 \in f^{-1}[\{0\}]$ and $1 \in f^{-1}[\{1\}]$ are disjoint non-empty open sets

of $\mathbb{R}$ such that $\mathbb{R} = f^{-1}[\{0\}] \cup f^{-1}[\{1\}]$, but it is contradiction to $\mathbb{R}$ is connected. ⊠

Note: Let $r: X \to A$ be a retraction and $i: A \to X$ be an inclusion.
Then, $A = \{x \in X \mid (i \circ r)(x) = x\}$
$= (id_X, (i \circ r))^{-1}[\Delta_X]$ where $\begin{cases} (id_X, i \circ r): X \to X \times X; x \mapsto (x, (i \circ r)(x)) \\ \Delta_X = \{(x, x) \mid x \in X\}. \end{cases}$

2023. Fall.

14. If X is Lindelöf and Y is compact, then $X\times Y$ is Lidelöf. Prove.

Proof. Let $\mathcal{O}$ be the open cover for $X\times Y$. Let $x_0\in X$ be fixed.
Since $\{x_0\}\times Y\to Y : (x_0,y)\mapsto y$ is 1-homeomorphism, $\{x_0\}\times Y\cong Y$ is compact.
Since $\mathcal{O}$ covers $X\times Y$, it also covers $\{x_0\}\times Y$, so there is a finite subcover $F_0\subset\mathcal{O}$, which satisfies $\{x_0\}\times Y\subset\bigcup F_0\subseteq X\times Y$
Given $y\in Y$, $(x_0,y)\in\bigcup F$, open, so there exist open sets $U_y\times V_y\subseteq X\times Y$ such that $(x_0,y)\in U_y\times V_y\subseteq\bigcup F$. The compactness gives that the open cover $\{U_y\times V_y \mid y\in Y\}$ of $\{x_0\}\times Y$ has a finite subcover
$$\{U_{y_1}\times V_{y_1},\cdots,U_{y_n}\times V_{y_n}\}.$$ Let $W_0=U_{y_1}\cap\cdots\cap U_{y_n}$
Then, $x_0\in W_0$ since $x_0\in U_{y_i}$ for all $1\le i\le n$, so that $\{x_0\}\times Y\subset W_0\times Y$, and W_0 is open, being finite intersection of open sets. Moreover,
$W_0\times Y\subseteq\bigcup F$, because $(x,y)\in W_0\times Y\iff x\in U_{y_1}\cap\cdots\cap U_{y_n}$ and $y\in V_{y_k}$ $\exists k$,
$$\Rightarrow (x,y)\in U_{y_k}\times V_{y_k}\ \exists k$$
$$\Rightarrow (x,y)\in\bigcup F.$$
Since $x_0\in X$ was arbitrary, we can consider open cover $\{W_x \mid x\in X\}$ for X, and since X is Lindelöf, it has a Countable subcover $\{W_{x_k}\mid k\in\mathbb{N}\}$,
So, $$X\times Y=\left(\bigcup_{k\in\mathbb{N}}W_{x_k}\right)\times Y=\bigcup_{k\in\mathbb{N}}W_{x_k}\times Y\subseteq\bigcup_{k\in\mathbb{N}}\bigcup F_k,$$
where $F_k\subset\mathcal{O}$ is the finite subcover determined when $x_k\in X$ chosen, thus $\mathcal{O}$ has a Countable subcover ∎

Debris

Let $\mathcal{O}=\{U_\lambda\mid\lambda\in\Lambda\}$ be an open cover of $X\times Y$.
Given $x_0\in X$, since the projection $\{x_0\}\times Y\to Y:(x_0,y)\mapsto y$ is Homeomorphism, So $\{x_0\}\times Y\cong Y$ is compact. Therefore, there is a finite subcollection
$$\mathcal{O}_{x_0}\subseteq\mathcal{O}\quad\text{satisfying}\quad\{x_0\}\times Y\subseteq\bigcup\mathcal{O}_{x_0}.$$
For each $x\in X$, define $W_x=\bigcup\mathcal{O}_x$. Let $x_0\in X$ be fixed.
Since W_{x_0} is open, being union of open sets, so given $(x,y)\in W_{x_0}$, there exist open sets $U_x\subseteq X$ and $V_x\subseteq Y$ such that
$$(x,y)\in U_x\times V_x\subseteq W_{x_0}$$
Then, $\{U_x\times V_x\mid x\in W_{x_0}\}$ is an open cover of $\{x_0\}\times Y$, so there is finite subcover $\{U^1_{x_0}\times V^1_{x_0},\cdots,U^n_{x_0}\times V^n_{x_0}\}$. Take $A_{x_0}=U^1_{x_0}\cap\cdots\cap U^n_{x_0}$.
Then, since $x_0\in A_{x_0}$, it is non-empty open set satisfying
$$\{x_0\}\times Y\subseteq A_{x_0}\times Y\subseteq W_x$$

2023 Fall.

15. Let (X,d) be a compact metric space.
Suppose that $f: X \to X$ is a continuous function satisfying

$$d(x,y) = d(f(x), f(y)) \text{ for all } x, y \in X.$$

a). Show that f is bijective.
b). Show that f is a Homeomorphism.

Proof. a). Since $f(x) = f(y) \Rightarrow d(f(x), f(y)) = d(x,y) = 0 \Rightarrow x = y$, f is injective. And, assume $X \setminus f[X]$ is not-empty set. Let $x \in X \setminus f[X]$. Define a sequence in X as: $x_0 = x$ and $x_{n+1} = f(x_n)$ $(n \geq 0)$. Since f is continuous and X is compact, $f[X]$ is also compact and since X is Hausdorff, $f[X]$ is closed set. Therefore, for some $\varepsilon > 0$,

$$x \in B_\varepsilon(x) \subseteq X \setminus f[X].$$

Meanwhile, for any $m > n$,

$$d(x_m, x_n) = d(f(x_{m-1}), f(x_{n-1})) = d(x_{m-1}, x_{n-1}) = \cdots = d(x_{m-n}, x_0) \geq \varepsilon \quad \cdots (*)$$

Since $x_{m-n} \in f[X]$. But, every sequence in a compact metric space has a convergent subsequence, it contradicts to $(*)$. So $f[X] = X$.

b). By a), f is bijective, so the inverse $f^{-1}: X \to X$ exists. And, given $x, y \in X$, $x = f(a)$, $y = f(b)$ for some $a, b \in X$ and so that

$$d(x,y) = d(f(a), f(b)) = d(a,b) = d(f^{-1}(f(a)), f^{-1}(f(b))) = d(f^{-1}(x), f^{-1}(y)).$$

Now, let $\varepsilon > 0$ be given. Take $\delta = \varepsilon$, then

$$d(x,y) < \delta \Rightarrow d(f^{-1}(x), f^{-1}(y)) = d(x,y) < \delta = \varepsilon,$$

So f^{-1} is continuous, so f is Homeomorphism. ∎

[2024, Spring.
1. Let $A, B \in M_{m\times n}(F)$. Show that $\operatorname{rank}(A+B) \le \operatorname{rank} A + \operatorname{rank} B$. [5m]

Proof. By definition,

$\operatorname{rank}(A+B) = \dim \operatorname{Im}(A+B)$ and $\operatorname{rank} A = \dim \operatorname{Im} A$, $\operatorname{rank} B = \dim \operatorname{Im} B$.

Given $\alpha \in \operatorname{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \operatorname{Im} A + \operatorname{Im} B$ for some $\beta \in F^n$, therefore $\operatorname{Im}(A+B) \subseteq \operatorname{Im} A + \operatorname{Im} B$. Now,

$\operatorname{rank}(A+B) \le \dim(\operatorname{Im} A + \operatorname{Im} B) \le \dim \operatorname{Im} A + \dim \operatorname{Im} B = \operatorname{rank} A + \operatorname{rank} B$. ∎

[2024 spring.
2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.
Pf. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ such that $P(\alpha_i) = c_i \alpha_i$ for some $c_i > 0$, for each i.
Let $1 \le i \le n$ be fixed. Then, by the assumption,

$$P^2 \alpha_i = c_i^2 \alpha_i = Q^2 \alpha_i \Rightarrow (Q^2 - c_i^2 I)\alpha_i = (Q - c_i I)(Q + c_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - c_i I)\alpha_i = 0$, because; if it is not zero, then

$$(Q - c_i I)(Q + c_i I)\alpha_i = (Q + c_i I)(Q - c_i I)\alpha_i = 0,$$

So $-c_i$ is negative eigenvalue of Q, it contradicts to Q is positive definite. Therefore, for each $1 \le i \le n$,

$$P\alpha_i = c_i \alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. □

2024. Spring

3. Let $T:\mathbb{R}^5\to\mathbb{R}^5;\ x\mapsto Ax$ where

$$A=\begin{pmatrix}1&1&0&0&0\\0&1&0&0&0\\0&0&1&1&0\\0&0&0&1&1\\0&0&0&0&1\end{pmatrix}$$

a). There exists T-invariant subspace $V\subseteq\mathbb{R}^5$ s.t $\dim V=4$?

b). // s.t $\dim V=2$, $T|_V$ is diagonalizable?

c) // $\dim V=3$, //

Pf.

a). Let $V=\langle e_1,e_2,e_3,e_4\rangle$. Then,

$Ae_1=e_1,\ Ae_2=e_1+e_2,\ Ae_3=e_3,\ Ae_4=e_3+e_4$

are all contained in V, so $T[V]\subseteq V$. (clearly $\dim V=4$)

b). Let $V=\langle e_1,e_3\rangle$. Then $Ae_1=e_1$, $Ae_3=e_3$ are contained in V, and

$[T|_V]_{\{e_1,e_3\}}=I_2$. So $T|_V$ is diagonalizable.

c). Observe characteristic polynomial of T is $(t-1)^5$.
If such a subspace V exists, then the characteristic polynomial of $T|_V$ is $(t-1)^3$. Since $T|_V$ is diagonalizable iff and only the minimal polynomial splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$. That is, $T|_V=I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim\ker(A-I)=\dim\ker\begin{pmatrix}0&1&0&0&0\\0&0&0&0&0\\0&0&0&1&0\\0&0&0&0&1\\0&0&0&0&0\end{pmatrix}=2.$$

2024 Spring.

4. (10 pts.) Let R be a ring. Denote by $GL_3(R)$ the set of all 3×3 invertible matrices with entries in R. Let G be the subset of $GL_3(R)$ consisting of upper-triangular matrices.

(a) Show that G is a subgroup of $GL_3(R)$.

(b) Show that G is not abelian.

(c) Show that G is solvable.

Proof. a). Given $A, B \in G$,

$(AB)_{ij} = \sum_{k=1}^{3} A_{ik} B_{kj}$. If $i > j$, then $(i,j) \in \{(2,1), (3,1), (3,2)\}$, so

$= A_{i1} B_{1j} + A_{i2} B_{2j} + A_{i3} B_{3j}$

$= 0 + 0 + 0$

$= 0$, by just casework, so AB is also upper-triangular.

Moreover, since A and B are invertible, so $AB \cdot B^{-1}A^{-1} = I$, thus AB is invertible with the inverse $B^{-1}A^{-1}$. Thus, $AB \in G$.

Clearly $I \in G$, and, given $A \in G$, there is $A^{-1} \in GL_3(\mathbb{R})$. Denote $B = A^{-1}$, then

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ 0 & A_{22} & A_{23} \\ 0 & 0 & A_{33} \end{pmatrix} \cdot \begin{pmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then, $A_{33} B_{31} = 0$ and $A_{33} B_{33} = 1$, but every unit cannot be a zero divisor, so $B_{31} = 0$. Similarly $B_{32} = 0$, And, now, $A_{22} B_{22} = 1$ but $A_{22} B_{21} = 1$, so $B_{21} = 0$. Therefore, $B = A^{-1}$ is upper-triangular, so $A^{-1} \in G$,

It proves G is a group.

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ & & \end{pmatrix}$$

b). Define a 3 by 3 matrix over $\mathbb{R}$ as:

$$(E_{ab})_{ij} = \begin{cases} 0 & (a,b) \neq (i,j) \\ 1 & (a,b) = (i,j) \end{cases}$$

Let $A = I + E_{12}$. Then, $A \cdot (I - E_{12}) = I^2 + E_{12} - E_{12} - E_{12}E_{12} = I$,

and let $B = I + E_{23}$. Then similarly $B \cdot (I - E_{23}) = I$, so $A, B \in G$.

Moreover, $AB = I + E_{12} + E_{23} + E_{12}E_{23} = I + E_{12} + E_{23} + E_{13}$

$BA = I + E_{12} + E_{23} + E_{23}E_{12} = \quad // \quad + 0$

So $AB \neq BA$, thus G is non-abelian.

observe that $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ but

$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, so

these are not commute. And, these have the determinant ± 1, units in $\mathbb{R}$, so contained in GL, and clearly upper-triangles, so G.

c). Let $GL(\mathbb{R}) \to \mathbb{R}$; $A \mapsto \det A$

Then, det is a Homomorphism with kernel $SL(\mathbb{R}) \trianglelefteq GL(\mathbb{R})$.

Now $G/(SL(\mathbb{R}) \cap G)$

2024, Spring.

5. Let K be a field, and $L_1, L_2 \subseteq \overline{K}$ be finite extensions of K. 4점

a). Prove $L = L_1L_2$ is finite extension.

b). If L_1 and L_2 are Galois over K, then L/K is Galois.

a), Since L_1 and L_2 both are finite, denote

$L_1 = K(\alpha_1, \cdots, \alpha_n)$ and $L_2 = K(\beta_1, \cdots, \beta_r)$ where $\alpha_i, \beta_j \in \overline{K}$ are algebraic over K.

Now, $L = L_1L_2 = \bigcap_{\substack{E \leq \overline{K} \\ L_1, L_2 \leq E}} E = K(\alpha_1, \cdots, \alpha_n, \beta_1, \cdots, \beta_r)$

is finitely generated by algebraic elements α_i, β_j, so L is finite. 1점.

b). Since L_1 and L_2 are splitting field of K, let f_1 and f_2 be polynomials over K such that L_i is the splitting field of f_i, $i = 1, 2$.
It is possible, because L_1, L_2 both are finite extension.
Now let E be the splitting field of f_1f_2. Then,
$E \supseteq L_1L_2$, because E is the smallest field containing all roots of f_1f_2,
$L_1L_2 \supseteq E$ because E contains all roots of f_1 and f_2. So, $L = E_1E_2$,
therefore L is splitting field over K. 1점.

Since L_1 and L_2 both are Galois, for some separable f_1 and f_2 over K,
L_1 and L_2 are splitting field of f_1 and f_2 respectively.

L_1L_2 가 f_1f_2 의 splitting field $\Rightarrow$ ok
근데 f_1f_2 가 separable 인건 보장 X.
중복 linear factor 생기면 f_1f_2 가 여러번씩 over K 인가?

Revised in the Algebra/Field Theory/

[2024 Spring.

7. Let $f:(-1,1)\to\mathbb{R}$ be a differentiable function satisfying $f(0)=0$ and let $g:(-1,1)\to\mathbb{R}$: $\begin{cases} f(x)/x & \text{if } x\neq 0 \\ f'(0) & \text{if } x=0. \end{cases}$

a). If $f\in C^2$, then $g\in C^1$ (prove)

b), If $f\in C^1$ but $\notin C^2$, then $g\in C^1$ may not hold.

Proof. Notice that g is continuous, since

$$\lim_{x\to 0} g(x) = \lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = f'(0).$$

a) For $x\neq 0$,

$$g'(x) = \left(\frac{f(x)}{x}\right)' = \frac{xf'(x)-f(x)}{x^2}$$ exists, By the L'Hospital Theorem,

$$\lim_{x\to 0} g'(x) = \lim_{x\to 0} \frac{f'(x)+xf''(x)-f'(x)}{2x} = \frac{1}{2}\cdot \lim_{x\to 0} f''(x) = \frac{1}{2}f''(0).$$

(Since $x^2\to 0$ and $xf'(x)-f(x)\to 0$ as $x\to 0$, we can use the theorem).

And, $g'(0) = \lim_{x\to 0} \frac{g(x)-g(0)}{x-0} = \lim_{x\to 0} \frac{f(x)/x - f'(0)}{x-0} = \lim_{x\to 0} \frac{f(x)-xf'(0)}{x^2} = \frac{1}{2}f''(0)$.

Since the L'Hôpital theorem again, so $g\in C^1$.

b). Let $f:(-1,1)\mapsto \begin{cases} x\cdot|x|. & \text{if } x\neq 0 \\ 0 & \text{if } x=0. \end{cases}$

2024 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$. For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f,g) := \sup_{x\in\mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n\to\infty} d(f, f_n) = 0$

b). Prove that $\lim_{n\to\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x)| \right) = \sup_{x\in\mathbb{R}} |f(x)|$.

Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis, Since $\mathbb{R}$ is complete, $\lim_{n\to\infty} f_n(x)$ exists, Put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \to 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x\in\mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$.

So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \to \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$.

Since x was arbitrary, $\sup_{x\in\mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.

Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$ and since f_n was bounded, f is bounded. And, the argument above proves $\sup_{x\in\mathbb{R}} |f - f_n| \to 0$ as $n \to \infty$, So since f_n is continuous and

$f_n \to f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left| \sup_{x\in\mathbb{R}} |f_n(x)| - \sup_{x\in\mathbb{R}} |f(x)| \right| \leq \sup_{x\in\mathbb{R}} |f_n(x) - f(x)| \to 0$ as $n \to \infty$,

proves directly.

[2024 Spring]

12. Compute the Gauss curvature at $p=(0,0,0)$ of the smooth surface given by $z=y^2-x^2$.

Proof. Consider the Monge patch of the given surface

$$X: \mathbb{D} \to \mathbb{R}^3 : (u,v) \mapsto (u, v, v^2-u^2)$$

Then, $X_u=(1,0,-2u)$ and $X_v=(0,1,2v)$, so the first fundamental form $E=1+4u^2$, $F=-2uv$, $G=1+4v^2$

so $\det\begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1+4u^2+4v^2+16u^2v^2-4u^2v^2 = 1$ at $(u,v)=(0,0)$

and $\begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ at $(u,v)=(0,0)$.

Meanwhile, the normal of the given surface is

$X_u \times X_v = (2u, -2v, 1)$ gives $n=(0,0,1)$ at $(u,v)=(0,0)$.

Meanwhile, $X_{uu}=(0,0,-2)$, $X_{uv}=(0,0,0)$, $X_{vv}=(0,0,2)$ so

$L_{uu}=-2$, $L_{uv}=0$, $L_{vv}=2$.

$$L^u_u = L_{uu}\, g^{uu} + L_{vu} \cdot g^{vu} = 2$$

$$L^v_u = L^u_v = L_{uu} \cdot g^{uv} + L_{vu} \cdot g^{vv} = -4$$

$$L^v_v = L_{uv}\, g^{uv} + L_{vv} \cdot g^{vv} = 4,$$

Therefore, the Gaussian Curvature is

$$\det\begin{pmatrix} 2 & -4 \\ -4 & 4 \end{pmatrix} = 24,$$

Note: Gaussian Curvature is $\det \mathrm{II} / \det \mathrm{I}$.

2024. spring.

14. Let X be a compact hausdorff space. Prove that

X is metrizable if and only if X has a countable basis.

Proof Suppose that X is metrizable by the metric $d: X \times X \to \mathbb{R}$.
Given $n \in \mathbb{N}$, $\{B_{\frac{1}{n}}(x) \mid x \in X\}$ is an open cover of X, so the compactness gives that $\mathcal{B}_n = \{B_{\frac{1}{n}}(x_i) \mid i \in F_n\}$, the finite subcover exists with finite index set F_n. Now, put $\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{B}_n$.

Let $x \in X$ be given and let $B_r(y)$ be an open ball containing x.
Then, for $\varepsilon = r - d(x,y)$, $x \in B_\varepsilon(x) \subseteq B_r(y)$, because

$$a \in B_\varepsilon(x) \Leftrightarrow d(a,x) < \varepsilon = r - d(x,y) \Rightarrow d(a,y) = d(a,x) + d(x,y) < r.$$

Now, let $N \in \mathbb{N}$ such that $\frac{1}{N} < \frac{\varepsilon}{2}$. Then, since each $\mathcal{B}_n$ covers X, for some x_i, $x \in B_{\frac{1}{N}}(x_i)$, and

$$z \in B_{\frac{1}{N}}(x_i) \Leftrightarrow d(z, x_i) < \frac{1}{N} \Rightarrow d(z,x) \le d(z,x_i) + d(x_i,x) < \frac{1}{N} + \frac{1}{N} < \varepsilon.$$

So, $x \in B_{\frac{1}{N}}(x_i) \subseteq B_\varepsilon(x)$, it proves $\mathcal{B}$ is a basis for (X,d),

and since $\mathcal{B}$ is countable union of finite sets, so it is countable.

Conversely, since compact Hausdorff space is normal, so it is regular.
By the Urysohn metrization theorem, second countable regular space is metrizable, so proved.

2024. Fall.

1. Let V be a finite-dimensional vector space over a field F. Let $T: V \to V$ be a linear operator such that $T^2 = T$. Prove

a). $V = \ker T \oplus \operatorname{Im} T$

b). T is diagonalizable.

Proof. a). Given $\alpha \in V$, $\alpha = \alpha - T(\alpha) + T(\alpha)$. And, $T(\alpha - T(\alpha)) = T(\alpha) - T^2(\alpha) = 0$ since $T^2 - T = 0$, so $\alpha - T(\alpha) \in \ker T$. Clearly $T(\alpha) \in \operatorname{Im} T$, so $V = \ker T + \operatorname{Im} T$. Meanwhile, let $\alpha \in \ker T \cap \operatorname{Im} T$. Then, $\alpha = T(\beta)$ for some $\beta \in V$, and $\alpha \in \ker T$ so that

$$0 = T(\alpha) = T(T(\beta)) = T(\beta) = \alpha,$$

thus $\alpha = 0$, $\ker T \cap \operatorname{Im} T = \{0\}$.

It proves a),

b). Since for any $\alpha \in \operatorname{Im} T$, $\alpha = T(\beta)$ for some $\beta \in V$ and so that $T(\alpha) = T^2(\beta) = T(\beta) = \alpha$, thus $T(\alpha) - \alpha = 0$. Therefore, $\operatorname{Im} T \subseteq \ker(T - I)$, and given $\alpha \in \ker(T - I)$, $T(\alpha) = \alpha$, so $\alpha \in \operatorname{Im} T$. Now, $V = \ker T \oplus \ker(T - I)$,

therefore T is diagonalizable. □

[2024 Fall.

2. (10 pts) Find all 2×2 matrices A with entries in $\mathbb{R}$ such that $AA^t = I_2$ and that A is diagonalizable over $\mathbb{R}$, that is, $Q^{-1}AQ$ is a diagonal matrix for some invertible 2×2 matrix Q with real entries.

22m

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then,

$A^t = A^{-1} \Leftrightarrow \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \frac{1}{\det A} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ with $\det A \neq 0$.

So, $\det A \cdot a = d$ and $\det A \cdot d = a$, so that $(\det A)^2 = 1$.

If $\det A = 1$, then $a = d$ and $b = -c$, so that

$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, so $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ for some θ.

And, since a matrix is diagonalizable if and only if the minimal polynomial splits and separable,

$\det(A - tI) = t^2 - 2\cos\theta \cdot t + 1 = (t - (\cos\theta + \sqrt{-\sin^2\theta}))(t + (\cos\theta - \sqrt{-\sin^2\theta}))$

So that the characteristic splits only when $\theta = 0$ or π, so the minimal polynomial splits only $\theta = 0$ or π, and in such a case, it is separable,

So $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

If $\det A = -1$, then $a = -d$ and $b = c$, so that

$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$,

and $\det(tI - A) = t^2 - \cos^2\theta - \sin^2\theta = (t-1)(t+1)$,

so A is always diagonalizable, so Consequently

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ or $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ $(\theta \in [0, 2\pi))$

are we desired.

2024 Fall

3. **(a)** An element a of a ring R is nilpotent if $a^n = 0$ for some integer $n > 0$. Show that the collection of all nilpotent elements in a commutative ring R is an ideal. Note that this ideal is called the **nilradical** of R.

(b) Find the nilradical of the ring $\mathbb{Z}_{12}$.

Proof. a). Let N be the nilradical of R. Let $a, b \in N$ be given. Then, $a^n = b^m = 0$ for some $n, m \in \mathbb{N}$. Now,

$$(a-b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^{n+m-k} b^k = 0, \text{ so } a-b \in N.$$

And, $(ab)^{nm} = a^{nm} b^{nm} = 0$, so $ab \in N$. Therefore, N is a subring. moreover, given $r \in R$, $(ra)^n = r^n a^n = r^n 0 = 0$, so $ra \in N$.
Thus, $N \subset R$ is an ideal.

$2^2 \cdot 3$

b). The nilradical of $\mathbb{Z}_{12}$ is $\{0, 6\}$, because
$k \in \mathbb{Z}_{12}$ satisfies $k^n = 0$ for some $n \in \mathbb{N}$

$\Leftrightarrow$ for some $n \in \mathbb{N}$, $12 \mid k^n$.
So, $12 = 2^2 \cdot 3$, $k = 0$ or 6.

2024. Fall

4. (10 pts) Let G be a finite group. Show that if the number of solutions x of the equation $x^m = e$ in G is at most m for each positive integer m, then G is cyclic. (Hint: you can use the fact that $n = \sum_{d|n} \phi(d)$, where $\phi(d)$ is the Euler-phi function.)

Proof. Denote that G as a Partition

$$G = \bigcup_{C \le G \text{ cyclic}} \mathcal{G}(C) \quad \text{where} \quad \mathcal{G}(C) = \{x \in G \mid \langle x \rangle = C\}.$$

It is actually a partition, because: given $x \in G$, $x \in \langle x \rangle$ and so $x \in \mathcal{G}(C)$.
And for cyclic subgroups $C_1, C_2 \le G$,

$$x \in \mathcal{G}(C_1) \cap \mathcal{G}(C_2) \Rightarrow \langle x \rangle = C_1 = C_2, \quad \text{So the claim is proved..}$$

Meanwhile, by the Lagrange's theorem, for each subgroup of G has order of factor of order of G, Put $n = |G|$. For each divisor d of n, if there is an order d element $r \in G$, then $|\langle x \rangle| = d$, and clearly

$$\langle r \rangle \subseteq \{x \in G \mid x^d = e\}$$

But, by the assumption,

$$d = |\langle r \rangle| \le |\{x \in G \mid x^d = e\}| \le d$$

therefore $\langle r \rangle = \{x \in G \mid x^d = e\}$, it contains all order d elements, and so that it is the unique cyclic subgroup of order d.
Now, by the Lagrange's theorem again, every element of G has order of divisor for $|G|$, therefore for each $d|n$, $\#\{x \in G \mid |x| = d\} \in \{0, \varphi(d)\}$ so

$$n = |G| = \sum_{d|n} \#\{x \in G \mid |x| = d\} \le \sum_{d|n} \varphi(d) = n$$

forces for all $d|n$, $\#\{x \in G \mid |x| = d\} = \varphi(d)$, so G has $\varphi(n)$-numbers order n elements, so cyclic. ∎

Additionally, $\#\{x \in G \mid |x| = d\} \in \{0, \varphi(d)\}$ is given by:
if there is no such element $|x| = d$, then $\#\{x \in G \mid |x| = d\} = 0$,
if such an element exists, as $x' \in G$,

$$\#\{x \in G \mid |x| = d\} = \mathcal{G}(\langle x' \rangle) = \varphi(d).$$

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5. Let $K=\mathbb{Q}(\sqrt[4]{2}, i)$ be the splitting field in $\mathbb{C}$ of x^4-2 over $\mathbb{Q}$. Describe all subgroups H of $Gal(K/\mathbb{Q})$ of order 2 and corresponding subfields K_H.

Proof. Since $i \notin \mathbb{Q}(\sqrt[4]{2})$ and the minimal polynomial of i over $\mathbb{Q}$ is x^2+1,

$$[K:\mathbb{Q}] = [K:\mathbb{Q}(\sqrt[4]{2})]\cdot[\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}] = 2\cdot 4 = 8$$

So $|Gal(K/\mathbb{Q})| = 8$. And, x^4-2 is the minimal polynomial of $\sqrt[4]{2}$, so that

$$x^4-2 = (x^2-\sqrt{2})(x^2+\sqrt{2}) = (x-\sqrt[4]{2})(x+\sqrt[4]{2})(x+\sqrt[4]{2}i)(x-\sqrt[4]{2}i)$$

and the $\mathbb{Q}$-automorphism maps an element to the conjugate, the $\sigma \in Aut(K/\mathbb{Q})$ is determined by

$$\sigma_{s,k}: \begin{cases} i \mapsto \pm i \ (=i^{2s+1},\ s=0,1) \\ \sqrt[4]{2} \mapsto \sqrt[4]{2}\cdot i^k \ (k=0,1,2,3) \end{cases}$$

Since $(\sigma_{1,0})^2 = id$ and $|(\sigma_{0,1})| = 4$, and

$|(\sigma_{1,0}\circ\sigma_{0,1})| = 2$, because $\sigma_{1,0}\circ\sigma_{0,1} \neq id$ but $(\sigma_{1,0}\circ\sigma_{0,1})^2 = id$.

therefore $Gal(K/\mathbb{Q}) \cong D_8$, and all order 2 elements are

$\sigma_{1,0}$, $\sigma_{0,2}$, $\sigma_{1,2}$, $\sigma_{1,3}$, $\sigma_{1,4}$ so $H = \langle\sigma_{0,2}\rangle, \langle\sigma_{1,i}\rangle$ $i=0,1,2,3$.

And, observe that since the trace is invariant value, so

$Tr_{\langle\sigma_{1,0}\rangle}(\sqrt[4]{2}) = \sqrt[4]{2}+\sqrt[4]{2} = 2\cdot\sqrt[4]{2}$, so $K_{\langle\sigma_{1,0}\rangle} = \mathbb{Q}(\sqrt[4]{2})$

$Tr_{\langle\sigma_{0,2}\rangle}(\sqrt[4]{2}i) = i\sqrt[4]{2} + i(-\sqrt[4]{2}) = 2i$, and

$Tr_{\langle\sigma_{0,2}\rangle}(\sqrt{2}) = \sqrt{2}+\sqrt{2} = 2\sqrt{2}$, so $K_{\langle\sigma_{0,2}\rangle} = \mathbb{Q}(i,\sqrt{2})$

$Tr_{\langle\sigma_{1,2}\rangle}(\sqrt[4]{2}i) = \sqrt[4]{2}\cdot i + (-\sqrt[4]{2})\cdot -i = \sqrt[4]{2}i$, so $K_{\langle\sigma_{1,2}\rangle} = \mathbb{Q}(\sqrt[4]{2}i)$.

$Tr_{\langle\sigma_{1,1}\rangle}(\sqrt[4]{2}) = \sqrt[4]{2}+\sqrt[4]{2}\cdot i$, so $K_{\langle\sigma_{1,1}\rangle} = \mathbb{Q}(\sqrt[4]{2}(1+i))$

$Tr_{\langle\sigma_{1,3}\rangle}(\sqrt[4]{2}) = \sqrt[4]{2}-\sqrt[4]{2}i$, so $K_{\langle\sigma_{1,3}\rangle} = \mathbb{Q}(\sqrt[4]{2}(1-i))$

(The equality is obtained by the Galois theorem gives that $|H|=2 \Rightarrow [K:K_H]=2$.)

2024. Fall.

6. Show that the series $\sum_{k=1}^{\infty} \frac{\sin k}{k}$ converges.

Does this series converges absolutely?

Proof. observe that: given $N \in \mathbb{N}$,

$$\sin\frac{1}{2} \cdot \sum_{k=1}^{N} \sin k = \sum_{k=1}^{N} \sin\frac{1}{2} \cdot \sin k = \frac{1}{2} \cdot \sum_{k=1}^{N} \left[\cos\left(k-\frac{1}{2}\right) - \cos\left(k+\frac{1}{2}\right)\right]$$

$$= \frac{1}{2} \cdot \left(\cos\frac{1}{2} - \cos\left(N+\frac{1}{2}\right)\right)$$

so that the series

$$\left|\sum_{k=1}^{N} \sin k\right| \le \frac{1}{\sin\frac{1}{2}} \cdot \frac{1}{2} \cdot \left(\left|\cos\frac{1}{2}\right| + \left|\cos\left(N+\frac{1}{2}\right)\right|\right) \le \frac{1}{\sin\frac{1}{2}}$$

is bounded. Therefore, since $\frac{1}{k}$ is decreasing sequence converges to 0, by the Dirichlet test, the given series converges.

Meanwhile, since $|\sin x| \le 1$ for all x, so that $\sin^2 x \le |\sin x|$.

But, since $\sin^2 k = \frac{1-\cos 2k}{2}$, and the series

$\sum_{k=1}^{\infty} \frac{\cos 2k}{k}$ converges similarly a) and $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, so that

$\sum_{k=1}^{N} \frac{\sin^2 k}{k} = \sum_{k=1}^{N} \frac{1-\cos 2k}{2k}$ diverges. But, since

$$\sum_{k=1}^{N} \frac{\sin^2 k}{k} \le \sum_{k=1}^{N} \frac{|\sin k|}{k},$$ so it does not converges absolutely.

∎

Note:

$$\cos(\alpha-\beta) - \cos(\alpha+\beta) = \cos\alpha\cdot\cos\beta + \sin\alpha\cdot\sin\beta - (\cos\alpha\cdot\cos\beta - \sin\alpha\cdot\sin\beta)$$

$$= 2\cdot\sin\alpha\cdot\sin\beta$$

2024 Fall.

7. (10 pts) Let $(f_n(x))_{n\in\mathbb{N}}$ be a sequence of continuously differentiable functions on $[0,1]$.

(a) Suppose that the sequence $(f_n(x))_{n\in\mathbb{N}}$ converges pointwisely to a function f. Does $(f_n(x))_{n\in\mathbb{N}}$ converges uniformly to f? Justify your answer.

(b) Suppose that there is a uniform constant $C>0$ such that, for all $n=1,2,\cdots$,

$$\sup_{x\in[0,1]}(|f_n(x)|+|f_n'(x)|)<C.$$

Show that $(f_n(x))_{n\in\mathbb{N}}$ has a uniformly convergent subsequence in $[0,1]$.

Proof. a). Define for each $n\in\mathbb{N}$,

$$f_n:[0,1]\to\mathbb{R};\ x\mapsto x^n.$$

Then, clearly $f_n\in C^1$, and $f_n\to f$ pointwise where

$$f(x)=\begin{cases}1 & \text{if } x=1\\ 0 & \text{if } 0\le x<1\end{cases}$$

But for each $n\in\mathbb{N}$, $\lim_{x\to 1^-} f_n(x)=1$ so that $\sup_{x\in[0,1]}|f(x)-f_n(x)|=1$,

therefore f_n does not uniformly converges.

b). Let $n\in\mathbb{N}$ be given. Given $\varepsilon>0$, take $\delta=\frac{\varepsilon}{C}$. Then, for any $x,y\in[0,1]$,

$$|x-y|<\delta \Rightarrow |f_n(x)-f_n(y)|=|f_n'(t)|\cdot|x-y|\le C\cdot\frac{\varepsilon}{C}<\varepsilon$$

for some t between x and y, by the Mean-Value theorem, therefore the $\{f_n\}$ is equicontinuous, and pointwise-bounded by

$$\sup_{x\in[0,1]}|f_n(x)|\le\sup_{x\in[0,1]}(|f_n(x)|+|f_n'(x)|)<C.$$

Finally, by the Arzela-Ascoli theorem, the sequence of functions $\{f_n\}$ which is pointwise-bounded, equicontinuous on the compact set $[0,1]$ has a uniformly convergence subsequence. ∎

Similar problem. Rudin, PMA, Chapter 7.

18. Let $\{f_n\}$ be a uniformly bounded sequence of functions which are Riemann-integrable on $[a, b]$, and put

$$F_n(x) = \int_a^x f_n(t)\,dt \qquad (a \le x \le b).$$

Prove that there exists a subsequence $\{F_{n_k}\}$ which converges uniformly on $[a, b]$.

2024 Fall.

8. Solve the followings: $x \in \mathbb{R}$, (8)

a). $x''(t) - 2\cdot x'(t) + x(t) = 0$

b). $x''(t) - 2x'(t) + x(t) = te^t + 4$.

Proof.

a). Since the characteristic polynomial of the equation is

$$r^2 - 2r + 1 = (r-1)^2,$$

the general solution is

$$x_g(t) = c_1 e^t + c_2 \cdot te^t \quad \text{where } c_1, c_2 \in \mathbb{R}.$$

b). Calculate the Wronskian:

$$W(e^t, te^t) = \begin{vmatrix} e^t & te^t \\ e^t & e^t + te^t \end{vmatrix} = e^{2t} + te^{2t} - te^{2t} = e^{2t}.$$

Therefore, the particular solution is

$$\begin{aligned} Y(t) &= \int_{t_0}^{t} \begin{vmatrix} e^u & te^t \\ ue^u & te^t \end{vmatrix} \cdot b(u) \Big/ W(e^u, ue^u)\, du \\ &= \int_{t_0}^{t} (e^u te^t - ue^u e^t)\cdot(ue^u + 4)/e^{2u}\, du \\ &= \int_{t_0}^{t} (ue^{2u}te^t + 4e^u te^t - u^2 e^{2u}e^t - 4ue^u e^t)/e^{2u}\, du \\ &= \int_{t_0}^{t} u\cdot te^t + 4\cdot te^t \cdot e^{-u} - u^2\cdot e^t - 4e^t \cdot ue^{-u}\, du \end{aligned}$$

Now, $x_g + Y$ is the general solution.

2024, Fall,

9. For a fixed $\xi \in \mathbb{R}$, let

$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh(\pi z)} \qquad (z \in \mathbb{C}).$$

a). Show that $\alpha = i/2$ and $\beta = 3i/2$ are simple poles of f

b). Compute the residues of f at α and β.

c). Evaluate $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh(\pi x)}\, dx$.

Proof. a). Notice that $e^{-2\pi i z \xi}$ is entire and has no zero, we need to investigate pole of $1/\cosh(\pi z)$.

Since $\cosh(\pi z_0) = 0$ if $z_0 = \alpha$ or β, and

$$\lim_{z \to z_0} \frac{\cosh(\pi z) - \cos(\pi z_0)}{z - z_0} = \sinh(\pi z_0) \cdot \pi \neq 0, \quad \text{so}$$

$$\lim_{z \to z_0} (z - z_0)^1 \cdot \frac{1}{\cosh(\pi z)} = \frac{1}{\pi \sinh(\pi z_0)} \quad \text{exists, so } \alpha \text{ and } \beta \text{ are simple poles.}$$

b). Since α and β both are simple pole,

$$\operatorname{Res}(f, \alpha) = \lim_{z \to \alpha} (z - \alpha) \cdot f(z) = e^{-2\pi i \alpha \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \alpha)} = \frac{e^{\pi \xi}}{\pi \cdot \sinh(\pi i/2)} = \frac{e^{\pi \xi}}{\pi i}.$$

$$\operatorname{Res}(f, \beta) = e^{-2\pi i \beta \xi} \cdot \frac{1}{\pi \cdot \sinh(\pi \beta)} = \frac{e^{3\pi \xi}}{\pi \cdot (-i)} = -\frac{e^{3\pi \xi}}{\pi i}.$$

c). Let C_R be the rectangle $C_R = C_1 * C_2 * C_3 * C_4$ along four line segment $-R$ to R, R to $R+i$, $R+i$ to $-R+i$, $-R+i$ to $-R$.

And, observe that

$$f(x+i) = \frac{e^{-2\pi i x \xi + 2\pi \xi}}{\cosh(\pi x + \pi i)} = e^{2\pi \xi} \cdot -f(x)$$

Therefore, $\int_{C_1 + C_3} f = \left(\int_{-R}^{R} f(x)\, dx \right) \cdot \left(1 + e^{2\pi \xi}\right)$.

And, since $\left| f(R + it) \right| = \left| \frac{e^{-2\pi i R \xi + 2\pi t \xi}}{\cosh(\pi R + \pi i t)} \right| \to 0$ as $\left|\cosh(\pi R + \pi i t)\right| \to \infty$ as $R \to \infty$.

Therefore, $\int_{C_2} f$, $\int_{C_4} f \to 0$, so, by the Residue theorem,

$$\int_{-\infty}^{\infty} f(x)\, dx = \frac{1}{1 + e^{2\pi \xi}} \cdot \lim_{R \to \infty} \int_{C_R} f = \frac{1}{1 + e^{2\pi \xi}} \cdot 2e^{\pi \xi}.$$

2024 Fall.

10. (10 pts) Let $f, f_n : [a,b] \to \mathbb{R}$, $n \in \mathbb{N}$, be functions such that $(f_n)_{n\in\mathbb{N}}$ converges uniformly to f. Prove or disprove the following.

(a) *If f_n is Riemann integrable for all $n \in \mathbb{N}$, then f is Riemann integrable.*

(b) *If f and f_n, $n \in \mathbb{N}$, are all differentiable, then $(f_n')_{n\in\mathbb{N}}$ converges to f'.*

Proof. a). For each $n \in \mathbb{N}$, define $\varepsilon_n = \sup\{|f_n(x) - f(x)| \mid x \in [a,b]\}$.
Then, $\varepsilon_n \to 0$ as $n \to \infty$ and for all $x \in [a,b]$

$$f_n - \varepsilon_n < f(x) < f_n + \varepsilon_n, \quad \text{so that}$$

$$\int_a^b (f_n - \varepsilon_n) < \overline{\int_a^b} f < \overline{\int_a^b} (f_n + \varepsilon_n) \quad \text{and}$$

$$\int_a^b (f_n - \varepsilon_n) < \underline{\int_a^b} f < \underline{\int_a^b} (f_n + \varepsilon_n), \text{ and since } f_n \text{ is integrable,}$$

so $\underline{\int_a^b} f_n = \overline{\int_a^b} f_n = \int_a^b f_n$, therefore

$$\overline{\int_a^b} f - \underline{\int_a^b} f < 2\varepsilon_n \cdot [b-a] \to 0 \text{ as } n \to \infty,$$

So f is Riemann-integrable.

b) Let $f_n : \mathbb{R} \to \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sin(nx)$.

Then, $f_n \to 0$ uniformly. but, $f_n' = \cos(nx)$ does not converges to 0,
So it disproves b).

[2024 Fall,
12. Let $p: X \to Y$ be a quotient map
Show that if Y is connected and $p^{-1}[\{y\}]$ is connected for each $y \in Y$,
then X is connected.
Proof. Suppose that X is disconnected, with a separation $\{U, V\}$.
Given $y \in Y$, $p^{-1}[\{y\}] \subseteq X$ is connected, so either

$$p^{-1}[\{y\}] \subseteq U \text{ or } p^{-1}[\{y\}] \subseteq V.$$

(If not, then $\{U, V\}$ is a separation of $p^{-1}[\{y\}]$)
Meanwhile, observe that

$$Y = p[X] = p[U \cup V] = p[U] \cup p[V]$$

Let $a \in p[U]$ be given. Then, $a = p(b)$ for some $b \in U$.
And, $p^{-1}[\{a\}]$ contains b, so $p^{-1}[\{a\}] \cap U \neq \emptyset$, therefore $p^{-1}[\{a\}] \subseteq U$
Thus, $p^{-1}[p[U]] \subseteq U$ and $U \subseteq p^{-1}[p[U]]$ generally, so
$U = p^{-1}[p[U]]$ is open. Since p is quotient map, $p^{-1}[p[U]] = U$ is open
implies that $p[U]$ is also open set, similarly $p[V]$ is open
Moreover, $p[U] \cap p[V] = \emptyset$ because suppose that

for some $a \in Y$, $a = p(b) = p(c)$ for some $b \in U$ and $c \in V$.
Then, $p^{-1}[\{a\}] = p^{-1}[p[\{b\}]] = p^{-1}[p[\{c\}]]$, but

Since $b \in p^{-1}[p[\{b\}]] \cap U$ and $c \in p^{-1}[p[\{c\}]] \cap V$,
$p^{-1}[\{a\}] \subseteq U$ and V, but it contradicts to $U \cap V = \emptyset$.
So, $\{p[U], p[V]\}$ is a separation of Y, it is contradiction ⊠

Similar problem.

Thm. Let $p: X \to Y$ be a continuous, onto, closed map such that:

for all $y \in Y$, $p^{-1}[\{y\}] \subset X$ is compact set. (Such p is called perfect)

If Y is compact, then X is compact.

Proof. Let $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X. For each $y \in Y$,

$$p^{-1}[\{y\}] \subset p^{-1}[Y] = X = \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha.$$

By the hypothesis, $p^{-1}[\{y\}]$ is compact, so that there is a finite index set $\Lambda_y \subset \Lambda$ such that $p^{-1}[\{y\}] \subset \bigcup_{\alpha \in \Lambda_y} \mathcal{U}_\alpha$.

Now, since p is closed map,

$$V_y = Y \setminus p\left[X \setminus \bigcup_{\alpha \in \Lambda_y} \mathcal{U}_\alpha\right]$$

is open set containing y.

12. Let $p: X \to Y$ be a closed continuous onto map s.t. $\forall y \in Y$, $p^{-1}[\{y\}]$ is compact.

If Y is compact, then X is compact. (Such p is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X. For each $y \in Y$,

$$p^{-1}[\{y\}] \subseteq \bigcup_{y \in Y} p^{-1}[\{y\}] \subseteq \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha$$

And by the assumption, $p^{-1}[\{y\}]$ is compact, therefore there is a finite index $\Lambda_y \subseteq \Lambda$ s.t.

$$p^{-1}[\{y\}] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i.$$

Now, since p is closed map,

$$V_y = Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} \mathcal{U}_i\right)\right]$$

is an open set of Y, containing y. Now, since Y is compact, for some $y_1, \dots, y_n$,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^{n} V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\left[\bigcup_{j=1}^{n} V_{y_j}\right] = \bigcup_{j=1}^{n} p^{-1}[V_{y_j}]$$

Meanwhile, $p^{-1}[V_y] = p^{-1}\left[Y \setminus p\left[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\right]\right] = X \setminus p^{-1}\left[p\left[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\right]\right] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i$,

therefore $Y \subseteq \bigcup_{j=1}^{n} \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i$

2024 Fall.

13. Let X be any space and Y be Compact Hausdorff space.

$f: X \to Y$ is Continuous map if and only if (T_2, Compact) the graph $G_f = \{(x, f(x)) \mid x \in X\}$ is closed.

proof. Let $f: X \to Y$ be a continuous map.

Let $(x,y) \in X \times Y \setminus G_f$ be given. Then, $y \neq f(x)$, so there exist disjoint open sets U, V

$$U \cap V = \phi, \quad y \in U, \quad f(x) \in V$$

Since $f(x) \in V \overset{\text{def}}{\Leftrightarrow} x \in f^{-1}[V]$ and f is continuous, $f^{-1}[V]$ is an open set in X.

Now, $(x,y) \in f^{-1}[V] \times U$

And, $(f^{-1}[V] \times U) \cap G_f = \phi$, because if $(x, f(x)) \in f^{-1}[V] \times U$, then

$f(x) \in V$ and $f(x) \in U$, it is contradiction to disjointness.

Therefore, $(x,y) \in f^{-1}[V] \times U \subseteq X \times Y \setminus G_f$, So proved. $\Rightarrow$

Conversely, let G_f is closed in $X \times Y$, Under the Product Topology.

Let $C \subseteq Y$ be a closed set in Y. Then, $G_f \cap (X \times C)$ is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because $P_X: X \times Y \to X : (x,y) \mapsto x$ is closed map if Y is compact. □

Note: for $g = f \times i: X \times Y \to Y \times Y : (x,y) \mapsto (f(x), y)$,

$$g^{-1}[\Delta(Y)] = \{(x,y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f$$ $\Leftarrow$

2024. Fall.

14. Let $S \subset \mathbb{R}^3$ be a smooth surface and $P \in S$ be a point.

2025. Spring.

1. Let F be any field, and let $V = M_{n\times n}(F)$, and let

$$T: V \to V;\ A \mapsto A + A^t$$

a). Show that $V = \ker T \oplus \operatorname{Im} T$.

b). Is T diagonalizable? Find all eigenvalues of T with multiplicities.

Proof. Let $ch(F) \neq 2$.

a). Given $A \in V$,

$$A = \frac{1}{2}(A - A^t) + \frac{1}{2}(A + A^t)$$

And, since $-\left(\frac{1}{2}(A-A^t)\right)^t = -\frac{1}{2}(A^t - A) = \frac{1}{2}(A - A^t)$ implies $\frac{1}{2}(A-A^t) \in \ker T$ and $\frac{1}{2}(A+A^t) = T(\frac{1}{2}A) \in \operatorname{Im} T$, so $V = \ker T + \operatorname{Im} T$.

Meanwhile, given $A \in \ker T \cap \operatorname{Im} T$,

$$A + A^t = 0 \quad \text{and} \quad A = B + B^t \text{ for some } B \in V.$$

But, $0 = A + A^t = B + B^t + B^t + B = 2\cdot(B + B^t) = 2\cdot A$,

so $A = 0$. It proves $\ker T \cap \operatorname{Im} T = \{0\}$, thus $V = \ker T \oplus \operatorname{Im} T$.

b). Since $A \in \ker T \iff A = -A^t$, so, all diagonal entries are zero and if the upper entries are chosen, then the lower entries are determined. So $(n^2 - n)/2$ is the dimension of $\ker A$.

Meanwhile, given $A \in \operatorname{Im} T$, $A = B + B^t$ for some $B \in V$ and

$$T(A) = T(B) + T(B^t) = B + B^t + B + B^t = 2\cdot A.$$

So, $\operatorname{Im} T \subseteq \ker(T - 2I)$. Conversely, given $A \in \ker(T - 2I)$, $A + A^t - 2A = A^t - A = 0$, so $A = A^t$. Meanwhile,

$$T(\tfrac{1}{2}A) = \tfrac{1}{2}A + \tfrac{1}{2}A^t = A, \text{ so } A \in \operatorname{Im} T. \quad \text{Thus, } \operatorname{Im} T = \ker(T - 2I).$$

By the dimension theorem,

$$\dim \operatorname{Im} T = \frac{n^2 + n}{2}.$$

Therefore, 0 and 2 are eigenvalues of T with multiplicity $\frac{n^2-n}{2}$, $\frac{n^2+n}{2}$ respectively, and diagonalizable. ∎

2025 Spring.

3.

a). Prove that a polynomial ring $F[x]$ over a field F is P.I.D.

b) Find $f(x) \in \mathbb{C}[x]$ such that $(f(x)) = (x^5+2x^4+3x^3+2x^2+x, x^6-1)$.

Proof.

a). Let $I \subseteq F[x]$ be an ideal. If $I = F[x]$ or $I = \{0\}$, then $I = (1)$ or (0).

Now, assume that $\{0\} \subsetneq I \subsetneq F[x]$. Consider a map

$$\deg : F[x] \to \mathbb{N} \cup \{0\} : f(x) \mapsto \deg f(x)$$

Let $m = \min\{\deg[I \setminus \{0\}]\}$, it is guaranteed by the Well-ordering Principle, and $m \geq 1$, because I cannot contain a unit, by the assumption

Now, choose $p(x) \in \deg^{-1}[\{m\}]$ Now, by the division algorithm,

$$f(x) = p(x)q(x) + r(x) \text{ for some } q, r \in \mathbb{C}[x], \text{ with } r = 0 \text{ or } \deg r < \deg p$$

But, since I is Ideal, $r(x) = f(x) - p(x)q(x) \in I$, so the minimality of degree of p gives that $r = 0$, so $f = pq \in I$. Therefore, $I \subseteq (p(x))$

And, $(p(x)) \subseteq I$ is trivial, so $(p(x)) = I$.

b). Since $x^6 - 1 = (x-1)(x^5+x^4+x^3+x+1) = (x-1)(x^2+x+1)(x^3+1)$

and $x^5+2x^4+3x^3+2x^2+x = x(x^4+2x^3+3x^2+2x+1) = x(x^2+x+1)^2$

So, the greatest common divisor is (x^2+x+1), so x^2+x+1 is the generating polynomial we desired, because $(g, h) = (\gcd(g, h))$ in P.I.D. ∎

Tebris

b). By the division,

$$x^6 - 1 = (x^5+2x^4+3x^3+2x^2+x)(x-2) + x^4+4x^3+3x^2+2x$$

$$x^5+2x^4+3x^3+2x^2+x = (x^4+4x^3+3x^2+x)(x-2) + 8x^3+6x^2+5x$$

※ 생각을 해보자, $\mathbb{C}$ 는 Algebraic Closure 이고, 즉 각 Polynomial 들은 linear factor 로 쪼개진다. 즉, 나머지정리를 5~6번은 써야하겠지

3번째에 2인 아닌것.

자리리 공통근을 똑같게 다 어쩌지.

Let $I \subseteq F[x]$ be an ideal. If $I = F[x]$, then $(1) = I$.

If $I = \{0\}$, then $I = (0)$. Now, assume I is non-trivial proper Ideal.

Let $p(x) \in I \setminus \{0\}$ be given with $\deg p(x) \leq \deg h(x)$ for all $h(x) \in F[x] \setminus \{0\}$.

(It is possible by well-ordering Principle)

Now, let $f(x) \in I$ be given. Then, by the division algorithm,

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4. (10 pts)

(a) Show that a finite abelian group is not cyclic if and only if it contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p.

(b) For positive integers d_1, d_2, d_3, d_4 such that d_1 divides d_2 and d_3 divides d_4, assume that $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$ is isomorphic to $\mathbb{Z}_{d_3} \times \mathbb{Z}_{d_4}$. Show that $d_1 = d_3$ and $d_2 = d_4$.

Proof. Notice that i given $n, m \in \mathbb{N}$,

$\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{nm} \iff \gcd(n,m)=1$, because;

if $\gcd(n,m)=1$, then $(1_n, 1_m) \in \mathbb{Z}_n \times \mathbb{Z}_m$ satisfies that

$|(1_n, 1_m)| = \operatorname{lcm}(|1_n|, |1_m|) = mn$, so $\langle (1_n, 1_m) \rangle = \mathbb{Z}_n \times \mathbb{Z}_m$, so cyclic, and

if $\gcd(n,m) \neq 1$, then for any $(a,b) \in \mathbb{Z}_n \times \mathbb{Z}_m$,

$|(a,b)| = \operatorname{lcm}(|a|, |b|) \leq \operatorname{lcm}(n,m) < mn/\gcd(n,m)$ so that $\mathbb{Z}_n \times \mathbb{Z}_m \ncong \mathbb{Z}_{nm}$.

a). By the fundamental theorem of finitely generated abelian group, finite abelian group $G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}$, for some $n_i \geq 2$.

By the Proposition above,

G is cyclic $\iff \gcd(n_1, n_2, \cdots, n_k) = 1$

So that, taking the negative, a) is proved.

b). Let $\varphi: \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \to \mathbb{Z}_{d_3} \times \mathbb{Z}_{d_4}$ be an isomorphism.

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5. Consider $K = \mathbb{Q}(\sqrt[4]{2}, i)$. Find the Norm

$$N_{K/\mathbb{Q}}(1+\sqrt[4]{2}+i).$$

Proof Since $[\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}] = 4$ and $[K:\mathbb{Q}(\sqrt[4]{2})] = 2$, by the first is obtained by the eisenstein and the second is given by $\pm i \notin \mathbb{Q}(\sqrt[4]{2})$, thus $[K:\mathbb{Q}] = 8$.
Moreover, the minimal polynomial of $\sqrt[4]{2}$ over $\mathbb{Q}$ is x^4-2 and
" i over $\mathbb{Q}$ is x^2+1,
which split as $x^4-2 = (x-\sqrt[4]{2})(x-\sqrt[4]{2}\cdot\xi)(x-\sqrt[4]{2}\xi^2)(x-\sqrt[4]{2}\xi^3)$
and $x^2+1 = (x+i)(x-i)$
where $\xi = i$. Since the $\mathbb{Q}$-Automorphisms of K must map a root to the conjugate, and the K contains all roots of x^4-2 and x^2+1, thus the $\sigma \in Aut(K/\mathbb{Q})$ is determined by

$$\sigma(\sqrt[4]{2}) = \sqrt[4]{2}\cdot i^k \quad (k=0,1,2,3) \text{ and } \sigma(i) = \pm i.$$

And, the K is the splitting field of x^4-2, $K/\mathbb{Q}$ is Galois.
Since $[K:\mathbb{Q}] = 8$, these are full elemets of $Gal(K/\mathbb{Q})$. Now,

$$N_{K/\mathbb{Q}}(1+\sqrt[4]{2}+i)$$

$$= (1+\sqrt[4]{2}+i)(1+\sqrt[4]{2}i+i)(1-\sqrt[4]{2}+i)(1-\sqrt[4]{2}i+i)(1+\sqrt[4]{2}-i)(1+\sqrt[4]{2}i-i)(1-\sqrt[4]{2}-i)(1-\sqrt[4]{2}i-i)$$
$$= (2i-\sqrt{2})(2i+\sqrt{2})(-2i-\sqrt{2})(-2i+\sqrt{2})$$
$$= (-4-2)(-4-2)$$
$$= 36. \quad \square$$

$(1+i)^2 = 1+2i-1 = 2i$
$(1-i)^2 = -2i$

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6. Prove or disprove that

If $f:[0,1]\to\mathbb{R}$ is uniformly continuous, then it is Lipschitz continuous.

Proof. Let $f:[0,1]\to\mathbb{R}; x\mapsto\sqrt{x}$. Since f is continuous and $[0,1]$ is compact, thus f is continuously uniformly. But, given $L>0$,

$$|f(0)-f(x)|=|\sqrt{x}|>L\cdot|x| \text{ when } x<\frac{1}{L^2}$$

Therefore, f does not satisfies the Lipschitz condition, so the given statement is false. ▨

Note: Given $P>0$, Consider $f(x)=x^P$. $(x\geq 0$

If $0<P\leq 1$, assume that $x\geq y\geq 0$. Put $h=x-y\geq 0$. Then, we want to show

$$x^P-y^P=(y+h)^P-y^P\leq h^P=(x-y)^P.$$

Given $\varepsilon>0$, take $\delta=$. Then, for all $x,y\in[0,1]$,

$$|x-y|<\delta=\varepsilon \quad\Rightarrow |f(x)-f(y)|\leq|\sqrt{x}-\sqrt{y}|=\frac{|x-y|}{\sqrt{x}+\sqrt{y}}$$

[2025. Spring.

7. Let $f:\mathbb{R}\to\mathbb{R}$ be a uniformly continuous function.
Prove that there exist constants C and D such that

$$|f(x)|\le C+D|x| \quad \text{for all } x\in\mathbb{R}.$$

Proof. Let $\varepsilon=1$. Then, there is $\delta>0$ such that

$$|x-y|\le\delta \Rightarrow |f(x)-f(y)|\le\varepsilon=1.$$

Now, take $C=|f(0)|+1$ and $D=1/\delta$.
Then, for each $x>0$, for some $k=0,1,\cdots, x-k\delta\in(0,\delta]$. Now,

$$\begin{aligned}|f(x)-f(0)| &\le |f(x)-f(x-\delta)+f(x-\delta)-f(x-2\delta)+\cdots+f(x-k\delta)-f(0)|\\ &\le |f(x)-f(x-\delta)|+|f(x-\delta)-f(x-2\delta)|+\cdots+|f(x-k\delta)-f(0)|\\ &\le (k+1)\le \frac{1}{\delta}\cdot x+1\end{aligned}$$

so $|f(x)|\le \frac{1}{\delta}x+1+|f(0)| = C+D|x|$.

And, if $x<0$, we can show that same result similarly, so proved. ☐

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8. Let $U = \{z \in \mathbb{C} \mid |z| = 1\}$ be the unit circle in the complex plane. Prove or disprove that:

Every continuous function on U can be approximated uniformly by polynomials in the variable $z \in \mathbb{C}$.

Pf. Let $f: U \to \mathbb{C}; z \mapsto 1/z$ be a continuous map.

Directly, for a curve $C: t \mapsto e^{it}$ $(0 \le t \le 2\pi)$,

$$\oint_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} ie^{it} = 2\pi i.$$

Now, suppose that there is a sequence of polynomials $\{p_n\}$ converges uniformly to f. Then, for each $n \in \mathbb{N}$, p_n is entire and so

$$\int_C p_n = 0 \quad \text{by the Cauchy-Goursat theorem,}$$

therefore

$$\int_C f = \int_C \lim_{n\to\infty} p_n = \lim_{n\to\infty} \int_C p_n = 0$$

But it contradicts to $\int_C f = 2\pi i$. ∎

2025. Spring.

9. (10 pts) Find general solutions $x = x(t)$ to the following second-order ODEs:

(a) $t^2x'' + tx' + x = 0, \quad t > 0;$

(b) $t^2x'' + 5tx' + 4x = 0, \quad t > 0.$

Justify your answers to each problem.

Proof.

a). Put $x = t^r$. Then,

$t^2 \cdot x'' + t \cdot x' + x = 0 \Rightarrow t^2 \cdot (r \cdot (r-1) \cdot t^{r-2}) + t \cdot r \cdot t^{r-1} + t^r$

$= r(r-1)t^r + r \cdot t^r + t^r$

$= t^r (r^2 + 1).$

Therefore, if $r = \pm i$, $x = t^{\pm i}$ satisfies a). If $r = i$, then

$x = t^i = (e^{\log t})^i = e^{i \log t} = \cos(\log t) + i \cdot \sin(\log t),$

And if $r = -i$, similarly,

$x = \cos(\log t) - i \cdot \sin(\log t)$

therefore, the general solution is

$x(t) = C_1 \cdot \cos(\log t) + C_2 \cdot \sin(\log t), \quad C_1, C_2 \in \mathbb{R}.$

b) Similarly, Put $x = t^r$, then we obtain that

$r(r-1) \cdot t^{r-2} \cdot t^2 + 5t \cdot r \cdot t^{r-1} + 4 \cdot t^r = t^r (r^2 - r + 5r + 4) = t^r (r+2)^2.$

So, $r = -2$, that is, $x = 1/t^2$ is a solution, and for $x = \log t / t^2$,

$t^2 \cdot \left(\frac{6 \cdot \log t - 5}{t^4} \right) + 5t \cdot \left(\frac{-2 \cdot \log t}{t^3} + \frac{1}{t^3} \right) + 4 \cdot \frac{\log t}{t^2}$

$= \frac{1}{t^2} (6 \log t - 5 - 10 \cdot \log t + 5 + 4 \log t) = 0,$ so the x is a solution.

therefore, the general solution is $x = C_1 \cdot \frac{1}{t^2} + C_2 \cdot \frac{\log t}{t^2}$ ⊠

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10. (10 pts) Let f be a real-valued function defined on the interval $[a, b]$. Define

$$\mathrm{TV}_{[a,b]}(f) = \sup \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})|,$$

where the supremum is taken over all possible subdivisions $a = x_0 < x_1 < \cdots < x_k = b$ of $[a, b]$. If $\mathrm{TV}_{[a,b]}(f) < \infty$, we say that f is of bounded variation over $[a, b]$.

(a) Let $f : [-1, 1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0, & x = 0, \\ x^2 \sin\left(\frac{1}{x^2}\right), & x \neq 0. \end{cases}$$

Is f of bounded variation on $[-1, 1]$? Justify your answer.

(b) Let f be of bounded variation on $[a, b]$ and differentiable on (a, b). Show that

$$\int_a^b |f'(x)| dx \leq \mathrm{TV}_{[a,b]}(f).$$

Proof. a). It is not of bounded variation. Define a partition of $[-1,1]$

$$P_n = \left\{-1, 1, \frac{1}{\sqrt{n\pi + \pi/2}}, \cdots, \frac{1}{\sqrt{\pi+\pi/2}}\right\}, \text{ for } n \in \mathbb{N}.$$

Then, $\sum_{i=1}^{n+2} |f(x_i) - f(x_{i-1})|$ where $x_i \in P_n$, with ordering, satisfies

$$\geq \sum_{i=1}^{n} \frac{1}{\left(\sqrt{i\pi + \frac{\pi}{2}}\right)^2} \geq \frac{1}{\pi} \cdot \sum_{i=1}^{n} \frac{1}{i} \to \infty \text{ as } n \to \infty,$$

So it is not.

b). Let $\varepsilon > 0$ be given. Then, there is a partition $P = \{x_0, \cdots, x_n\}$ on $[-1,1]$ such that

[2025. Spring.
11. Given a continuous map $f:\mathbb{R}\to\mathbb{R}$, show that

$$\Gamma_f:\mathbb{R}\to\mathbb{R}^2:x\mapsto(x,f(x))$$

is continuous.
Proof. Let $B\subset\mathbb{R}^2$ be a basis element, as $B=B_1\times B_2$ where $B_1,B_2\subseteq\mathbb{R}$ are open sets. Now,

$$\begin{aligned}\Gamma_f^{-1}[B_1\times B_2]&=\Gamma_f^{-1}[(B_1\times\mathbb{R})\cap(\mathbb{R}\times B_2)]\\&=\Gamma_f^{-1}[B_1\times\mathbb{R}]\cap\Gamma_f^{-1}[\mathbb{R}\times B_2]\\&=B_1\cap f^{-1}[B_2]\end{aligned}$$

So, since f is continuous, it is finite intersection of open sets so open. Therefore, Γ_f is continuous. ▱

[2025 Spring.
13. Let D^2 be a closed unit disc in $\mathbb{R}^2$ centered at the origin.
a). Show that the quotient space $\mathbb{R}^2/D^2$ is homeomorphic to $\mathbb{R}^2$
b). Let $U = \text{int } D^2$. Determine $\mathbb{R}^2/U \cong \mathbb{R}^2$.
Proof. Let $\sim$ be a relation defined on $\mathbb{R}^2$ generated by:

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow (x_1, y_1), (x_2, y_2) \in D^2.$$

Then, it is clearly equivalent relation, and $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$, by definition.
a). Define a map

$$f: \mathbb{R}^2 \to \mathbb{R}^2 : (x, y) \mapsto \begin{cases} (0,0) & \text{if } x^2+y^2 \le 1 \\ \left(1 - \frac{1}{\sqrt{x^2+y^2}}\right) \cdot (x, y) & \text{if } x^2+y^2 > 1 \end{cases}$$

Then, given $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow (x_1, y_1), (x_2, y_2) \in D^2 \text{ or } (x_1, y_1) = (x_2, y_2)$$

$$\Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$$

and, it is onto continuous map, so f induces a Homeomorphism

$$\tilde{f}: \mathbb{R}^2/D^2 \to \mathbb{R}^2 : [(x, y)] \mapsto f(x, y).$$

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14. For $a, b, c > 0$, consider the ellipsoid

$$E_{a,b,c} = \left\{ (x,y,z) \in \mathbb{R}^3 \;\middle|\; \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

a). If $a=b$, then $E_{a,a,c}$ is a surface of revolution (Proved.)

b) Give a parametrization of $E_{a,a,c}$. Compute the first fundamental form of this parametrization and the Christoffel symbols

c). Show that the curve $E_{a,b,c} \cap \{x=0\}$ is the image of a geodesic.

a). Let

$$X: (0,2\pi)\times(0,2\pi) \to \mathbb{R}^3 ; (t,\theta) \mapsto (a\cdot\cos t\cdot\cos\theta,\ a\cdot\cos t\cdot\sin\theta,\ c\cdot\sin t)$$

It covers $E_{a,a,c}$ excepts a line, so $E_{a,a,c}$ is a surface of revolution.

b). The parametrization X is given in a). And,

$X_t = (-a\cdot\sin t\cdot\cos\theta,\ -a\cdot\sin t\cdot\sin\theta,\ c\cdot\cos t)$

$X_\theta = (-a\cos t\cdot\sin\theta,\ a\cdot\cos t\cdot\cos\theta,\ 0)$. So the first fundamental form is

$E = a^2\sin^2 t + c^2\cos^2 t$, $F=0$, $G = a^2\cdot\cos^2 t$. $\begin{bmatrix} E & F \\ F & G \end{bmatrix}^{-1} = \begin{bmatrix} E^{-1} & \\ & G^{-1} \end{bmatrix}$.

$X_{tt} = (-a\cos t\cdot\cos\theta,\ -a\cos t\cdot\sin\theta,\ -c\cdot\sin t)$

$X_{t\theta} = (+a\sin t\sin\theta,\ -a\sin t\cos\theta,\ 0)$

$X_{\theta\theta} = (-a\cdot\cos t\cdot\cos\theta,\ -a\cos t\cdot\sin\theta,\ 0)$. Now,

$$\Gamma^t_{tt} = \langle X_{tt}, X_t\rangle g^{tt} + \langle X_{tt}, X_\theta\rangle g^{\theta t} = (a^2-c^2)\cdot\sin t\cdot\cos t / (a^2\sin^2 t + c^2\cos^2 t)$$

$$\Gamma^t_{\theta\theta} = \langle X_{\theta\theta}, X_t\rangle \cdot g^{tt} + \langle X_{\theta\theta}, X_\theta\rangle \cdot g^{\theta t} = a^2\sin t\cos t / (a^2\sin^2 t + c^2\cos^2 t)$$

$$\Gamma^\theta_{t\theta} = \langle X_{t\theta}, X_t\rangle g^{t\theta} + \langle X_{t\theta}, X_\theta\rangle \cdot g^{\theta\theta} = -a^2\sin t\cos t / a^2\cos t = -\tan t.$$

And, $\Gamma^t_{t\theta} = \Gamma^\theta_{tt} = \Gamma^\theta_{\theta\theta} = 0$.

[2025. Fall

1. a). Show that $\phi : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p'(0)$ is linear functional. 8m

b). Find a dual basis corresponding to $\{1, t, \cdots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{n} b_i x^i$ and $c \in \mathbb{R}$ be given.

Then, $\phi(cf+g) = (cf+g)'\big|_{t=0} = c \cdot f'\big|_{t=0} + g'\big|_{t=0} = c \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p^{(k)}(0) \;\middle|\; k = 0, 1, \cdots, n \right\}$$

Then, for each $k = 0, 1, \cdots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \quad \text{and} \quad \frac{1}{k!} \cdot \phi_k(t^r) = 0 \quad \text{if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

[2025. Fall]
2. Let $A \in M_{m\times m}(\mathbb{R})$.
a). For $n\in\mathbb{N}$, $\operatorname{rank} A^n \le \operatorname{rank} A$.
b). Let $L_A:\mathbb{R}^m \to \mathbb{R}^m : x \mapsto Ax$.
Show that for $n\ge 2$,

$\operatorname{rank} A^n = \operatorname{rank} A$ implies $\ker L_A \cap \operatorname{Im} L_A = \{0\}$.

Sol) a). Given $x \in \operatorname{Im} A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.
So, $x = A^n y = A^{n-1}Ay \in \operatorname{Im} A$, for $n\ge 2$. Thus $\operatorname{Im} A^n \subseteq \operatorname{Im} A$, it follows $\operatorname{rank} A^n \le \operatorname{rank} A$. If $n=1$, clear.
b). Given $x \in \ker A$, $A^n x = A^{n-1}Ax = A^{n-1}0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$
Since $\operatorname{rank} A^n = \operatorname{rank} A$, the dimension theorem gives $\operatorname{nullity} A^n = \operatorname{nullity} A$.
Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.
Now, let $x \in \ker L_A \cap \operatorname{Im} L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.
Then, $A^{n-1}\cdot x = A^n y = 0$, because $Ax = 0$, so $A^{n-2}\cdot Ax = 0$, and this implies $y \in \ker A^n$.
Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$. So proved. ∎

[Similar problem by 박노준]
Let A be a n by n nilpotent matrix with $\operatorname{rank} A = r$.
Prove that $A^{r+1} = 0$.
Proof. Since A is nilpotent, let $k \in \mathbb{N}$ be an integer such that

$A^{k-1} \neq 0$, and $A^k = 0$.

By the dimension theorem,

$\ker A^k = n$, and $\ker A = n - r$.

Moreover, $\dim \ker A^m < \dim \ker A^{m+1}$ for each $1 \le m \le k-1$,

since if for some $m \le k-1$, $\dim \ker A^m = \dim \ker A^{m+1}$ then it contradicts to $A^{k-1} \neq 0$ but $A^k = 0$

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3. Find all $a, b, c \in \mathbb{R}$ such that $A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix}$ is diagonalizable. [6m]

Sol). Since the Charaterstic polynomial of A is regardless of a, b, c,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

Calculating gives $= \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c).

[2025. Fall.

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that

$$\det A = \pm \det B.$$

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol) Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

and $\det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$

so, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal ∎.

Define a map

$$\varphi : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to \mathbb{R}^\times : (A, B) \mapsto (\det A \cdot \det B^{-1})^2$$

Then, $\ker \varphi = N$ and φ is homomorphism, onto $(0, \infty)$, because the det preserves multiplication, so is φ, and ontoness is given by: fix $B = I$, and $A = \mathrm{diag}(\sqrt{r}, 1, 1, \cdots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times). \quad \blacksquare$$

Comment: φ 를 먼저 찾고 시작하라, hom 임을 보이면 앞 계산과정이 단순해짐!

2025. Fall

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$. (1 pt)

a) G is cyclic.

b1. For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G,

Prof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y.

Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \bmod 21 \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g, $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \bmod n$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solution

[2025. Fall.
6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}.$$

22m

a). Prove that $\{a_n\}^*$ is closed.
b). Show $\lim_{n\to\infty} \sup a_n = \sup\{a_n\}^*$
c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0,1]$?
Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.
Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.
Let a be a limit point of $\{a_n\}^*$. Then, given $k \in \mathbb{N}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
For $k=1$, choose such a $p_1 \in B_1(a)\setminus\{a\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{|p-a|}(p)$, then $a_{p_1} \in B_1(a)$.

For chosen $a_{p_1}, \cdots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{1}{k}}(a)\setminus\{a\}$.
Then, similarly, there is $a_{p_k} \in B_{\frac{1}{k}}(a)$ such that $p_k > p_{k-1}$,

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point a is contained in $\{a_n\}^*$, so it is closed. ∎
b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup\{a_n\}^* = \max\{a_n\}^* = a \text{ for some subsequential limit } a.$$

So, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < a + \varepsilon$ (since $a = \max\{a_n\}^*$)
and there exists infinitely many terms in $\{a_n\}$ such that $a - \varepsilon < a_n$ (since $a_{n_k} \to a$)
Thus $\lim_{n\to\infty} \sup a_n = a$.

c). Let b_n be defined as :

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \cdots$$

Then, given $b \in [0,1]$, there exists a subsequence in $\{b_n\}$ converges to b, because: for any $\varepsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \varepsilon,$$

So we can construct the subsequence. ∎

[2025, Fall.
7. Let E be a nonempty bounded subset of a Metric space (X, d). 10m
Define $\operatorname{diam} E := \sup\{d(x,y) \mid x, y \in E\}$

a). Prove $\operatorname{diam} E = \operatorname{diam} \overline{E}$
b). If E is compact, then $d(x', y') = \operatorname{diam} E$ for some $x', y' \in E$.
Proof.
a). Since $E \subset \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$ is clear.
To show the inverse inequality, using the property of supremum.,
Given $\varepsilon > 0$, there exists $x, y \in \overline{E}$ such that

$$\operatorname{diam} \overline{E} - \frac{\varepsilon}{2} < d(x,y) \leq \operatorname{diam} \overline{E}$$

And, since $\overline{E}$ is closure, there are $x', y' \in E$ such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now, $\operatorname{diam} \overline{E} < d(x,y) + \frac{\varepsilon}{2} < d(x,x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$, So proved.] 7m.
b), Since E is compact set, so $E \times E$ is also compact set.
Since the metric function d is continuous, it preserves compactness,
so $d[E \times E] \subseteq \mathbb{R}$ is compact set. Therefore,

$$\operatorname{diam} E = \max d[E \times E] \in d[E \times E],$$

So there is $(x', y') \in E \times E$ such that $d(x', y') = \max d[E \times E]$] 3m.

[2025. Fall.

8. Let $a>1$. Evaluate [5m]

$$\lim_{t\to 0+}\int_t^{at}\frac{\cos x}{x}\,dx$$

Proof. Consider the taylor expansion

$$\cos x=\sum_{n=0}^{\infty}(-1)^n\cdot\frac{x^{2n}}{(2n)!}$$

Therefore $\frac{\cos x}{x}=\frac{1}{x}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots$

Now, $\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\int_t^{at}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots\,dx$

$$=\ln a+\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}$$

(It is possible, because $\sum(-1)^n\cdot x^{2n-1}/(2n)!$ converges for all $x\in(0,)$

Now, $\lim_{t\to 0+}\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{2t^{2n-1}}{(2n)!}\,dx=0$, so

the given limit is $\ln a$.

Revised in next.

Observation for $\lim_{t\to 0+}\int_t^{at}\frac{\cos x}{x}\,dx$.

As we know, the power series representation

$$\frac{1}{x}\cdot\cos x=\frac{1}{x}\cdot\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}$$

is valid for all $x\neq 0$. Therefore, for any $t>0$,

$$\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

$$=\int_t^{at}\frac{1}{x}\,dx+\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

Meanwhile, the series converges uniformly in $[t,at]$ for small enough $t>0$, specifically, $t<\frac{1}{a}$, then for each $n\in\mathbb{N}$,

$$\left|(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\right|\le\left|\frac{1}{(2n)!}\right| \quad \text{and} \quad \sum_{n=1}^{\infty}\frac{1}{(2n)!}<\infty,$$

therefore the Weierstrass M-test gives the condition. $\frac{1}{2n}x^{2n}$

Thus, we can exchange the integral to the sum,

$$\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}(-1)^n\cdot\left(\frac{1}{2n\cdot(2n)!}\cdot(a^{2n}t^{2n}-t^{2n})\right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\lim_{t\to 0+}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{1}{2n\cdot(2n)!}\cdot t^{2n}(a^{2n}-1)$$

$$=\sum_{n=1}^{\infty}\lim_{t\to 0+}\quad //$$

$$=\sum_{n=1}^{\infty}0$$

$$=0$$

Consequently, the given integral is

$$\int_t^{at}\frac{1}{x}\,dx=\ln at-\ln t=\ln a.$$

Note: $\frac{\cos x}{x}=\frac{1}{x}+\frac{\cos x-1}{x}$, $\left|\int_t^{at}\frac{\cos x-1}{x}\,dx\right|\le\int_t^{at}\frac{x}{2}\to 0$ as $t\to 0$.

2025, Fall.

9. (10 pts) Consider the following differential equation with independent variable t:

$$(1+t^2)y'' + ty' + y = 0, \qquad \begin{cases} y(0) = 0, \\ y'(0) = 1. \end{cases}$$

Assume that a solution y has a power series expansion of the form

$$y(t) = \sum_{n=0}^{\infty} a_n t^n.$$

$a_0 + a_1 t + a_2 t^2 + \cdots$

(a) Determine the first five coefficients: a_0, a_1, a_2, a_3, a_4.

(b) Prove that the power series $y(t)$ converges for $|t| < 1$.

Proof. a). Since $y'(t) = \sum_{n=1}^{\infty} n a_n t^{n-1} = \sum_{n=0}^{\infty} (n+1)\cdot a_{n+1} \cdot t^n$ and

$y''(t) = \sum_{n=1}^{\infty} n(n+1)\cdot a_{n+1} \cdot t^{n-1} = \sum_{n=0}^{\infty} (n+1)(n+2)\cdot a_{n+2} \cdot t^n$,

$$(1+t^2)y'' + ty' + y = \sum_{n=0}^{\infty} (n+1)(n+2)\cdot a_{n+2} \cdot t^n + \sum_{n=2}^{\infty} n(n-1)\cdot a_n t^n + \sum_{n=1}^{\infty} n a_n t^n + \sum_{n=0}^{\infty} a_n t^n$$

$$= 2\cdot a_2 + a_0 + \left(6\cdot a_3 + 2a_1\right)\cdot t + \sum_{n=2}^{\infty} \left((n+1)(n+2)a_{n+2} + n(n-1)a_n + n a_n + a_n\right) t^n \Big)$$

And since $y(0)=0$, $a_0 = 0$ and $y'(0)=1$, $a_1 = 1$.

So that $a_2 = 0$, $a_3 = -\frac{1}{3}$, and

$3\cdot 4\cdot a_4 + 2a_2 + a_0 = 0$, so $a_4 = 0$. So the fives are determined.

b). Since $(n+1)\cdot(n+2)\cdot a_{n+2} + (n^2+1)\cdot a_n = 0$ for each $n \in \mathbb{N}$,

for each $k \in \mathbb{N}$, $a_0 = 0 = a_{2k}$. And, for odd $n \in \mathbb{N}$,

$$|a_{n+2}| = \frac{n^2+1}{(n+1)\cdot(n+2)} \cdot |a_n| = \frac{n^2+1}{n^2+3n+2} \cdot |a_n| < |a_n|,$$

and since $|a_3| = \frac{1}{3}$, $|a_n| < \frac{1}{3}$ for all odd n, so that

given $t \in (-1, 1)$,

$$\sum_{n=0}^{\infty} |a_n t^n| \le \sum_{n=0}^{\infty} |t|^n < \infty, \quad \text{so } \sum a_n t^n \text{ converges.} \quad \square$$

2025. Fall.

10. (10 pts)

(a) Show that for all $\xi \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx = e^{-\pi \xi^2}.$$

Proof. Let C_R be a contour as composition of; for $\xi > 0$,
$C_1: t \mapsto t : (-R \le t \le R)$, $C_2: t \mapsto i\cdot\xi t + R$ $(0 \le t \le 1)$
$C_3: t \mapsto t + i\xi$ $(-R \le t \le R)$, $C_4: t \mapsto i\xi t - R$ $(0 \le t \le 1)$,

$$C_R = C_1 * C_2 * (-C_3) * (-C_4).$$

And let $f(z) = e^{-\pi z^2}$

Since f is analytic, by the Cauchy-Goursat theorem,
$\int_{C_R} f = 0$, for any $R > 0$. And, observe that

$$\left|\int_{C_2} f\right| = \left|\int_0^1 e^{-\pi(-\xi^2 x^2 + 2iR\xi x + R^2)} \cdot i\xi\, dx\right|$$

$$\le \int_0^1 \left| e^{\xi^2 \pi x^2 - \pi R^2 - \pi\cdot 2iR\xi x} \cdot i\xi \right| dx$$

$$= \xi \int_0^1 e^{\xi^2 \cdot \pi \cdot x^2 - \pi R^2}\, dx \le \xi \cdot e^{\pi\xi^2 - \pi R^2} \to 0 \text{ as } R \to \infty.$$

And, similarly, $\int_{C_4} f \to 0$. Meanwhile,

$$\int_{C_3} f = \int_{-R}^{R} e^{-\pi\cdot(x^2 + 2xi\xi - \xi^2)} \cdot dx = e^{\pi\xi^2} \cdot \int_{-R}^{R} e^{-\pi x^2 - 2\pi i x\xi}\, dx.$$

Therefore,

$$0 = \lim_{R\to\infty} \int_{C_R} f = \int_{-\infty}^{\infty} e^{-\pi x^2} dx - e^{\pi\xi^2} \cdot \int_{-\infty}^{\infty} e^{-\pi x^2} \cdot e^{-2\pi i x\xi}\, dx$$

and since $\int_{-\infty}^{\infty} e^{-\pi x^2} = 1$, so proved, taking $e^{-\pi\xi^2}$ to both terms.

In case of $\xi < 0$, similar.

(b) Define the theta function

$$\theta(t) := \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}, \qquad t > 0.$$

Using the part (a), show that for all $t > 0$,

$$\theta(t) = \frac{1}{\sqrt{t}}\,\theta\left(\frac{1}{t}\right).$$

Prof.

2025. Fall.

11. (10 pts.) Let X be a topological space, and $A \subset X$ is a finite subset. Determine whether the closure $\overline{A}$ is also a finite set. (Give a proof or a counter-example.)

Proof. Let $X = \mathbb{Q}$ with indiscrete topology, and $A = \{0\}$.
Since the closure is the smallest closed set containing them, so there $\overline{A} = \mathbb{Q}$, so, A is finite but $\overline{A}$ is not. □

Note: $\supset$ or $?$.

[2025. Fall.

12. Define an equiv. rel on $\mathbb{R}$ as $x \sim y \iff x - y \in \mathbb{Q}$.

" as $x \mathcal{R} y \iff x = y$ or $x, y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\mathcal{R}$.

c). In $\mathbb{R}/\approx$, determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. a). Let $\mathcal{U} \subseteq \mathbb{R}/\sim$ be an open set. Then, for the canonical quotient map $\pi : \mathbb{R} \to \mathbb{R}/\sim$; $x \mapsto [x]_\sim$, $\pi^{-1}[\mathcal{U}] \subseteq \mathbb{R}$ is open. Given $y \in \mathbb{R}$, $[y] = y + \mathbb{Q}$ is dense in $\mathbb{R}$, so $[y] \cap \pi^{-1}[\mathcal{U}] \neq \emptyset$. But, since

$$\pi^{-1}[\mathcal{U}] = \{x \in \mathbb{R} \mid \pi(x) = [x] \in \mathcal{U}\} = \bigcup_{[x] \in \mathcal{U}} [x] = \bigcup_{[x] \in \mathcal{U}} [x],$$

it implies that $y \in \pi^{-1}[\mathcal{U}]$, so $\mathbb{R} = \pi^{-1}[\mathcal{U}]$.

Therefore, if $\mathcal{U} \subseteq \mathbb{R}/\sim$ is non-empty open set, then $\pi^{-1}[\mathcal{U}] = \mathbb{R}$, so that $\mathcal{U} = \mathbb{R}/\sim$. The topology on $\mathbb{R}/\sim$ is $\{\emptyset, \mathbb{R}/\sim\}$. ∎

b).

[2025. Fall.
13. Consider a regular plane curve $C: I \to \mathbb{R}^2$.
a). Give a formula for the normal n to C, and define the signed curvature.
b). Explain the relation between the total angle change and the signed curvature.
c). Assume C is a closed plane curve parametrized by arc-length, and $I=[0, L]$.
Prove or disprove that: there are $t, t' \in I$ such that $C'(t) = -C'(t')$.
and if $t=0$, there is such a $t' \in I$?
Moreover, show that $\int_0^L |k(s)| ds \geq 2\pi$ where $k(s)$ is the signed curvature.
Proof. Denote $C(t) = (x(t), y(t))$.
a). The tangent vector is

$$T(t) = \frac{C'(t)}{|C(t)|} = \frac{(x'(t), y'(t))}{\sqrt{x'(t)^2 + y(t)^2}}$$

So, the normal vector is

$$N(t) = \frac{(-y'(t), x'(t))}{\sqrt{x'(t)^2 + y(t)^2}}$$

And, since the curvature $k(s(t))$ satisfies $T'(t) = k(t) \cdot N(t)$,

$$k(t) = k(t) \langle N(t), N(t) \rangle = \langle T'(t), N(t) \rangle$$

Pebaris.

Denote $C = (x, y)$. Then,

$$T = (x', y') / |(x', y')| = \frac{(x', y')}{\sqrt{x'^2 + y'^2}}$$

is the tangent vector, and the normal vector is

$$N = \frac{(-y', x')}{\sqrt{x'^2 + y'^2}}.$$ Moreover, the signed curvature k satisfies $kN = T'$,

So $k = k \cdot \langle N, N \rangle = \langle T', N \rangle \cdot |C'(t)|^{-1}$ Meanwhile,

$$T' = \frac{(x'', y'')}{\sqrt{x'^2 + y'^2}} + (x', y') \cdot \left(-\frac{1}{2} \cdot \frac{2x'x'' + 2y'y''}{(x'^2 + y'^2)^{\frac{3}{2}}} \right)$$ so

$$k = \frac{1}{x'^2 + y'^2} \cdot (x'y'' - x''y') \cdot \frac{1}{\sqrt{x'^2 + y'^2}}.$$

TSNU. Qual.

Prove that $\lim_{b\to\infty} \int_0^b \frac{\sin x}{x}\,dx$ exists.

Proof. First, let $a_n = n\pi$ $(n\in\mathbb{N})$. Then, a sequence $\left\{\int_0^{a_n} \frac{\sin x}{x}\,dx\right\}$ converges, because: $\left|\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x}\,dx\right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x}\,dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi} \frac{\sin x}{x}\,dx = \sum_{k=1}^{n} \int_{a_{k+1}}^{a_k} \frac{\sin x}{x}\,dx$ is alternating series, so converges.

2026. Spring.

1. Find a 2×2 matrix A over $\mathbb{C}$ such that

$$A^{2025}=\begin{bmatrix}2 & 0\\ 2 & 5\end{bmatrix}$$

Proof. Since $\det(tI-A)=(t-2)(t-5)$, $\begin{bmatrix}2 & 0\\ 2 & 5\end{bmatrix}$ is diagonalizable to

$\begin{bmatrix}2 & 0\\ 2 & 5\end{bmatrix}=P\begin{bmatrix}2 & 0\\ 0 & 5\end{bmatrix}P^{-1}$ where $P=\begin{bmatrix}3 & 0\\ -2 & 1\end{bmatrix}$

Therefore, $A=P\begin{bmatrix}2^{1/2025} & 0\\ 0 & 5^{1/2025}\end{bmatrix}P^{-1}$ is the matrix we desired. ☒

$$\begin{bmatrix}3 & 0\\ -2 & 1\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 5\end{bmatrix}\cdot\frac{1}{3}\cdot\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}=\frac{1}{3}\begin{bmatrix}6 & 0\\ -4 & 5\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}=\frac{1}{3}\cdot\begin{bmatrix}6 & 0\\ 6 & 15\end{bmatrix}=\begin{bmatrix}2 & 0\\ 2 & 5\end{bmatrix}.$$

$$\ker\begin{bmatrix}0 & 0\\ 2 & 3\end{bmatrix}=\operatorname{Span}\left\{\begin{pmatrix}3\\ -2\end{pmatrix}\right\},\quad \ker\begin{bmatrix}-3 & 0\\ 2 & 0\end{bmatrix}=\operatorname{Span}\left\{\begin{pmatrix}0\\ 1\end{pmatrix}\right\},$$

2026. Spring.

2. Let W be the subspace of $\mathbb{R}^4$ spanned by all solutions of

$$x_1 + x_2 - x_3 - x_4 = 0.$$

a). Find a basis for $W^\perp$.

b). Find $V_1 \in W$ and $V_2 \in W^\perp$ such that $(1,1,0,1) = V_1 + V_2$.

Proof Let $T: \mathbb{R}^4 \to \mathbb{R}; \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$. Then, $W = \ker T$.

Since $(\ker T)^\perp = \operatorname{Im} T^*$, because: given $\alpha \in \operatorname{Im} T^*$, and $\beta \in \ker T$,

$$\langle \alpha, \beta \rangle = \langle T^*(\gamma), \beta \rangle = \langle \gamma, T(\beta) \rangle = \langle \gamma, 0 \rangle = 0 \quad \text{for some } \gamma,$$

So that $\operatorname{Im} T^* \subseteq (\ker T)^\perp$, and since $\operatorname{rank} T^* = \operatorname{rank} T$ and $\dim (\ker T)^\perp = \operatorname{rank} T$, so $\operatorname{Im} T^* = (\ker T)^\perp$. Now, for

$$T^*: \mathbb{R} \to \mathbb{R}^4; x \mapsto \begin{pmatrix} x \\ x \\ -x \\ -x \end{pmatrix},$$

$\operatorname{Im} T^* = \operatorname{span}\begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} = (\ker T)^\perp$, so $\{(1,1,-1,-1)\}$ is the basis we desired.

b). Since $\{(1,1,-1,-1)\}$ spans $(\ker T)^\perp$, so orthogonal projection of $(1,1,0,1)$ onto $(\ker T)$ is

$$(1,1,0,1) - \frac{1}{4} \cdot 1 \cdot (1,1,-1,-1) = \left(\frac{3}{4}, \frac{3}{4}, \frac{1}{4}, \frac{5}{4}\right) \in (\ker T) = W$$

Therefore, $V_1 = \frac{1}{4}(3,3,1,5)$ and

$$V_2 = \left(\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right)$$

are we desired. ☒

2026. Spring.

3. Let V be an n-dimensional vector space over $\mathbb{C}$. 2nm

Let $T: V \to V$ be a linear operator such that $T^3 = T$.

a). Find all possible characteristic polynomials and corresponding minimal polynomial for T.

b). Prove $V = \ker T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.

$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$m(t) = t$ or $t-1$ or $t+1$ or $t(t-1)$ or $t(t+1)$ or $(t+1)(t+1)$ or $t(t-1)(t+1)$.

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3}$$ where r_1, r_2, r_3 are non-negative integers and $r_1 + r_2 + r_3 = n$.

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is

$$(t-1)(t+1)$$

and other cases are similar.

b). ~~Given $\alpha \in \ker T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$, so $\alpha - T\alpha^2 \in \ker T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover, if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$, Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = T T^2\beta = \alpha = 0$, so $V = \ker T \oplus \operatorname{Im} T$. ☒

Note: Consider a case of $T^2 + T + I = 0$. Does still hold $V = \ker T \oplus \operatorname{Im} T$?

[Modified.
Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.
a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.
b). Describe $T|_{\operatorname{Im} T^2}$.
c). Prove that T is diagonalizable $\Leftrightarrow \ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

So $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. So, $T\alpha = T^3\beta$, Multiflying T, then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$. Therefore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$, $\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$, So, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$, moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable. That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$,

$x^4 + 2x + 3$ over $\mathbb{F}_7$

2026 Spring

5. Let $f(x) = x^4 + 2x + 3 \in \mathbb{F}_7[x]$ and $K = \mathbb{F}_7[x]/(f)$.
Let $\varphi : K \to K ; a \mapsto a^7$.

a). Prove K is a field and φ is Automorphism.

b). Let $G = \langle \varphi \rangle \leq Aut(K)$. Determine the number of $a \in K$ such that $|Orb(a)| = 2$.

Proof

a). Since $f(0) = 3$, $f(1) = 6$, $f(2) = 2$, $f(3) = 6$, $f(4) = 1$, $f(5) = 1$, $f(6) = 2$,
so $f(x)$ has no linear factor. So, if $f(x)$ is reducible,

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) \text{ for some } a, b, c, d \in \mathbb{F}_7.$$

Then,
$$\begin{cases} a + c = 0 \\ ac + b + d = 0 \\ ad + bc = 2 \\ bd = 3 \end{cases}$$

If $a = 0$ or $c = 0$, then $a = c = 0$, so $0 = 2$.
If $b = 0$ or $d = 0$, then $0 = 3$,
Therefore, a, b, c, d are non-zeros.
Reduce the system:

$$\Rightarrow \begin{cases} b + d = a^2 \\ a(d - b) = 2 \\ bd = 3 \end{cases}$$

If $b = d$, then $b^2 = 3$, but $\sqrt{3} \notin \mathbb{F}_7$, so $b \neq d$.
And, $3 = 1 \cdot 3$ or $2 \cdot 5$ or $4 \cdot 6$, so
$a^2 = 4$ or 0 or 3. Since $\sqrt{3} \notin \mathbb{F}_7$ and $a \neq 0$, $a^2 = 4$, so $a = 2$ or 5,
with $\{b, d\} = \{1, 3\}$. But,
$2 \cdot (1 - 3) = 3 \neq 2$,
$2 \cdot (3 - 1) = 4 \neq 2$,
$5 \cdot (1 - 3) = 4 \neq 2$
$5 \cdot (3 - 1) = 3 \neq 2$
So contradiction. Therefore, f must be an irreducible.
Since $\mathbb{F}_7$ is a field, $\mathbb{F}_7[x]$ is P.I.D., so (f) is maximal, so $\mathbb{F}_7[x]/(f)$ is a field. And, since the characteristic of K is 7, for any $a, b \in K$,

$$(a + b)^7 = \sum_{k=0}^{7} \binom{7}{7-k} a^{7-k} b^k = a^7 + b^7$$

and $(ab)^7 = a^7 b^7$, so φ is Homomorphism. Moreover,
$\varphi(a) = 0 \iff a^7 = 0 \iff a = 0$, so $\ker \varphi$ is trivial, so φ is 1-1.
Now, since $[K : \mathbb{F}_7] = \deg f = 4$, so $\#K = 7^4$, finite, so φ is bijective.

b). Since every finite field is perfect, so K is Galois over $\mathbb{F}_q$,
So, $|Aut(K)| = \deg f = 4$,
Consider an Action

$$G \times K \to K : (\varphi^n, \alpha) \mapsto \varphi^n(\alpha).$$

So. $|Orb(\alpha)| = 2 \iff \varphi^2(\alpha) = \alpha$ and $\alpha \neq 0 \iff \alpha^{4q} = \alpha$ and $\alpha \neq 0$
So, the fixed field by $\langle \varphi^2 \rangle$, $K^{\langle \varphi^2 \rangle} \setminus \{0\}$ is we desired.

Revised in the next

Revised.

Prove that $f(x)=x^4+2x+3$ is irreducible over $\mathbb{F}_7$.

And describe the Galois group of $f(x)$.

1). $f(x)$ has no linear factor.

$f(0)=3\neq 0$. And, since $\mathbb{F}_7^\times$ is a group of order 6 under the multiplication.

Therefore, for any $a\in\mathbb{F}_7^\times$, $a^6=(a^3)^2=1$ implies $a^3=\pm 1$.

If $a^3=1$, then $a^4=a$, so $f(a)=a^4+2a+3=3(a+1)=0$ implies $a=-1$,

but $(-1)^3=-1=6\neq 1$, so $a^3\neq 1$.

If $a^3=-1$, then $a^4=-a$, so $f(a)=-a+3=0$ implies $a=4$,

but $4^3=64=1\neq -1$, so $a^3\neq -1$. Now, $f(x)$ has no root in $\mathbb{F}_7^\times$.

2). $f(x)$ has no factor of degree 2.

Suppose that $f(x)=(x^2+ax+b)(x^2+cx+d)$. Then, we have a system:

$$\begin{cases} a+c=0 \\ ac+b+d=0 \\ ad+bc=2 \\ bd=3. \end{cases} \Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} (d-b)^2=(b+d)^2-4bd=a^4-12=a^4+2 \\ a^2(d-b)^2=4 \end{cases}$$

$a\neq 0$, so $a^6=1$

$\Rightarrow a^6+2a^2=1+2a^2=4 \Rightarrow a^2=5$. But, $\{x^2 \mid x\in\mathbb{F}_7\}=\{0,1,4,2\}$,

so $\sqrt{a}\notin\mathbb{F}_7$.

Now, f is irreducible $\Rightarrow \mathbb{F}_7[x]/(f)$ is a field.

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7. Let $f:(a-\beta, a+\beta)\to\mathbb{R}$ be an analytic function given by

$$f(x)=\sum_{n=0}^{\infty} c_n(x-a)^n \qquad (a, c_n\in\mathbb{R}).$$

with radius of convergence $\beta>0$.

a). Derive the power series for

$$F(x)=\int_a^x f(t)\,dt.$$

b). Prove that F is analytic on $(a-\beta, a+\beta)$ and determine its radius of conver:

Proof. Let $f_N=\sum_{n=0}^{N} c_n(x-a)^n$.

a). Given $x\in(a-\beta, a+\beta)$, $f_N(x)$ converges and for all $|a-t|\le|a-x|$,

$$|c_n|\cdot|t-a|^n\le|c_n|\cdot|x-a|^n \text{ and } \sum_{n=0}^{\infty}|c_n|\cdot|x-a|^n \text{ converges}$$

So the weierstrass M-test gives $f_N\to f$ uniformly on $|a-t|\le|a-x|$.

Therefore, by the theorem for integrability of uniformly convergence functions,

$$\int_a^x f(x)\,dx=\int_a^x \lim_{N\to\infty} f_N(x)\,dx=\lim_{N\to\infty}\int_a^x f_N(x)\,dx=\sum_{n=0}^{\infty}\frac{c_n}{n+1}\cdot(x-a)^{n+1}$$

b). Since $\lim\limits_{n\to\infty}\sup\left(\frac{|c_n|}{n+1}\right)^{\frac{1}{n+1}}=\lim\limits_{n\to\infty}\sup|c_n|^{\frac{1}{n}}$,

the Radius of Convergence of F equals to f, and thus it is also analytic on $(a-\beta, a+\beta)$.

[2026 Spring]

8. Let $\mathbb{D}$ be the unit open disc in $\mathbb{C}$. Prove or disprove:

Every holomorphic bijection $f: \mathbb{D} \to \mathbb{D}$ has a point $w \in \mathbb{D}$ such that $f(w) = w$.

proof. Let $f: \mathbb{D} \to \mathbb{D}$ be a holomorphic bijection.

Suppose that for all $w \in \mathbb{D}$, $f(w) \neq w$. Then, a map

$$z \mapsto \frac{1}{f(z) - z} \quad (z \in \mathbb{D})$$ is well-defined,

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13. (10 pts.) Let $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ and X a topological space.

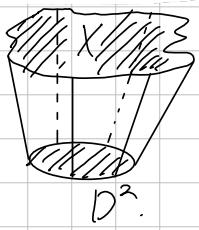
(a) Let $f, g : S^1 \to X$ be homotopic maps.
Show that the spaces $X \cup_f D^2$ and $X \cup_g D^2$ obtained by attaching a disc D^2 to X using f and g, respectively, are homotopy equivalent.

(b) Let $h : S^1 \to S^1$ be given by $h(e^{2\pi i s}) = e^{8\pi i s}$ for $0 \le s \le 1/2$ and $h(e^{2\pi i s}) = e^{-4\pi i s}$ for $1/2 \le s \le 1$. Show that $S^1 \cup_h D^2$ is contractible.

Proof. Define a map

$h: X \cup_f D^2 \to X \cup_g D^2$:

$$: [\cdot] \mapsto \begin{cases} [x]_g & \text{if } [\cdot] = [x]_f \text{ where } x \in X \\ [2r \cdot e^{i\theta}]_g & \text{if } 0 \le r \le \frac{1}{2} \\ [F(e^{i\theta}, 2-2r)]_g & \text{if } \frac{1}{2} \le r \le 1 \end{cases} \quad \text{if } [\cdot] = [r \cdot e^{i\theta}]_f$$



Observe: $i: S^1 (\subset D^2) \to D^2 : z \mapsto z$ is an inclusion.

$f \simeq g \Leftrightarrow$ A Homotopy $F: S^1 \times I \to X$ s.t. $F(z,0) = f(z)$ and $F(z,1) = g(z)$ exists.

$S^1 \not\simeq D^2$

$$\begin{array}{ccccc} X & \xleftarrow{g} & S^1 & \xrightarrow{f} & X \\ \downarrow{\scriptstyle C_X} & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle C_X} \\ X \cup_g D^2 & \longleftarrow & D^2 & \longrightarrow & X \cup_f D^2 \end{array}$$

2026. Fall.

1. Let A be an $n \times m$ real matrix and $b \in \mathbb{R}^n$. Prove the following.

   (a) If $\text{rank}(A) < \text{rank}([A\ b])$, then $Ax = b$ has no solution.

   (b) For any $b \in \mathbb{R}^n$, $A^TAx = A^Tb$ has a solution.

   (c) If columns of A are linearly independent, then $A^TAx = A^Tb$ has a unique solution.

Prof. a). Observe that:

$Ax=b$ has a solution $\Leftrightarrow$ for some $x_0 \in \mathbb{R}^m$, $Ax_0 = b$

$\Leftrightarrow b \in \text{Im}\, A$

$\Leftrightarrow \text{Span}(A_1, \cdots, A_m) = \text{Span}(A_1, \cdots, A_m, b)$

$\Leftrightarrow \text{rank}\, A = \text{rank}([A\ b])$

where A_i is ith column vector of A, so proved.

b. Given $b \in \mathbb{R}^n$, since $\text{Im}A \subseteq \mathbb{R}^n$ has finite dimension, so $\mathbb{R}^n = \text{Im}A \oplus (\text{Im}A)^\perp$, thus $b = Ax_0 + z$ for some $x_0 \in \mathbb{R}^m$, $z \in (\text{Im}A)^\perp$.

Now, $A^T(Ax_0 - b) = -A^Tz$ and since $z \in (\text{Im}A)^\perp$,

$\langle z, Ay \rangle = 0$ for all $y \in \mathbb{R}^m \Rightarrow \langle A^Tz, y \rangle = 0\ \forall y \Rightarrow A^Tz = 0$

So $A^TAx_0 - A^Tb = 0$, proved.

c). First, given $x \in \ker A$, $A^TAx = A^T0 = 0$. And, given $x \in \ker A^TA$,

$A^TAx = 0 \Rightarrow 0 = \langle 0, x \rangle = \langle A^TAx, x \rangle = \langle Ax, Ax \rangle = \|Ax\|^2 \Rightarrow Ax = 0$

thus $\ker A = \ker A^TA$, so Columns of A are independent implies by dimension thm, $\text{rank}\, A = m = \text{rank}\, A^TA \Rightarrow A^TA$ invertible $\Rightarrow x = (A^TA)^{-1}A^Tb$ is the unique solution. [$\langle \cdot \rangle$ is a standard inner product in $\mathbb{R}^n$]

[2026 Fall.

2. Let $A \in M_{4\times 4}(\mathbb{R})$ be a symmetric with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$.

Let $P = \frac{1}{2}\cdot\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$.

a) Prove that P is an orthogonal projection.

b) prove $tr(AP) \leq \lambda_1 + \lambda_2$

c) $\max_{\substack{x,y\in\mathbb{R}^4 \\ \|x\|=\|y\|=1 \\ \langle x,y\rangle = 0}} x^TAx + y^TAy = \lambda_1 + \lambda_2$.

a). By just calculation, $P^2 = P = P^T$, and in $\mathbb{R}$, $P^T = P^*$, so P is an orthogonal projection, rank $P = 2$, because the row vectors space has dimension 2.

b). Since P is orthogonal projection, $\mathbb{R}^4 = \mathrm{Im}\,P \oplus \ker P$, with $(\ker P)^\perp = \mathrm{Im}\,P$, and rank P = nulity $P = 2$. Since A is symmetric, there are four eigenvectors $\{\alpha_1, \cdots, \alpha_4\}$ corresponding to $A\alpha_i = \lambda_i\alpha_i$ $(i = 1, \cdots, 4)$ and $\{\alpha_1, \cdots, \alpha_4\}$ is orthonormal.

And, let $\{V_1, V_2\}$ be a basis for $\mathrm{Im}P$ and $\{V_3, V_4\}$ be a basis for $\ker P$, satisfying $\{V_1, \cdots, V_4\}$ is orthonormal.

Since the trace is invariant under the similar, for $B = (V_1 \cdots V_4)$,

$$tr(AP) = tr(B^TAPB) = V_1^TAPV_1 + \cdots + V_4^TAPV_4 = V_1^TAV_1 + V_2^TAV_2$$
$$= \langle AV_1, V_1\rangle + \langle AV_2, V_2\rangle$$

Now, since $V_1, V_2 \in \langle \alpha_1, \cdots, \alpha_4\rangle$ and $\|V_1\| = \|V_2\| = 1$ and $\langle V_1, V_2\rangle = 0$, $V_1 = a_1\alpha_1 + \cdots + a_4\alpha_4$ and $V_2 = b_1\alpha_1 + \cdots + b_4\alpha_4$ such that $a_1^2 + \cdots + a_4^2 = b_1^2 + \cdots + b_4^2 = 1$ and $a_1b_1 + \cdots + a_4b_4 = 0$. Now,

$$\langle AV_1, V_1\rangle + \langle AV_2, V_2\rangle = a_1^2\lambda_1 + \cdots + a_4^2\lambda_4 + b_1^2\lambda_1 + \cdots + b_4^2\lambda_4$$
$$= \lambda_1(a_1^2 + b_1^2) + \cdots + \lambda_4(a_4^2 + b_4^2)$$

But, for each $1 \leq i \leq 4$, $a_i^2 + b_i^2 \leq 1$ and $a_1^2 + \cdots + a_4^2 + b_1^2 + \cdots + b_4^2 = 2$, therefore $\leq \lambda_1 \cdot 1 + \lambda_2 \cdot 1 = \lambda_1 + \lambda_2$. It proves b)

c) Given $x, y \in \mathbb{R}^4$ such that $\langle x, y\rangle = 0$, with $\|x\| = \|y\| = 1$, $x^TAx + y^TAy = \langle Ax, x\rangle + \langle Ay, y\rangle \leq \lambda_1 + \lambda_2$, by the argument in b). And, take $x = \alpha_1$ and $y = \alpha_2$, we obtain c)

2026 Fall.

3. Find all prime and Maximal ideals of $\mathbb{Z} \times \mathbb{Z}$

Proof.

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4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T: G \to S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\ker T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

a). Given $g_1, g_2 \in G$, $T(g_1 g_2)(aH_1, bH_2) = (g_1 g_2 aH_1, g_1 g_2 bH_2)$

$= T(g_1) \circ T(g_2)(aH_1, bH_2)$

for each $a, b \in G$, so it preserves the operation.

And, $g \in \ker T \iff T_g = id \iff$ for all $a, b \in G$, $T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$

$\iff$ for all $a, b \in G$, $a^{-1} g a \in H_1$ and $b^{-1} g b \in H_2$

$\Rightarrow g \in H_1 \cap H_2$ (take $a = b = 1_G$)

b). By the first-isomorphism theorem,

$$G/\ker T \cong \operatorname{Im} T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the Codomain of T is finite set.

Now, $|G : H_1 \cap H_2| \leq |G : \ker T| \leq \# S_{G/H_1 \times G/H_2}$

because $\ker T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G : A|$ is finite, then

$$G/A \to G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \subseteq B$,

$g_1 A = g_2 A \iff g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \iff g_1 B = g_2 B$

and surjectiveness is clear.

2026 Fall,

5. Let $F = \mathbb{Q}(\sqrt{2+\sqrt{2}})$.

a). Prove that $F = \mathbb{Q}(\cos\frac{\pi}{8})$

b) Prove that F is normal extension of $\mathbb{Q}$ and find $Gal(F/\mathbb{Q})$.

Proof.

a). observe that $\cos\frac{\pi}{4} = \cos^2\frac{\pi}{8} - \sin^2\frac{\pi}{8} = 2\cos^2\frac{\pi}{8} - 1$, so that

$$\cos\frac{\pi}{8} = \frac{1}{\sqrt{2}}\cdot\sqrt{\frac{\sqrt{2}}{2}+1} = \sqrt{\frac{1}{2}+\frac{1}{\sqrt{2}}} = \frac{1}{2}\cdot\sqrt{2+\sqrt{2}}$$

Therefore, $F = \mathbb{Q}(\cos\frac{\pi}{8})$.

b). Let $\zeta = \exp(2\pi i/16)$. Then, $\cos\frac{\pi}{8} = (\zeta+\zeta^{-1})/2$. Let $E = \mathbb{Q}(\zeta)$.

Since ζ is 16th primitive root of unity, E is the splitting field over $\mathbb{Q}$ of

$$(x-\zeta)(x-\zeta^3)\cdots(x-\zeta^{15}) \in \mathbb{Q}[x]$$

and $Gal(E/F) \cong (\mathbb{Z}/16\mathbb{Z})^\times$, abelian group.

Since $(\zeta+\zeta^{-1})/2 \in \mathbb{Q}(\zeta)$, so that $\mathbb{Q} \subset F \subset E$. Meanwhile, observe

$$x^2 - (\zeta+\zeta^{-1})x + \zeta\cdot\zeta^{-1} = x^2 - (\zeta+\zeta^{-1})x + 1 = (x-\zeta)(x-\zeta^{-1}) \in F[x]$$

is irreducible over F because $\zeta, \zeta^{-1} \notin F$, so E is the splitting field of $x^2 - 2\cos\frac{\pi}{8}+1$ over F, so that $[E:F]=2$, thus $[F:\mathbb{Q}]=4$.

And, since $Gal(E/\mathbb{Q}) = \{\sigma_a \mid 1\le a\le 16,\ \gcd(a,n)=1\}$

where $\sigma_a(\zeta_n) = \zeta_n^a$, and $\sigma_{15}(\zeta+\zeta^{-1}) = \zeta^{-1}+\zeta$, therefore

$$F = \mathbb{Q}(\zeta+\zeta^{-1}) = E^{\langle\sigma_{15}\rangle}.$$

Now, since $Gal(E/F)$ is abelian, $F/\mathbb{Q}$ is normal extension and

$$Gal(F/\mathbb{Q}) = Gal(E^{\langle\sigma_{15}\rangle}/\mathbb{Q}) \cong Gal(E/\mathbb{Q})/\langle\sigma_{15}\rangle \cong \mathbb{Z}/4\mathbb{Z}. \quad \square$$

Note: $\Phi_n(x) = \prod_{\substack{1\le a\le n \\ \gcd(a,n)=1}} (x-\zeta^a)$ so that $x^n-1 = \prod_{d\mid n}\Phi_d(x)$.

2026 Fall.

6. Let $f:\mathbb{R}\to\mathbb{R}$ be a twice-differentiable function. Suppose that $\lim_{x\to\infty} f(x)=0$ and there exists $M>0$ such that $|f''(x)|\le M$ for all $x\in\mathbb{R}$. Prove that

$$\lim_{x\to\infty} f'(x)=0.$$

Proof. Let $a\in\mathbb{R}$ be fixed. Given $h>0$, by the Taylor theorem,

$$f(a+2h)=f(a)+f'(a)\cdot 2h+\frac{f''(\xi)}{2!}\cdot(2h)^2 \quad \text{for some } \xi\in(a,a+2h).$$
$$=f(a)+f'(a)\cdot 2h+f''(\xi)\cdot 2h^2.$$

Therefore,

$$f'(a)=\frac{f(a+2h)-f(a)}{2h}+f''(\xi)\cdot h, \quad \text{taking absolute value:}$$

$$|f'(a)|\le \frac{1}{2h}\cdot\left(|f(a+2h)|+|f(a)|\right)+h\cdot|f''(\xi)|$$
$$\le \frac{1}{2h}\cdot 2\cdot\sup_{x\ge a}|f(x)|+h\cdot\sup_{x\ge a}|f''(x)|=\sup|f|\cdot\frac{1}{h}+\sup|f''|\cdot h.$$

Now, take $h=\sqrt{\frac{\sup|f|}{\sup|f''|}}$ where the supremum is taken on $x\ge a$, then

$$|f'(a)|\le 2\cdot\sqrt{\sup|f|\cdot\sup|f''|}\le 2\sqrt{M}\cdot\sqrt{\sup|f|}$$

Finally, since $f(x)\to 0$ as $x\to\infty$, letting $a\to\infty$, $|f'(a)|\to 0$. ☐

Note: $\frac{A}{h}+Bh=2\sqrt{AB}$ if $h=\frac{\sqrt{A}}{\sqrt{B}}$.

[2026 Fall]

8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x<0 \\ f(x) & \text{if } 0 \le x \le 1 \\ f(1) & \text{if } x>1 \end{cases}$$

For each $n \ge 1$, define

$$\rho_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} \rho_n(x-y)\, \tilde{f}(y)\, dy, \quad x \in \mathbb{R}.$

a). Show that $T_n f \to f$ on $[0,1]$.

b). Determine $T_n f \to f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \iff -\frac{1}{n}+x \le y \le \frac{1}{n}+x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y)\, dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x,t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x)-f(t)| < \varepsilon$.

Now, given $x \in [0,1]$ and $n > \frac{1}{\delta}$,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot \tilde{f}(y)\, dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot f(x)\, dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot [\tilde{f}(y) - f(x)]\, dy \right| \\ &\le \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)|\, dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore, $\sup_{[0,1]} |T_n f - f| \to 0$ as $n \to \infty$, so converges uniformly.

[2026 Fall]

9. Let $f \in C([0,\pi])$ and define

$$(Tf)(x) := \int_0^\pi \min\{x,y\} \cdot f(y)\, dy, \qquad 0 \le x \le \pi$$

(a). Show that Tf is twice differentiable on $(0,\pi)$ and satisfies

$$(Tf)'(x) = \int_x^\pi f(y)\, dy, \qquad (Tf)''(x) = -f(x)$$

(b). Suppose that $f \neq 0$ and $\lambda \neq 0$ satisfy $Tf = \lambda f$.
Show that $\lambda > 0$ and that f satisfies

$$-f''(x) = \frac{1}{\lambda} \cdot f(x), \qquad 0 < x < \pi$$

With the boundary conditions $f(0) = 0$, $f'(\pi) = 0$.

(c) Determine all $\lambda > 0$ and all non-trivial functions f satisfying the differential equation and the boundary conditions in (b).

Proof. Observe that; given $0 \le x \le \pi$,

$$\int_0^\pi \min\{x,y\} \cdot f(y)\, dy = \int_0^x \min\{x,y\} \cdot f(y)\, dy + \int_x^\pi \min\{x,y\} \cdot f(y)\, dy$$

$$= \int_0^x y f(y)\, dy + x \cdot \int_x^\pi f(y)\, dy$$

(a). By the Fundamental Theorem of Calculus and f is Continuous, both $\int_0^x y f(y)\, dx$ and $x \cdot \int_x^\pi f(y)\, dy$ are differentiable, so

$$(Tf)'(x) = x f(x) + \int_x^\pi f(y)\, dy - x f(x) = \int_x^\pi f(y)\, dy$$

And, FTC again, $(Tf)'$ is differentiable, and $(Tf)''(x) = -f(x)$.

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a,

Proof. b). To show $\lambda > 0$, observe that: Since $Tf \in C^2$, so $(Tf)'' = -f$ implies $(Tf)'' = (\lambda f)'' = \lambda f'' = -f$, so that $\lambda f'' + f = 0$.

And, since the observation in a),

$$f(0) = \frac{1}{\lambda} \cdot Tf(0) = 0 \quad \text{and} \quad f'(\pi) = \frac{1}{\lambda} \cdot \int_{\pi}^{\pi} f(y)\,dy = 0.$$

If $\lambda < 0$, then since the characteristic of the equation is $(t^2 + \frac{1}{\lambda})$, therefore $f(x) = C_0 e^{x/\sqrt{-\lambda}}$ and $f(x) = C_1 \cdot e^{-x/\sqrt{-\lambda}}$ forms the basis of solution set, but these do not satisfies the boundary conditions. Thus $\lambda > 0$.

c). Since $\lambda > 0$, the general solution is: $t = \pm i/\sqrt{\lambda}$ implies

$$f(x) = C_0 \cdot \cos\left(\frac{x}{\sqrt{\lambda}}\right) + C_1 \cdot \sin\left(\frac{x}{\sqrt{\lambda}}\right), \quad C_0, C_1 \in \mathbb{R}.$$

Apply the boundary conditions, then,

$$f(0) = C_0 = 0,$$

$$f'(\pi) = -\frac{C_0}{\sqrt{\lambda}} \sin\left(\frac{\pi}{\sqrt{\lambda}}\right) + \frac{C_1}{\sqrt{\lambda}} \cos\left(\frac{\pi}{\sqrt{\lambda}}\right) = 0,$$

that is, $C_0 \sin\left(\frac{\pi}{\sqrt{\lambda}}\right) = C_1 \cos\left(\frac{\pi}{\sqrt{\lambda}}\right) = 0$,

Thus, $\lambda \in \left\{\left(\frac{2}{2k+1}\right)^2 \mid k = 0, 1, 2, \cdots\right\}$ and

$$f(x) = C \cdot \sin\left(\frac{x}{\sqrt{\lambda}}\right) \quad (C \in \mathbb{R}).$$

is the general solution. ☐

2026. Fall.

14. (10 pts.) Group the following spaces by their homotopy type. Justify your answer.

(a) The projective 3-space $\mathbb{R}P^3$ with one point deleted.

(b) The torus T with one point deleted.

(c) The infinite cylinder $S^1 \times \mathbb{R}$ with one point deleted.

(d) The projective plane $\mathbb{R}P^2$ with two points deleted.

(e) The 3-sphere S^3 with three points deleted.

Proof.

[Another problems. 26/09/08]

1. Let $T \in End(V)$ where V is fin. dim v.s over F with $ch(F) \neq 2$.

a) Prove that $V = \ker T \oplus \mathrm{Im}\, T$.

b) Prove that $T|_{\mathrm{Im}\, T} : \mathrm{Im}\, T \to \mathrm{Im}\, T : \alpha \mapsto T(\alpha)$ is an Automorphism.

c) In addition, if $T^3 = T$, prove T is diagonalizable and find all possible eigenvalue.

Proof.

a). Given $\alpha \in \ker T$, $T^2(\alpha) = T(T(\alpha)) = T(0) = 0$, so $\ker T \subseteq \ker T^2$.

Meanwhile, by the dimension theorem and by the assumption,

$$\dim V = \mathrm{rank}\, T + \mathrm{nullity}\, T = \mathrm{rank}\, T^2 + \mathrm{nullity}\, T = \mathrm{rank}\, T^2 + \mathrm{nullity}\, T^2$$

So, $\dim \ker T = \dim \ker T^2$, and $\dim \ker T \leq \dim V < \infty$ gives $\ker T = \ker T^2$. Let $\alpha \in \ker T \cap \mathrm{Im}\, T$. Then, $\alpha = T(\beta)$ for some $\beta \in V$ and $T(\alpha) = T^2(\beta) = 0$. So, $\beta \in \ker T^2 = \ker T$, it gives $\alpha = T(\beta) = 0$.

And, let $\{\alpha_1, \cdots, \alpha_k\}$ and $\{\alpha_{k+1}, \cdots, \alpha_n\}$ be the basis for $\ker T$ and $\mathrm{Im}\, T$. Then, $\{\alpha_1, \cdots, \alpha_n\}$ is linearly independent, because $\ker T \cap \mathrm{Im}\, T = \{0\}$, which proves $V = \ker T \oplus \mathrm{Im}\, T$.

b) Since $\mathrm{Im}\, T \supseteq \mathrm{Im}\, T^2$ and $\mathrm{rank}\, T = \mathrm{rank}\, T^2$, $\mathrm{Im}\, T = \mathrm{Im}\, T^2$ and so

$$T[\mathrm{Im}\, T] \subseteq \mathrm{Im}\, T$$

And, since $V = \mathrm{Im}\, T \oplus \ker T$, $\ker T|_{\mathrm{Im}\, T} = \{0\}$, so the restriction is injective, and also surjective by $\dim V < \infty$. So it is automorphism.

c). $T^3 = T$ means $x(x^2-1)$ is a multiple of the minimal polynomial, but $x(x-1)(x+1)$ is separable and splits over F, so the minimal polynomial also splits and is separable, thus T is diagonalizable. And the possible eigenvalues are $0, \pm 1$.

Note: Given $T : V \to V$, (not necessary finite-dimensional)

$V = \ker T \oplus \mathrm{Im}\, T$ if and only if $T|_{\mathrm{Im}\, T}$ is an Automorphism.

In such the case, given $\alpha \in V$,

$$\alpha = \underbrace{\left[\alpha - (T|_{\mathrm{Im}\, T})^{-1}(T(\alpha))\right]}_{\ker T} + \underbrace{(T|_{\mathrm{Im}\, T})^{-1}(T(\alpha))}_{\mathrm{Im}\, T}.$$

2. Let V, W be fin. dim real inn. prod space, and let $T: V \to W$ be injection.

a). Prove that $T^* \circ T: V \to V$ is invertible.

b). Prove that $T \circ (T^* \circ T)^{-1} \circ T^*: W \to W$ is an orthogonal proj of W onto $\mathrm{Im}\, T$.

Proof:

a). T is 1-1 $\Leftrightarrow \ker T = \{0\}$, since $\dim V < \infty$.

Meanwhile, observe that given $\alpha \in V$,

$$T^*T(\alpha) = 0 \Rightarrow \langle T^*T(\alpha), \alpha \rangle = 0$$

$$\Rightarrow \langle T(\alpha), T(\alpha) \rangle = 0$$

$$\Rightarrow T(\alpha) = 0$$

So $\ker T^*T \subset \ker T = \{0\}$, thus $\ker T^*T = \{0\}$.

Since $\dim V < \infty$, T^*T is 1-1 and onto.

b) observe that

$P^2 = T(T^*T)^{-1}T^*T(T^*T)^{-1}T^* = T(T^*T)^{-1}T^* = P$

$P^* = P$, since $(ABC)^* = (BC)^*A^* = C^*B^*A^*$ and $(T^{-1})^* = (T^*)^{-1}$.

So, $P^2 = P^* = P$, P is orthogonal projection.

And, since T is 1-1, so is T^*, thus T^* is onto, so

$$\mathrm{Im}\, P = T(T^*T)^{-1}T^*[W] = T[W] = \mathrm{Im}\, T.$$

So proved.

1 term.

1.2

3. Let G be a group of order 56. 9.5 80
Prove that G is non-simple.

proof. Notice that $56 = 2^3 \cdot 7$, and denote n_p is number of sylow p-subgroups.
By the sylow theorem, $n_7 \in \{1, 8\}$ and $n_2 \in \{1, 7\}$.
Either $n_7 = 1$ or $n_2 = 1$, then the sylow subgroup is normal, so we are done. Suppose that $n_7 = 8$ and $n_2 = 7$.
by the Lagrange's theorem, all non-identity elements in sylow 7-subgroup has order of 7, since $n_7 = 8$, there are $8 \cdot 6 = 48$ distinct elements of order 7 in G. Meanwhile, since $n_2 = 7$, let $P, Q \in Syl_2(G)$.
Then $|P \cap Q| \leq 4$, so there are $7 + 4 = 11$ district elemones whose order is multiple of 2, but $48 + 11 = 59 > 56$, it contrudicts. □

1 on

4. Let $R = \mathbb{Z}[x]/(6, x^2+x+1)$.

a) Decompose R as a product over $\mathbb{F}_2$ and $\mathbb{F}_3$.

b) Identify the two factors up to isomorphism as explicitly as possible.

c) Determine the number of units, nilpotent, and idempotent.

Proof: a), Notice that x^2+x+1 is irreducible, because $0, \pm 1$ are only possibilities for root, but $0^2+0+1 \neq 0$, $1^2+1+1 = 3 \neq 0$, $1+-1+1 = 1 \neq 0$. So, the degree 2 polynomial x^2+x+1 has no root in $\mathbb{Z}$, so irreducible.

Denote $f = x^2+x+1$. Meanwhile, observe that

$$(6, f) = (2, f) \cap (3, f) \quad \text{and} \quad (2, f) + (3, f) = (1) = R, \text{ since } 3-2=1.$$

Now, by the Chinese remainder theorem,

$$\mathbb{Z}[x]/(6, f) \cong \mathbb{Z}[x]/(2, f) \times \mathbb{Z}[x]/(3, f)$$

Since 2 is prime and f is irreducible, so $\mathbb{Z}[x]/(2, f) \cong \mathbb{F}_{2^2}$.

But, f is not irreducible over $\mathbb{Z}/3\mathbb{Z}$, as $f = (x+2)^2$, so

$$\mathbb{Z}[x]/(3, f) \cong (\mathbb{Z}/3\mathbb{Z})[x]/(\overline{x+2})^2.$$

Consequently, $\mathbb{Z}[x]/(6, f) \cong \mathbb{F}_{2^2} \times \left(\mathbb{F}_3[x]/(\overline{x}+2)^2\right)$

Since $\mathbb{F}_{2^2}^{\times}$ is a cyclic group, thus only nilpotent element in $\mathbb{F}_{2^2}$ is 0.

And, since $(\overline{x}+2)^2 = 0$ in $\mathbb{F}_3[x]/(\overline{x}+2)^2$, so

$(\overline{x}+2) \cdot a$ $(a = 0, 1, 2)$ are nilpotent elements in $\mathbb{F}_3[x]/(\overline{x}+2)^2$.

Now, all nilpotent in R is $(0, a\overline{x} + 2a)$, $a \in \mathbb{F}_3$.

35. Let I, J, K be ideals of R. (with 1)

(a) Prove that $I(J+K) = IJ + IK$ and $(I+J)K = IK + JK$.

(b) Prove that if $J \subseteq I$ then $I \cap (J+K) = J + (I \cap K)$.

R: Commute, with 1

If $I + J = R$, then $IJ = I \cap J$.

$(3, f) + (2, f) = \mathbb{Z}[x]$, since $3 - 2 = 1$.

$$\begin{aligned}(3, f) \cap (2, f) = (3, f)(2, f) &= [(3)+(f)] \cdot [(2)+(f)] \\ &= (3)(2) + (3)(f) + (2)(f) + (f)(f) \\ &= (6) + (3f, 2f, f^2) = (3, 2, f)(f) \\ &= (6) + (f) \\ &= (6, f).\end{aligned}$$

$(6, f) = (6) + (f) = (3)(2) + (f)$

5. Let $\alpha = \sqrt[4]{3}$ and let K be the splitting field of $x^4 - 3$ over $\mathbb{Q}$.

a). Prove $K = \mathbb{Q}(\alpha, i)$ and show $[K : \mathbb{Q}] = 8$.

b). Define $r, s \in Gal(K/\mathbb{Q})$ as

$r(\alpha) = i\cdot\alpha$, $r(i) = i$, $s(\alpha) = \alpha$, $s(i) = -i$. Prove $Gal(K/\mathbb{Q}) \cong D_8$.

c). Determine $K^{\langle r \rangle}$ and $K^{\langle s \rangle}$

Proof. a).

6. Let (f_n) be a sequence in $C^1([0,1])$ such that

$\sup_n \|f_n'\|_\infty \le 1$ and $\int_0^1 |f_n(x)|\,dx \to 0$. Prove $f_n \to 0$ uniformly.

Proof.

Additional Problems.

Let $A=\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$ over $\mathbb{R}$ or $\mathbb{C}$.

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}\begin{bmatrix} v_1 & v_2 & \cdots \end{bmatrix}$$

Find the characteristic polynomial, minimal polynomial, eigenvalues, spectrum. Decide whether A is diagonalizable over $\mathbb{R}$ or $\mathbb{C}$.

Proof. Denote χ_A and m_A as characteristic and minimal polynomial, respectively. First,

$$\chi_A = \det(tI-A) = (t-2)^2\cdot(t+1)^2.$$

And, since χ_A and m_A have same roots, so

$$m_A = (t-2)^i\cdot(t+1)^j \quad (i,j\in\{1,2\}).$$

Observe $(A-2I)\cdot(A+I) = \begin{bmatrix} 0 & 1 & & \\ & 0 & & \\ & & -3 & 1 \\ & & & -3 \end{bmatrix}\cdot\begin{bmatrix} 3 & 1 & & \\ & 3 & & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

And $(A-2I)^2\cdot(A+I)\neq 0$, $(A-2I)(A+I)^2\neq 0$, so Cayley-Hamilton theorem gives that $\chi_A = m_A$. And, the eigenvalues 2 and -1.

Since the minimal polynomial is not separable, thus it is not diagonalizable, both $\mathbb{R}$ and $\mathbb{C}$.

The eigenspaces are;

$$\ker(A-2I) = \operatorname{Span}\{e_1\},\quad \ker(A+I) = \operatorname{Span}\{e_3\}.$$

Another problem set, 1.

1. Let V be a finite-dim. inn. space, and let $P: V \to V$ be a linear map. Prove that if P is idempotent and normal, then it is Hermitian.

Proof. Observe that for any normal operator T, given $x \in V$,

$$T(x)=0 \Leftrightarrow \langle T(x), T(x)\rangle = 0 \Leftrightarrow \langle T^*(x), T^*(x)\rangle = 0 \Leftrightarrow T^*(x)=0.$$

And, since $P^*P = PP^*$,

$$(I-P)\circ(I-P^*) = I - P - P^* + P\circ P^* = I - P - P^* + P\circ P^* = (I-P^*)\circ(I-P)$$

and $(I-P)^* = I - P^*$, so $I-P$ is normal. Now, since $P^2 = P$, given $x \in V$,

$$P\circ(I-P)(x)=0 \Rightarrow P^*\circ(I-P)(x)=0 \Rightarrow P^* = P^*\circ P$$

and

$$(I-P)\circ P(x)=0 \Rightarrow (I-P^*)\circ P(x)=0 \Rightarrow P = P^*\circ P.$$

Therefore, $P = P^*$. ☑

6. Let T be a linear operator on an 8-dimensional complex vector space. Suppose $\chi_T(x)=(x-1)^5\cdot(x+1)^3$, $m_T(x)=(x-1)^2(x+1)$ and $\operatorname{rank}(T-I)=4$. Determine the Jordan Canonical form of T.

Proof. By the dimension theorem, $\dim \ker(T-I)=4$.

And, by the theorem for Jordan form, $\bigcup_{k=1}^{\infty}\ker(T-I)^k=\ker(T-I)^2$ since m_T has multiplicity 2 of $(x-1)$, and $\dim\ker(T-I)^2=5$, since χ_T has multiplicity 5 of $(x-1)$.

Now, we obtain a block for eigenvalue 1,

$$J_1(T)=\begin{bmatrix} I_3 & & \\ & 1 & 1 \\ & & 1 \end{bmatrix}=[1]\oplus[1]\oplus[1]\oplus J_2(1).$$

Meanwhile, $(x+1)$ has multiplicity 1 in m_T, therefore

$$J_{-1}(T)=\begin{bmatrix} -1 & & \\ & -1 & \\ & & -1 \end{bmatrix}.$$

Finally, we obtain the Jordan form

$$J_T=\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & 1 & & & \\ & & & & 1 & & & \\ & & & & & -1 & & \\ & & & & & & -1 & \\ & & & & & & & -1 \end{bmatrix}.$$

$\chi_T(c)=0$
$\Leftrightarrow \det(cI-T)=0$
$\Leftrightarrow \dim\ker(cI-T)\geq 1$

Note:

[Practice sets, by Lee.

1. (10 pts.) Let $A = \mathrm{diag}(1, 1, 3)$ and let $T : M_3(\mathbb{C}) \to M_3(\mathbb{C})$ be defined by $T(X) = AX - XA$.

(a) Find all eigenvalues of T and a basis for each eigenspace.

(b) Determine the characteristic and minimal polynomials of T. Is T diagonalizable?

Proof. a). Notice that: given $X=(x_{ij}) \in M_{3\times 3}(\mathbb{C})$,

$$AX - XA = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ 3x_{31} & 3x_{32} & 3x_{33} \end{bmatrix} - \begin{bmatrix} x_{11} & x_{12} & 3x_{13} \\ x_{21} & x_{22} & 3x_{23} \\ x_{31} & x_{32} & 3x_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -2x_{13} \\ 0 & 0 & -2x_{23} \\ 2x_{31} & 2x_{32} & 0 \end{bmatrix}$$

Therefore, $T(X) = AX - XA = \lambda X$ for some $\lambda \in \mathbb{C}$ and non-zero $X \in M_{3\times 3}(\mathbb{C})$

$\Leftrightarrow$

$$\begin{cases} \lambda = 0 \\ X \in \mathrm{span}\{e_{11}, e_{12}, e_{21}, e_{22}, e_{33}\} \setminus \{0\} \end{cases}$$

or $\begin{cases} \lambda = 2 \\ X \in \mathrm{span}\{e_{31}, e_{32}\} \setminus \{0\} \end{cases}$

or $\begin{cases} \lambda = -2 \\ X \in \mathrm{span}\{e_{13}, e_{23}\} \setminus \{0\} \end{cases}$

where $\beta = \{e_{ij} \mid 1 \le i, j \le 3\}$ is the standard basis for $M_{3\times 3}(\mathbb{C})$.

So, we obtained all eigenvalues and the basis for corresponding eigenspaces.

b). By observation in a), the matrix representation of T is with order $\beta = \{e_{11}, e_{12}, e_{21}, e_{22}, e_{33}, e_{31}, e_{32}, e_{13}, e_{23}\}$,

$$[T]_\beta = \begin{bmatrix} 0 & & & & & & & & \\ & 0 & & & & & & & \\ & & 0 & & & & & & \\ & & & 0 & & & & & \\ & & & & 0 & & & & \\ & & & & & 2 & & & \\ & & & & & & 2 & & \\ & & & & & & & -2 & \\ & & & & & & & & -2 \end{bmatrix}$$

, so it is already diagonal, and the characteristic is

$$\chi_T(t) = t^5 \cdot (t-2)^2 \cdot (t+2)^2$$

and since T is diagonalizable, the minimal polynomial is separable,

$$m_T(t) = t(t-2)(t+2)$$

□

Note: for $A = \mathrm{diag}(a_1, a_2, a_3)$, $T(e_{ij}) = (a_i - a_j) \cdot e_{ij}$.

2. (10 pts.) Let P and Q be orthogonal projections on a finite-dimensional real or complex inner product space V.

(a) Prove that PQ is an orthogonal projection if and only if $PQ = QP$.

(b) Under this condition, determine $\mathrm{im}(PQ)$ in terms of $\mathrm{im}\, P$ and $\mathrm{im}\, Q$.

a). Suppose that PQ is orthogonal projection. Then, PQ is Hermitian, so

$(PQ)^\dagger = Q^\dagger P^\dagger = QP = PQ$.

Conversely, suppose P and Q are Commute Then,

$(PQ)^2 = PQPQ = PPQQ = PQ$

and

$(PQ)^\dagger = Q^\dagger P^\dagger = QP = PQ$, so proved.

(Notice that: T is orthogonal projection $\Leftrightarrow T$ is linear and $T = T^2 = T^\dagger$).

b). Notice that

$\mathrm{Im}\, PQ = PQ[V] \subseteq P[V] = \mathrm{Im}\, P$, and since $PQ = QP$, Similarly,

$\mathrm{Im}\, QP = \mathrm{Im}\, PQ \subseteq \mathrm{Im}\, Q$. Thus, $\mathrm{Im}\, PQ \subseteq \mathrm{Im}\, P \cap \mathrm{Im}\, Q$.

Let $\alpha \in \mathrm{Im}\, P \cap \mathrm{Im}\, Q$. Then, $P\alpha = Q\alpha = \alpha$ and so that $\alpha = PQ\alpha \in \mathrm{Im}\, PQ$, thus $\mathrm{Im}\, PQ = \mathrm{Im}\, P \cap \mathrm{Im}\, Q$. ☒

3. (10 pts.) Let G be a group of order 45.

(a) Prove that G is abelian.

(b) Determine all possible isomorphism types of G.

Proof. a). By the Sylow theorem, the number of Sylow p-subgroup, n_p is determined as; $45 = 3^2 \cdot 5$, so

$n_3 = 1$ and $n_5 = 1$

Therefore, there exist two normal subgroup $H, K \trianglelefteq G$ with $|H| = 9, |K| = 5$. Moreover, by the Lagrange's theorem, $H \cap K = \{e\}$, since every non identity element of H has order of 3^1 or 3^2, and G has order of 1 or 5. Moreover, $HK \leq G$ and $|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 45$, so that $HK = G$.

And, since group of order of prime squares is abelian, H is abelian, and K is cyclic, so abelian. Now, let $x \in H$ and $y \in K$. Then, $[x,y] = x^{-1}y^{-1}xy \in H \cap K$ since both are normal so that $[x,y] = e$. That is, $xy = yx$. Now, given $a, b \in G = HK$, $a = x_1y_1$, $b = x_2y_2$ for some $x_1, x_2 \in H$ and $y_1, y_2 \in K$,

$ab = x_1y_1x_2y_2 = x_2y_2x_1y_1 = ba$, so G is abelian.

b). Let $\varphi : H \times K \to HK (= G)$; $(h,k) \mapsto hk$. It is Homomorphism, because

$\varphi((h_1,k_1)\cdot(h_2,k_2)) = \varphi(h_1h_2, k_1k_2) = h_1h_2k_1k_2 = h_1k_1h_2k_2 = \varphi(h_1,k_1)\cdot\varphi(h_2,k_2)$.

And, $(h,k) \in \ker\varphi \Rightarrow hk = e \Rightarrow k = h^{-1} \Rightarrow h, k \in H \cap K$, so $|\ker\varphi| = \{(e,e)\}$, injective. φ is surjective by $HK = G$. Now, $H \times K \cong HK = G$, so that

$G \cong C_3 \times C_5$ or $C_3 \times C_3 \times C_5$. ∎

4. (10 pts.) Consider the ring $R = \mathbb{F}_3[x]/(x^4 - 1)$.

(a) Express R as a product of fields by giving an explicit ring isomorphism.

(b) Find all maximal ideals of R and determine the number of units in R.

Proof. a). Notice that:

$$x^4-1=(x^2+1)(x+1)(x-1)$$

and x^2+1 is irreducible, since $\{0^2, 1^2, 2^2\} = \{0,1\}$ has no $-1=2$. And, since F_3 is a field, so $F_3[x]$ is an euclidean domain, therefore x^2+1, $x+1$, $x-1$ are mutually relatively prime implies these are comaximal mutually. Now, define

$$\varphi: F_3[x] \to \left(F_3[x]/(x^2+1)\right) \times \left(F_3[x]/(x+1)\right) \times \left(F_3[x]/(x-1)\right)$$
$$: f(x) \mapsto (f(x)+(x^2+1),\ f(x)+(x+1),\ f(x)+(x-1))$$

is surjective Ring Homomorphism, whose kernel is

$$\ker \varphi = (x^2+1)(x+1)(x-1) = (x^4-1)$$

Now, the irreducibility gives that

$$R = F_3[x]/\ker\varphi \cong \operatorname{Im}\varphi \cong F_{3^2} \times F_3 \times F_3.$$

b). Since $I \subset R$ is maximal ideal $\Leftrightarrow$ R/I is a field, so that

$I \subset F_{3^2} \times F_3 \times F_3$ is maximal $\Leftrightarrow$ $I = F_{3^2} \times F_3 \times \{0\}$

or $F_{3^2} \times \{0\} \times F_3$

or $\{0\} \times F_3 \times F_3$.

And, $(x,y,z) \in F_{3^2} \times F_3 \times F_3$ is unit $\Leftrightarrow$ $x,y,z \neq 0$

So the number of units is $8 \times 2 \times 2 = 32$. ☐

5. (10 pts.) Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $\alpha = \sqrt{2} + \sqrt{3}$.

(a) Find the minimal polynomial of α over $\mathbb{Q}$, and show that $K = \mathbb{Q}(\alpha)$.

(b) Find every field L such that $\mathbb{Q} \subsetneq L \subsetneq K$.

a). Since $\sqrt{2}+\sqrt{3} \in \mathbb{Q}(\sqrt{2},\sqrt{3})$, $\mathbb{Q}(\alpha) \subseteq K$. And,

$1 = 3-2 = (\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})$, so that $\sqrt{3}-\sqrt{2} = \frac{1}{\alpha} \in \mathbb{Q}(\alpha)$,

It implies that $(\sqrt{3}+\sqrt{2}+(\sqrt{3}-\sqrt{2}))/2 = \sqrt{3} \in \mathbb{Q}(\alpha)$, and

$(\sqrt{3}+\sqrt{2}-(\sqrt{3}-\sqrt{2}))/2 = \sqrt{2} \in \mathbb{Q}(\alpha)$, so $K = \mathbb{Q}(\alpha)$.

Meanwhile, observe that

$$x = \sqrt{2}+\sqrt{3} \Rightarrow (x-\sqrt{2})^2 = x^2 - 2\sqrt{2}x + 2 = 3$$

$$\Rightarrow (x^2-1)^2 = x^4 - 2x^2 + 1 = 8x^2 \Rightarrow x^4 - 10x^2 + 1 = 0$$

So that $x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$ has α as a root, and since $[\mathbb{Q}(\alpha):\mathbb{Q}] = [K:\mathbb{Q}] = [K:\mathbb{Q}(\sqrt{2})]\cdot[\mathbb{Q}(\sqrt{2}):\mathbb{Q}] = 4$, so that $x^4 - 10x^2 + 1$ is the minimal polynomial of α over $\mathbb{Q}$.

b) By the tower Lemma, every intermidate field $\mathbb{Q} \subsetneq L \subsetneq K$ satisfies that $[L:\mathbb{Q}] = 2$. Therefore, the subfields are

$\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, and $\mathbb{Q}(\sqrt{6})$.

And, by the Galois theorem, $Gal(K/\mathbb{Q}) \cong \mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$, but both have proper non-trivial subgroups at most three, so these are all subfields, we desired ☐

Note: Classic.

6. (10 pts.) Let $f : [0, 1] \to \mathbb{R}$ be continuous and suppose that

$$\int_0^1 f(x)x^{2n}\,dx = 0 \qquad (n = 0, 1, 2, \ldots).$$

Prove that f is identically zero.

Proof. Let $g:[0,1]\to\mathbb{R}; x\mapsto f(\sqrt{x})$. Since $x\mapsto\sqrt{x}$ is continuous, so g is also continuous on $[0,1]$.

Therefore, by the Weierstrass-Approximation theorem, there exists a sequence of polynomial $\{P_n\}$ such that

$$P_n(x)\to g(x) \text{ uniformly on } [0,1],$$

that is, $\sup|P_n(x)-g(x)|\to 0$ as $n\to\infty$, and so that

$$\sup_{x\in[0,1]}|P_n(x^2)-g(x^2)| = \sup_{x\in[0,1]}|P_n(x^2)-f(x)|\to 0, \text{ that is,}$$

$P(x^2)\to f(x)$ uniformly on $[0,1]$.

Now, by the theorem for integral of uniformly convergence functions,

$$\int_0^1 (f(x))^2\,dx = \int_0^1 f(x)\cdot\lim_{n\to\infty}P_n(x^2)\,dx = \lim_{n\to\infty}\int_0^1 f(x)\cdot P_n(x^2)\,dx$$

(because since f is bounded, because $f[[0,1]$ is compact and so that

$$\sup_{x\in[0,1]}|f(x)^2-f(x)\cdot P_n(x)| \le \sup_{x\in[0,1]}|f(x)|\cdot\sup_{x\in[0,1]}|f(x)-P_n(x)|\to 0 \text{ as } n\to\infty,)$$

And, by the assumption, for any $n\in\mathbb{N}$,

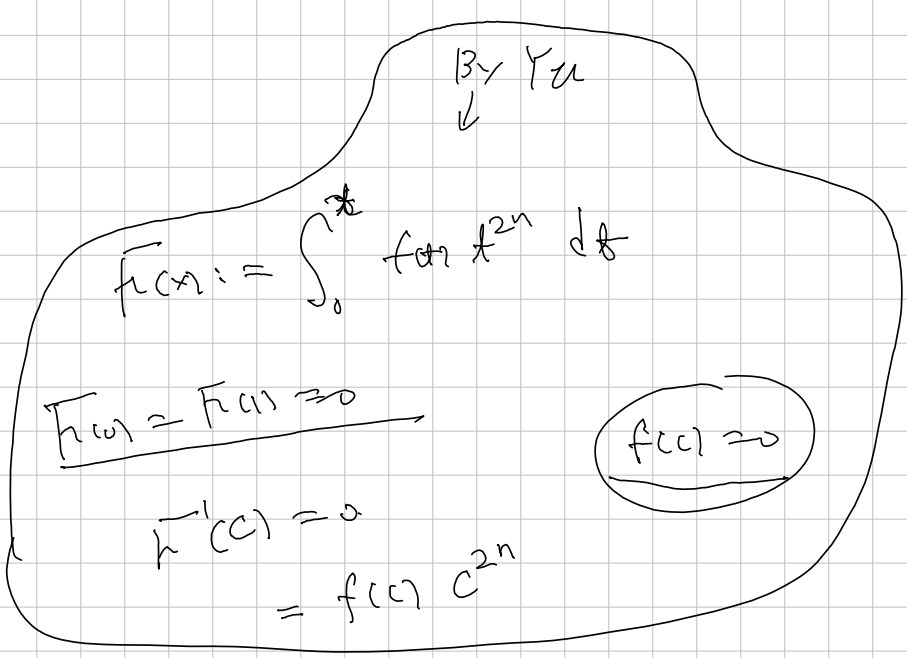
$$\int_0^1 f(x)\cdot P_n(x^2)\,dx = 0, \text{ since } P_n(x^2)\in \operatorname{Span}\{x^{2n-2}\mid n\in\mathbb{N}\},$$

Consequently, $\int_0^1 f^2\,dx = 0$, so that $f^2=0$, since $f^2\ge 0$, so $f=0$. ∎

Debris

Since f is continuous on the compact set $[0,1]$, by the stone-weierstrass theorem, there exists a sequence of polynomial $\{P_n\}$ such that

$P_n\to f$ uniformly.



7. (10 pts.) For $a \in \mathbb{R}$, define $f_n(x) = n^a x(1-x^2)^n$ on $[0,1]$.

(a) Determine the pointwise limit of f_n, and find exactly those a for which the convergence is uniform.

(b) Determine $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx$ for every a, allowing the value $+\infty$.

a). For $x=0$ or 1, $f_n(x)=0$. Let $x\in(0,1)$ be given. Then, for any a,

$f_n(x)=n^a\cdot x\cdot(1-x^2)^n \to 0$ as $n\to\infty$, since $0<|-x^2<1$.

Therefore $f_n\to 0$ pointwise on $[0,1]$. Now, observe that

$$\frac{d}{dx}\left(x(1-x^2)^n\right)=(1-x^2)^n+nx\cdot(1-x^2)^{n-1}\cdot(-2x)=(1-x^2)^{n-1}\cdot(1-(2n+1)x^2).$$

So, f_n has a maximum value at $x=1/\sqrt{2n+1}$. Therefore

$$\sup_{x\in[0,1]}\left|n^a x(1-x^2)^n\right|=n^a\cdot\frac{1}{\sqrt{2n+1}}\cdot\left(\frac{2n}{2n+1}\right)^n \to 0 \text{ as } n\to\infty \text{ if and only if}$$

$a<\frac{1}{2}$, since $\left(\frac{2n}{2n+1}\right)^n=\left(1+\frac{1}{2n}\right)^{2n\cdot-\frac{1}{2}}\to e^{-\frac{1}{2}}$ as $n\to\infty$.

Now, $a<\frac{1}{2}$ is the condition we desired to uniformly convergence.

b). Given $n\in\mathbb{N}$, let $1-x^2=u$, then

$$\int_0^1 n^a x(1-x^2)^n\,dx=\int_1^0 n^a\cdot\left(-\frac{1}{2}\right)\cdot u^n\,du=\frac{n^a}{2}\cdot\int_0^1 u^n\,du=\frac{n^a}{2}\cdot\frac{1}{n+1}.$$

So, the limit is

$$\lim_{n\to\infty}\int_0^1 f_n(x)\,dx=\begin{cases}0 & \text{if } a<1\\ 1/2 & \text{if } a=1\\ \infty & \text{if } a>1.\end{cases}$$

☑

8. (10 pts.) Determine, counting multiplicities, the number of zeros of

$$F(z) = z^5 + 4z + e^z$$

in each of the regions $|z| < 1$ and $1 < |z| < 2$. Justify that no zero lies on either boundary circle.

Proof. Let $f(z) = z^5$ and $g(z) = 4z + e^z$. Then, on $|z| = 2$,
$|f| = 32$ and $|g| \le |4z| + |e^z| \le 8 + e^2 < 32$,
So that $f + g = F$ has five zeros inside $|z| = 2$.

Now, let $f(z) = z^5 + 4z$ and $g(z) = e^z$. Then
$|f| \ge ||4z| - |z^5|| = 3$ and $|g| = |e^z| \le e < 3$,
so $f + g = F$ has only one zero inside $|z| = 1$.

Moreover, on $|z| = 1$, $|F(z)| \ge 4 - e - 1 > 0$,
So that F has no zero on $|z| = 1$.

Finally, the number of roots is:

$$\begin{cases} 1 & |z| < 1 \\ 4 & 1 < |z| < 2. \end{cases}$$

□

9. (10 pts.) Let $f \in C([0,\pi],\mathbb{R})$. Determine a necessary and sufficient condition on f for the boundary value problem

$$y'' + y = f(x), \qquad y(0) = y(\pi) = 0$$

to have a solution $y \in C^2([0,\pi])$. When the condition holds, find all solutions.

Since the characteristic is $r^2+1=0$, the homogeneous solution is

$$Y_h(x) = C_1 \cdot \cos x + C_2 \cdot \sin x$$

Calculate the Wronskian: $W(\cos x, \sin x) = 1$.

Now, the particular solution is

$$Y_p(x) = \int_0^x \det\begin{pmatrix} \cos t & \cos x \\ \sin t & \sin x \end{pmatrix} \cdot f(t) \cdot \frac{1}{W}\, dt$$

$$= \sin x \cdot \int_0^x \cos t \cdot f(t)\, dt - \cos x \cdot \int_0^x \sin t \cdot f(t)\, dt$$

Applying the boundary condition for $Y = Y_h + Y_p$:

$$Y(0) = C_1 + 0 + 0 = 0 \Rightarrow C_1 = 0$$

$$Y(\pi) = 0 \cdot (-1) + C_2 \cdot 0 + 0 + \int_0^\pi \sin t \cdot f(t)\, dt = 0$$

Therefore, the solution exists if and only if

$$\int_0^\pi \sin t\, f(t)\, dt = 0.$$

And, the general solution is

$$Y(x) = C \cdot \sin x + Y_p(x) \quad (C \in \mathbb{R}).$$

10. (10 pts.) Define $F : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$F(x, y) = (e^x \cos y, e^x \sin y).$$

(a) Prove that F is a local diffeomorphism at every point, but is not injective.

(b) Determine the image of $\mathbb{R} \times (-\pi, \pi)$ and prove that the restriction of F to this set is a diffeomorphism onto its image.

a). The Jacobian of F is

$$DF = \begin{pmatrix} e^x \cos y & e^x \sin y \\ e^x \cdot -\sin y & e^x \cos y \end{pmatrix}$$

and $\det DF = e^{2x} \cdot (\cos^2 y + \sin^2 y) = e^{2x} \neq 0$ for all $(x, y) \in \mathbb{R}^2$

so by the inverse function theorem, F is locally diffeomorphic.

But, $F(0, 0) = F(0, 2\pi)$, so it is not injective.

b)

11. (5 pts.) Let X be a compact Hausdorff space and let $A \subset X$ be nonempty and closed. Let X/A be the quotient obtained by identifying all points of A to one point and making no other identifications. Prove that X/A is Hausdorff.

Proof. Let $[a], [b] \in X/A$ be given with $[a] \neq [b]$.
If $a, b \in A$, then $[a] = A = [b]$, so contradiction.
Assume $a, b \in X \setminus A$. Then, $[a] = \{a\}$ and $[b] = \{b\}$.
Meanwhile, for a canonical projection $\pi : X \to X/A ; x \mapsto [x]$,

$U \subseteq X/A$ is open $\Leftrightarrow \pi^{-1}[U] = \{x \in X \mid \pi(x) = [x] \in U\}$ is open

Since $a \neq b$ and X is hausdorff, there exist two disjoint open set
$W, V \subset X$ such that $a \in W$, $b \in V$.
Since A is closed, $W \cap (X \setminus A)$ and $V \cap (X \setminus A)$ are open sets,
and since $\pi|_{X \setminus A}$ is one-to-one,

$\pi[W \cap (X \setminus A)]$, $\pi[V \cap (X \setminus A)]$ are disjoint open sets
containing $[a]$ and $[b]$, respectively.

WLOG, assume $a \in X \setminus A$ and $b \in A$.
Since compact Hausdorff space is normal, so also regular,
there are disjoint open sets C, D such that

$a \in C$ and $b \in A \subset D$, so that $C \cap A = \emptyset$.

Now, since $\pi^{-1}[\pi[C]] = \bigcup_{y \in C} [y] = C$ and
$\pi^{-1}[\pi[D]] = D$, so that $\pi[C]$, $\pi[D]$ are open.
Moreover, $\pi[C] \cap \pi[D] \neq \emptyset$, and so X/A is Hausdorff. ∎

1. (10 pts.) Let $V = P_3(\mathbb{R})$ and define $T : V \to V$ by $(Tp)(x) = p(x+1) - p(x)$.

(a) Find the characteristic and minimal polynomials of T.

(b) Find an ordered basis of V in which the matrix of T is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$(x+1)^2 - x^2$
$= 2x+1$

Proof. a). Notice that: $\beta = \{1, x, x^2, x^3\}$ is a basis for V.
The matrix representation is

$$[T]_\beta = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$T(1) = 0$
$T(x) = x+1-x = 1$
$T(x^2) = (x+1)^2 - x^2 = 2x+1$
$T(x^3) = 3x^2+3x+1$

So, the characteristic polynomial is t^4, and since $[T]_\beta^3 \neq 0$, the minimal polynomial is also t^4.

b). observe that:

$$[T]_\beta^2 = \begin{bmatrix} 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad [T]_\beta^3 = \begin{bmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, $\ker T^3 = \operatorname{Span}\{1, x, x^2\}$, $\ker T^2 = \operatorname{Span}\{1, x\}$.
Take $\alpha = x^3$. Then,

$T(\alpha) = 3x^2+3x+1$, $T^2(\alpha) = 6x+6$ $T^3(\alpha) = 6$

Therefore, $\beta' = \{T^3(\alpha), T^2(\alpha), T(\alpha), \alpha\}$ satisfies

$$[T]_{\beta'} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left(= [T - \lambda I]_{\beta'} + \lambda [I]_{\beta'} \right).$$

☑

2. (10 pts.) Let T be a normal operator on a finite-dimensional complex inner product space V, and let $W \subseteq V$ be a T-invariant subspace.

(a) Prove that W is also T^*-invariant.

(b) Prove that $W^\perp$ is T-invariant and that $T|_W$ is normal.

Proof. a). Since T is normal, by the spectral theorem,

$T = c_1 T_1 + \cdots + c_k T_k$ where c_i's are distict eigenvalues of T,
T_i is orthogonal projection onto $\ker(T - c_i)$

Now, let $f(x) = \sum_{i=1}^{k} \prod_{j \neq i} \overline{c_i} \cdot \frac{(x - c_j)}{(c_i - c_j)}$.

Then, f is a polynomial such that $f(c_i) = \overline{c_i}$ for all $1 \le i \le k$.

Now, $T^* = \overline{c_1} T_1^* + \cdots + \overline{c_k} \cdot T_k^* = f(c_1) T_1 + \cdots + f(c_k) T_k = f(T)$

So $T^*[W] = f(T)[W] \subseteq W$.

b). Given $\alpha \in W^\perp$, $T(\alpha) \in W^\perp$

$\Leftrightarrow \forall \alpha \in W^\perp, \forall \beta \in W, \langle T(\alpha), \beta \rangle = 0$

And, by a), W is T^*-invariant and so that given $\alpha \in W^\perp$, $\beta \in W$,

$\langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle = 0$, so $W^\perp$ is T-invariant.

Meanwhile, let $\alpha \in W$ be given. Then,

$T \circ T^*(\alpha),\ T^* \circ T(\alpha) \in W$ and $T T^*(\alpha) = T^* T(\alpha)$,

So $T|_W$ is normal. ☑

Note: let $W \subset V$ is T-invariant, then $(T|_W)^* = P_W \circ T^*|_W$, because given $x, y \in W$,

$\langle T|_W(x), y \rangle = \langle T(x), y \rangle = \langle x, T^*(y) \rangle = \langle x, \underbrace{P_W T^*(y)}_{\in W} + \underbrace{(T^*(y) - P_W T^*(y))}_{\in W^\perp} \rangle$

$= \langle x, P_W T^*(y) \rangle = \langle x, (T|_W)^*(y) \rangle$

Note:

Let T be a linear operator on a finite-dimensional vector space V, and let W be a T-invariant subspace of V. Suppose that $v_1, v_2, \ldots, v_k$ are eigenvectors of T corresponding to distinct eigenvalues. Prove that if $v_1 + v_2 + \cdots + v_k$ is in W, then $v_i \in W$ for all i. *Hint:* Use mathematical induction on k.

Prove that the restriction of a diagonalizable linear operator T to any nontrivial T-invariant subspace is also diagonalizable. *Hint:* Use the result of Exercise 23.

If $k=1$, then we are done. If for $k=n$, the statement is hold.
Suppose that $v_1, \cdots, v_{n+1}$ are eigenvectors corresponding to distinct eigenvalues and $v_1 + \cdots + v_n + v_{n+1} \in W$. Since W is T-invariant,

$$T(v_1) + \cdots + T(v_{n+1}) = \lambda_1 v_1 + \cdots + \lambda_{n+1} v_{n+1} \in W. \quad (\lambda_i \text{ is the eigen-})$$

And, $\lambda_{n+1} v_1 + \cdots + \lambda_{n+1} v_{n+1} \in W$, so that

$$(\lambda_1 - \lambda_{n+1}) v_1 + \cdots + (\lambda_n - \lambda_{n+1}) v_n + \underbrace{(\lambda_{n+1} - \lambda_{n+1})}_{=0} \cdot v_{n+1} \in W,$$

so by the hypothesis of induction, $v_1, \cdots, v_n \in W$ and thus $v_{n+1} \in W$. ②

Let $\alpha \in W$ be given. Then,

$$\alpha = \alpha_1 + \cdots + \alpha_k \quad \text{for some } \alpha_i \in \ker(T - c_i I)$$

By the lemma above, each $\alpha_i \in W$, so that $\alpha_i \in W \cap \ker(T - c_i I)$.
Therefore,

$$W = \sum W \cap \ker(T - c_i I)$$
$$= \bigoplus W \cap \ker(T - c_i I)$$

4. (10 pts.) The group S_4 acts on the three partitions

$$P_1 = \{\{1,2\},\{3,4\}\}, \quad P_2 = \{\{1,3\},\{2,4\}\}, \quad P_3 = \{\{1,4\},\{2,3\}\}$$

by permuting the labels.

(a) Determine the kernel and image of the resulting homomorphism $S_4 \to S_3$.

(b) Describe the stabilizer of P_1 and determine its order.

Proof. a) Consider the afforded Homomorphism by the action

$$\varphi : S_4 \to S_{\{P_1,P_2,P_3\}} ; \sigma \mapsto \{\{\{a,b\},\{c,d\}\} \mapsto \{\{\sigma(a),\sigma(b)\},\{\sigma(c),\sigma(d)\}\}$$

Then, $\sigma \in \ker \varphi$ if and only if $\varphi(\sigma)(P_i) = P_i$ for all $1 \le i \le 3$.

Meanwhile, $\varphi(\sigma)(P_1) = P_1 \iff \sigma[\{1,2\}] = \{1,2\}$ or $\{3,4\}$.

$\varphi(\sigma)(P_2) = P_2 \iff \sigma[\{1,3\}] = \{1,3\}$ or $\{2,4\}$

$\varphi(\sigma)(P_3) = P_3 \iff \sigma[\{1,4\}] = \{1,4\}$ or $\{2,3\}$.

So, the possible σ are: $\sigma = id$ and

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

thus, since $|\ker \varphi| = 4$, by the first-isomorphism theorem, the Homomorphism φ satisfies $S_4/\ker \varphi \cong \operatorname{Im} \varphi = S_3$.

b). $\operatorname{Stab}(P_1) = \{\sigma \in S_4 \mid \varphi(\sigma)(P_1) = P_1\}$.

As observation in a), the possible σ are

$\sigma = id$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$,

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$, $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$

it has order of 8. ☐

Note: $|G : \operatorname{Stab}(\alpha)| = |\operatorname{orb}(\alpha)|$, and since φ is onto, so $|\operatorname{orb}(\alpha)| = 3$.

5. (10 pts.) Let $F = \mathbb{F}_2[t]/(t^3 + t + 1)$ and let $\alpha = [t] \in F$.

(a) Prove that F is a field, and find all roots of $x^3 + x + 1$ in F.

(b) Show that $\mathrm{Tr}(u) = u + u^2 + u^4$ belongs to $\mathbb{F}_2$ for every $u \in F$, and list every element of its kernel.

a). Since $0^3+0+1=1\neq 0$, and $1^3+1+1=1\neq 0$, so that t^3+t+1 is irreducible over $\mathbb{F}_2$, so the quotient $F_2[t]/(t^3+t+1)$ is a field. Meanwhile, since $\alpha^3+\alpha+1=t^3+t+1+(t^3+t+1)$, so $\alpha\in F$ is a root of x^3+x+1 in F. And, since the Frobenious map

$$\sigma: F\to F: a\mapsto a^2$$ is automorphism, if $x^3+x+1=0$, then

$$0=\sigma(0)=\sigma(x^3+x+1)=(\sigma(x))^3+\sigma(x)+1$$

So that α^2 and $\alpha^4=\alpha^2+\alpha$ are also root of x^3+x+1. and $\alpha, \alpha^2, \alpha^2+\alpha$ are distinct, because

$$f(t)+(t^3+t+1)=0\in F \Leftrightarrow t^3+t+1 \mid f(t),$$

so $\alpha-\alpha^2, (\alpha^2+\alpha)-\alpha, (\alpha^2+\alpha)-\alpha^2$ are not zero. Since x^3+x+1 has degree 3, these are all zero of the polynomical.

b). Since $(\mathrm{Tr}\,u)^2=u^8+u^4+u^2$ and since $F^\times\cong C_7$, $u^8=1\cdot u=u$,

so $$=u+u^4+u^2=\mathrm{Tr}(u).$$

Therefore, $\mathrm{Tr}\,u\in\{0,1\}=\mathbb{F}_2$. And,

$$\mathrm{Tr}(u)=u+u^2+u^4=u(1+u+u^3)=0$$

and since the field has no zero divisor,

$$\ker \mathrm{Tr}=\{0,\alpha,\alpha^2,\alpha^2+\alpha\}.$$ ☑

8. (10 pts.) Let f be entire and suppose there is $C > 0$ such that

$$|f(z)| \le C(1 + |z|^{3/2}) \qquad (z \in \mathbb{C}).$$

(a) Prove that f is a polynomial of degree at most 1.

(b) Would the conclusion follow if this bound were assumed only for real z? Justify your answer.

Proof. a), Since f is entire, we can write

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot z^n \quad (a_n \in \mathbb{C})$$

is valid. Meanwhile, by the Cauchy's inequality,

$$|f^{(n)}(0)| = \frac{n!}{2\pi} \cdot \left| \int_{|z|=r} \frac{f(z)}{z^{n+1}} \, dz \right|$$

$$\le \frac{n!}{2\pi} \cdot 2\pi r \cdot \frac{1}{r^{n+1}} \cdot \sup_{|z|=r} |f(z)|$$

$$= \frac{n!}{r^n} \cdot \sup_{|z|=r} |f(z)|$$

So that $\left| \frac{f^{(n)}(0)}{n!} \right| \le C \cdot \left(\frac{1}{r^n} + r^{\frac{3}{2}-n} \right) \to 0$ as $r \to \infty$, when $\frac{3}{2} - n < 0$.

Therefore, $f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n = f(0) + f(1) \cdot z$.

b). Let $f(z) = \sin z$. Then, for all $x \in \mathbb{R}$,

$$|f(x)| = |\sin x| \le 1$$

but it is <u>not</u> a polynomial. ☒

1. (10 pts.) Let A be an $m \times n$ complex matrix and B an $n \times m$ complex matrix.

(a) Prove that $\det(I_m + tAB) = \det(I_n + tBA)$ for every $t \in \mathbb{C}$.

(b) Deduce that AB and BA have the same nonzero eigenvalues, with the same algebraic multiplicities.

Proof. a). Let $M = \begin{pmatrix} I_m & tA \\ -B & I_n \end{pmatrix}$. Then,

$\begin{pmatrix} I_m & -tA \\ 0 & I_n \end{pmatrix} \cdot \begin{pmatrix} I_m & tA \\ -B & I_n \end{pmatrix} = \begin{pmatrix} I_m + tAB & 0 \\ -B & I_n \end{pmatrix}$, so $\det M = \det(I_m + tAB)$,

$\begin{pmatrix} I_m & tA \\ -B & I_n \end{pmatrix} \begin{pmatrix} I_m & -tA \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & 0 \\ -B & tBA + I_n \end{pmatrix}$, so $\det M = \det(I_n + t \cdot BA)$.

So, $\det(I_m + tAB) = \det M = \det(I_n + tBA)$.

b). Since $\mathbb{C}$ is algebraically closed, every polynomial over $\mathbb{C}$ splits, so

$\det(I_m + tAB) = \prod_{i=1}^{m} (\lambda_i t + 1)$ for some $\lambda_i \in \mathbb{C}$ and

$\det(I_n + tBA) = \prod_{k=1}^{n} (\mu_k t + 1)$ for some $\mu_k \in \mathbb{C}$.

Since $\det(I_m + tAB) = \det(I_n + tBA)$, these are same polynomial,

So b) is proved. ☺

[Kaist. 2021 spring.
Let $A \in M_{n\times n}(\mathbb{R})$. Prove that

$$\lim_{h\to 0} \frac{\det(A+h\cdot B)-\det(A)}{h} = \det A \cdot \operatorname{trace}(A^{-1}\cdot B).$$

Proof. Since A is invertible, given $|h|>0$,

$$\frac{\det(A+hB)-\det(A)}{h} = \det(A)\cdot \frac{\det(I+h\cdot A^{-1}\cdot B)-\det(I)}{h}.$$

Meanwhile, consider the characteristic polynomial

$$\det(I+h\cdot A^{-1}\cdot B) = a_n\cdot h^n + a_{n-1}h^{n-1} + \cdots + a_1\cdot h + a_0,$$

Since $a_0 = \det(I+0\cdot A^{-1}B) = \det I = 1$, so that

$$\lim_{h\to 0} \frac{\det(I+hA^{-1}B)-\det(I)}{h} = a_1.$$

And since $a_1 = \operatorname{trace}(A^{-1}B)$, because: given $C\in M_{n\times n}(\mathbb{R})$,

$$\det(I+t\cdot C) = \det(e_1+t\cdot Ce_1,\ e_2+t\cdot Ce_2,\ \cdots,\ e_n+t\cdot Ce_n)$$
$$= 1+\operatorname{tr}(C)\cdot t + \cdots$$

So, proved.

3. (10 pts.) Let S and T be diagonalizable operators on a finite-dimensional complex vector space V. Suppose that $ST = TS$. Prove that V has a basis consisting of common eigenvectors of S and T. No inner product is assumed.

Proof. Since T is diagonalizable, for distinct eigenvalues $c_1, \dots, c_k$,

$$V = W_1 \oplus \cdots \oplus W_k \quad \text{where } W_i = \ker(T - c_i I), \ \forall i.$$

Let $1 \le i \le k$ be fixed. Then, given $\alpha \in W_i$,

$c_i \cdot S(\alpha) = S(c_i \alpha) = S(T(\alpha)) = TS(\alpha)$, so that $S(\alpha) \in \ker(T - c_i I) = W_i$

Therefore, each W_i is S-invariant, by $TS = ST$.

Now, since S is diagonalizable, each restriction

$S|_{W_i}$ is also diagonalizable, since $m_{S|_{W_i}} \mid m_S$,

So that $S = S|_{W_1} + \cdots + S|_{W_k}$

is also diagonalizable, therefore take eigenvectors in W_i for $S|_{W_i}$, the union of these is diagonalizing eigenvectors for S and also for T. ☐

1. (10 pts.) Let $T : \mathbb{C}^6 \to \mathbb{C}^6$ be nilpotent, with $\operatorname{rank} T = 4$ and $\operatorname{rank}(T^2) = 2$.

(a) Determine all possible minimal polynomials of T and the corresponding values of $\operatorname{rank}(T^3)$.

(b) Give a matrix realizing each possibility.

Proof. a), By the dimension theorem, $\dim \ker T = 2$, $\dim \ker T^2 = 4$. Since T is nilpotent, so $T^6 = 0$, thus T has only Jordan block corresponding to 0. Since $\dim \ker T = 2$, T has 2 numbers Jordan block to 0, and since $\dim \ker T^2 = 4$, both two blocks have size bigger or equals to 2, So the possible cases of block sizes are $4+2$ or $3+3$.

In the case of $4+2$,

$$T \sim \begin{bmatrix} 0 & 1 & & & & \\ & 0 & 1 & & & \\ & & 0 & & & \\ & & & 0 & 1 & \\ & & & & 0 & \end{bmatrix} \oplus \begin{bmatrix} 0 & 1 \\ & 0 \end{bmatrix} = A,$$ and since $A^4 = 0$ but $A^3 \neq 0$, the minimal polynomial of T is x^4.

And, $\dim \ker T^3 = 5$, so $\operatorname{rank} T^3 = 1$.

In the case of $3+3$,

$$T \sim \begin{bmatrix} 0 & 1 & & & & \\ & 0 & 1 & & & \\ & & 0 & & & \\ & & & 0 & 1 & \\ & & & & 0 & 1 \\ & & & & & 0 \end{bmatrix} = B,$$ and since $B^3 = 0$ but $B^2 \neq 0$, So the minimal polynomial is x^3.

And, $\dim \ker T^3 = 6$, so $\operatorname{rank} T^3 = 0$.

It also proves b). ∎

2. (10 pts.) Let T be a positive definite self-adjoint operator on a finite-dimensional complex inner product space V.

(a) Prove that there is a unique positive definite self-adjoint operator S such that $S^2 = T$.

(b) Prove that every linear operator A satisfying $AT = TA$ also satisfies $AS = SA$.

Proof. a). Since T is Hermitian, there is an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ such that $[T]_\beta$ is diagonal. And, for each eigenvalue $\lambda_i \in \mathbb{C}$ corresponding to α_i, $\langle T\alpha_i, \alpha_i \rangle = \langle \lambda_i \alpha_i, \alpha_i \rangle = \lambda_i > 0$, since T is positive definite.
Now, let $S: V \to V$ be a linear map such that

$[S]_\beta = \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix}$. Then, $[S]_\beta^2 = [S^2]_\beta = [T]_\beta$, so the existence is proved.

To show that the uniqueness, let R be a positive definite operator such that $R^2 = T = S^2$. Then, for each $\alpha_i \in \beta$,

$R^2\alpha_i = S^2\alpha_i = (\sqrt{\lambda_i})^2\alpha_i$, so that $(R + \sqrt{\lambda_i} I) \circ (R - \sqrt{\lambda_i} I)\alpha_i = 0$.

If $(R - \sqrt{\lambda_i} I)\alpha_i \neq 0$, then $\ker(R + \sqrt{\lambda_i} I) > \{0\}$, so R has negative eigenvalue, but it is contradiction to R is positive definite. So.
$R\alpha_i = \sqrt{\lambda_i}\alpha_i = S\alpha_i$ for all α_i, so $R = S$.

b). Given $\alpha \in \ker(T - \lambda_i I)$, $AT\alpha = A\lambda_i\alpha = \lambda_i A\alpha = TA\alpha$, so $A\alpha \in \ker(T - \lambda_i I)$.
And, since $T - \lambda_i I = (S + \sqrt{\lambda_i} I) \cdot (S - \sqrt{\lambda_i} I)$, $\ker(S - \sqrt{\lambda_i} I) \subseteq \ker(T - \lambda_i I)$ and if $(T - \lambda_i I)\alpha = 0$ but $(S - \sqrt{\lambda_i} I)\alpha \neq 0$, then

$0 = (T - \lambda_i I)\alpha = (S + \sqrt{\lambda_i} I)(S - \sqrt{\lambda_i} I)\alpha$ so that S has a negative eigenvalue, so contradiction. Therefore, $\ker(S - \sqrt{\lambda_i} I) = \ker(T - \lambda_i I)$.
Now, for each α_i,

$AS\alpha_i = A\sqrt{\lambda_i}\alpha_i$ and $SA\alpha_i = \sqrt{\lambda_i} A\alpha_i$,

So $AS = SA$. ∎

[Similar problem.
Let S, T be positive definite such that $ST = TS$.
Prove that ST is also positive definite.
Proof. By the spectral decomposition,
$T = \lambda_1 T_1 + \cdots + \lambda_k T_k$ where λ_i's are distinct eigenvalues of T,
T_i's are orthogonal projection onto $\ker(T - \lambda_i I)$.
Since T is positive definite, each λ_i is positive, so we can define
$\sqrt{T} = \sqrt{\lambda_1}\, T_1 + \cdots + \sqrt{\lambda_k}\, T_k$ and it satisfies that $(\sqrt{T})^2 = T$ and $\sqrt{T} = \sqrt{T}^*$.
Meanwhile, since $ST = TS$, given $\alpha \in \ker(T - \lambda_i I)$,
$TS\alpha = ST\alpha = S\lambda_i\alpha = \lambda_i S\alpha$, so $S\alpha \in \ker(T - \lambda_i I)$. Now,
given $\alpha \in \ker(T - \lambda_i I)$,
$S\sqrt{T}\alpha = S \cdot \sqrt{\lambda_i}\alpha = \sqrt{\lambda_i} S\alpha = \sqrt{T} S\alpha$, so that $S\sqrt{T} = \sqrt{T}S$,
($S\alpha \in \ker(T - \lambda_i I) = \ker(\sqrt{T} - \sqrt{\lambda_i} I)$)
Finally, for each $\alpha \in V \setminus \{0\}$,
$\langle TS\alpha, \alpha \rangle = \langle \sqrt{T} S\alpha, \sqrt{T}\alpha \rangle = \langle S\sqrt{T}\alpha, \sqrt{T}\alpha \rangle > 0$ since S is positive definite,
and $\sqrt{T}\alpha \neq 0$, so proved. ∎

[Debris.
Since S and T both are Hermitian,
$(ST)^* = T^* S^* = TS = ST$, so ST is also Hermitian.
Meanwhile, let $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ be a basis for V such that
$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$ for some positive values $\lambda_1, \cdots, \lambda_n$.
Then, for each α_i,
$TS\alpha_i = ST\alpha_i = S\lambda_i\alpha_i = \lambda_i S\alpha_i$, so that $S\alpha_i \in \ker(T - \lambda_i I)$,
so that $\ker(T - \lambda_i I)$ is S-invariant. Therefore, since S is diagonalizable,
$S|_{\ker(T - \lambda_i I)}$ is diagonalizable, so that each α_i,
$ST\alpha_i = S\lambda_i\alpha_i = \lambda_i c_i \alpha_i$ where c_i is eigenvalue for α_i of S,
and since S is positive definite, $c_i > 0$ and so that $\lambda_i c_i > 0$,
so proved. ∎

4. (10 pts.) In $\mathbb{Z}[x]$, consider the ideals

$$I = (2, x^2 + x + 1), \qquad J = (2, x^2 + x).$$

(a) Prove that I is maximal and that J is not prime.

(b) Find all prime ideals of $\mathbb{Z}[x]$ containing J.

Proof. a). Define a map

$\varphi : \mathbb{Z}[x] \to \mathbb{Z}_2[x]/(x^2+x+1) ; \sum a_n x^n \mapsto \sum (a_n \bmod 2) \cdot x^n + (x^2+x+1)$.

KAIST, 2027 Spring

1. Let $f: \mathbb{C} \to \mathbb{C}$ be an entire function which has a pole at $z=\infty$. Prove that f is a polynomial.

Proof. Suppose that $f(1/z)$ has a pole of order m at $z=0$.
That is, $\lim_{z\to 0} z^m \cdot f\left(\frac{1}{z}\right) = A$, for some non-zero $A \in \mathbb{C}$.

Now, given $\varepsilon = 1$, there exists $\delta > 0$ such that

$$|z| < \delta \Rightarrow \left| z^m \cdot f\left(\frac{1}{z}\right) - A \right| < \varepsilon = 1,$$

Put $z = \frac{1}{z'}$, then equivalently

$$|z'| > \frac{1}{\delta} \Rightarrow \left| f(z')/z'^m - A \right| < 1$$

$$\Rightarrow |f(z')| \le (1+|A|) \cdot |z'|^m.$$

Meanwhile, by the Cauchy's inequality, if $n > m$,

$$|f^{(n)}(0)| \le \frac{n!}{2\pi i} \cdot \frac{1}{R^n} \cdot \sup_{|z|=R} |f(z)|$$

$$\le n! \cdot R^{m-n} \to 0 \quad \text{as } R \to \infty$$

So $f = \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{m} a_n z^n$ is a polynomial of degree at most m. ∎

KAIST 2021, Spring

Prove that

$$\int_0^{2\pi} \frac{1}{1-2\alpha\cdot\cos\theta+\alpha^2}\,d\theta = \frac{2\pi}{|\alpha^2-1|} \quad (\alpha \neq \pm 1, \text{ real}).$$

Proof. Let $z(\theta) = e^{i\theta}$ $(0 \le \theta \le 2\pi)$. Then, $dz = i\cdot e^{i\theta}\,d\theta$.

And, since $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} = \frac{1}{2}\left(z+\frac{1}{z}\right)$, therefore

$$\int_0^{2\pi} \frac{1}{1-2\alpha\cos\theta+\alpha^2}\,d\theta = \int_{|z|=1} \frac{1}{1-\alpha\left(z+\frac{1}{z}\right)+\alpha^2}\cdot\left(-\frac{i}{z}\right)dz$$

$$= \int_{|z|=1} \frac{i}{\alpha z^2-(1+\alpha^2)z+\alpha}\,dz$$

$$= \int_{|z|=1} \frac{i}{z^2-\left(\alpha+\frac{1}{\alpha}\right)z+1}\,dz \cdot \frac{1}{\alpha}$$

$$= \frac{i}{\alpha}\cdot\int_{|z|=1} \frac{1}{(z-\alpha)\left(z-\frac{1}{\alpha}\right)}\,dz.$$

Now, if $0<|\alpha|<1$, then by the Residue theorem,

$$= \frac{i}{\alpha}\cdot 2\pi i\cdot\frac{1}{\alpha-\frac{1}{\alpha}} = 2\pi\cdot\frac{1}{1-\alpha^2}.$$

And, if $|\alpha|>1$, the similarly

$$= \frac{i}{\alpha}\,2\pi i\cdot\frac{1}{\frac{1}{\alpha}-\alpha} = 2\pi\cdot\frac{1}{\alpha^2-1}$$

And, if $\alpha=0$, then trivial. So, proved. ☐

$0<r<1$ $[0,1]$ $f_n(x)=f(x^n)$ $\to f(0)$ $\to f(1)$

$A\in M_{p\times q}(F)$, $B\in M_{q\times m}$

$\operatorname{rank}(AB)\le \operatorname{rank}(B)$

2) $=\|\beta^n\|_{R^2}$ $e_i^T A=(A^T e_i)^T$ C_i

3) col

2.

3. $Ax=0$ x^n

4 $E(G)$ $|2G|$ G $|G|$

5. R $\forall a\ \exists b$ $aba=a$

6. a) $f(x_n)$

b) $C(\mathbb{R}^2)$ $\mathbb{R}$

7. $\int_0^1 g=1$, $g(x+1)=g(x)$ $2\le a<b\le 1$

$\lim_{n\to\infty}\int_a^b g(nx)\,dx=b-a$

b) $\lim_{n\to\infty}\int_0^1 h(x)g(nx)=\int_0^1 h\,dx$

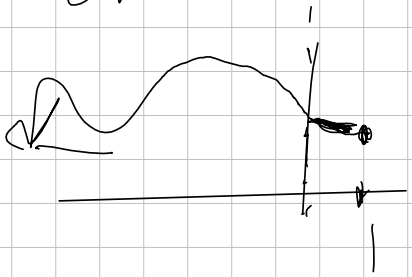
$Ax=0$

$k<n$ $\frac{[nb]}{n}\to b$

$I^{-1}Jz$

$z=\frac{x^2}{a^2}-\frac{y^2}{b^2}$

$\int_0^1 |h(x)||g(nx)-1|\,dx$



1. Let F be an arbitrary field.
a). Given $A \in F^{p\times q}$ and $B \in F^{q\times r}$, prove $\text{Row}(AB) \subseteq \text{Row}(B)$.
b). If $A \in F^{q\times q}$ and invertible, prove $\text{Row}(AB) = \text{Row}(B)$.
c). Find A, B such that A is invertible but $\text{Col}(AB) \neq \text{Col}(B)$.

2). Let P_1, P_2 be orthogonal projection onto $W_1, W_2 \subseteq \mathbb{R}^n$.
a). Prove that $W_1 \perp W_2 \iff P_1P_2 = 0$.
b). Prove that $\text{Im}\, P_1P_2 = W_1 \cap W_2$. if $P_1P_2 = P_2P_1$.

3). Let $A \in M_{n\times n}(F)$. Prove that $A^k = 0$ for some k, then
$$\det(xI_n - A) = x^n.$$

4. Prove that:
G is non-abelian group of order pq, then G has a trivial center.

5. Let R be a ring satisfying
$\forall a \neq 0$, $\exists! b \in R$ s.t. $aba = a$. Prove:
a). R has no zero divisor.
b). $bab = b$
c). R has unity 1_R.
d). R is division Ring.

6. Let $f: (M_1, d_1) \to (M_2, d_2)$ be a uniform conti. map.
a). If (x_n) is cauchy, then $(f(x_n))$ is cauchy.
b). Let $M_1 \subseteq \mathbb{R}^d$ be bounded,
$M_2 = \mathbb{R}$.
Prove f is bounded.

7. Let $y' = y(y+2)\cdot x$, $y(0) = y_0$

Find general solution and determine maximum interval the solution defined

8. Let $f : [0,1] \to \mathbb{R}$ be cont'i. map.
Define $f_n : [0,1] \to \mathbb{R} ; x \mapsto f(x^n)$.

a). Find limit function of f_n.
b). Given $r \in (0,1)$, $f_n \to F$ uniformly on $[0,r]$, F is a map in a).
c). Determine all cont'i. map $g : [0,1]$ s.t

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t $m, n \geq N \Rightarrow |g(x^m) - g(x^n)| < \varepsilon$.

9. Let $g : \mathbb{R} \to \mathbb{R}$ be cont'i. s.t

$\int_0^1 g = 1$, $g(x+1) = g(x)$.

a). Given $0 < a < b < 1$,

$\lim_{n \to \infty} \int_a^b g(nx)\, dx = b - a$.

b). Prove; given cont'i $h : [0,1] \to \mathbb{R}$,

$\lim_{n \to \infty} \int_0^1 h(x)\, g(nx)\, dx = \int_0^1 h$

10. ?

11. Infinite subset of compact. S.P. has a limit point.
12. Find fd group of;
a). $\mathbb{R}^3 \setminus S^1$ b; $T \setminus \{a, b\}$,
13. a) $f : X \to Y$ H.e. then the induced map is iso.
b).
14. Gauss. Mecm Curvature